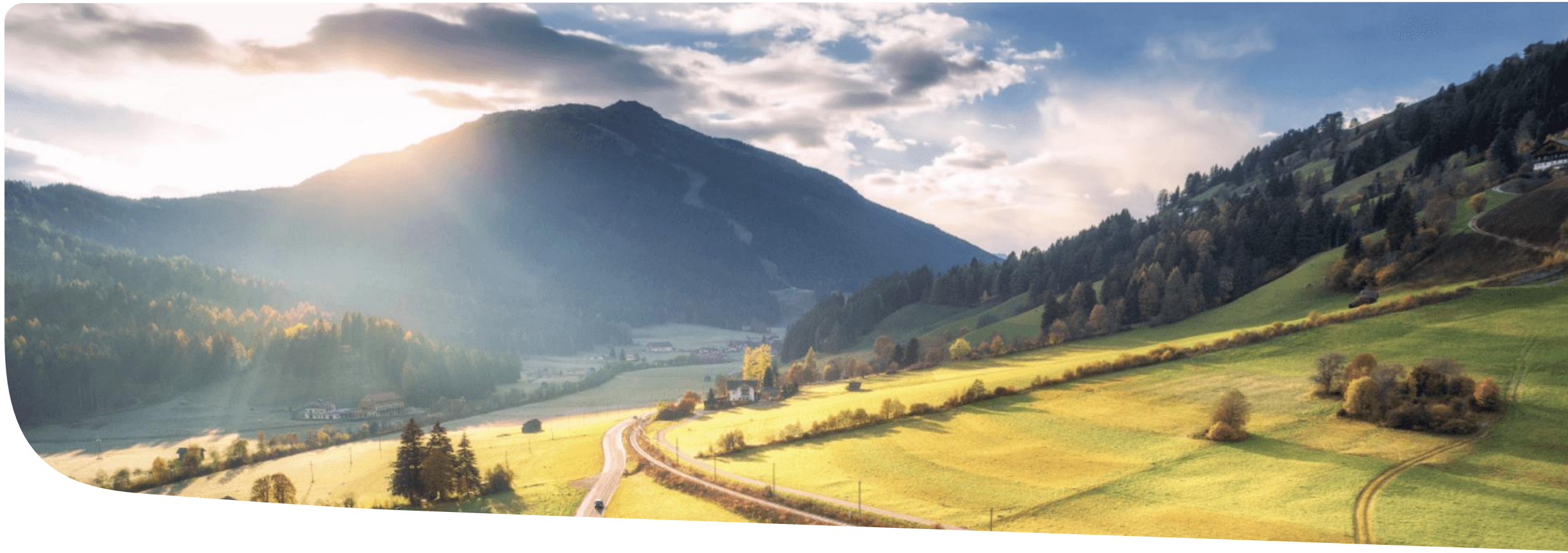
UWB-Training



UWB-Training
Basics

Basics

1. UWB Introduction
2. ToF Distance Measurement
3. CCC (MAC/UWB Frames, STS, SFD, KDF, Configuration, Pulse Shapes, Time0, Hopping, ….)
4. System Architecture/Overview
5. Ranger5 Concept (API, Radio Settings, Sequences)
6. Ranger4 (SMASTD/FS19)
7. Setup Tools

UWB-Training

Basics
Ranger5

1. NXP Delivery
2. NXP Examples
    - CW mode
    - Ranging
    - Crypto
3. NXP Tools
    - Radio Setting Generator
    - DSP Toolbox
    - Debugger Scripts)
4. NXP Demoboard
5. UWB measurements (Debugger, Spectrum analyzer)

UWB-Training
Basics FBD6 (Explanation feature, where is it implemented?)

- UWB Configuration (Channel, Sync Code, Pulse Shape Combo
- Box2Box (Initiator, Responder)
- Ranging with 1 smartphone (Responder)
- Loopback
- Testmodes (CW mode, CF mode, RX mode, XTAL Pin)
- Multiranging
- UWB Window Calculation
- Temperature Compensation
- Hopping Mode
- AoA (Angle of Arrival)
- First Path Information
- Ranging without PrePoll
- Distance Calculation with Final Data message (FDD) over CAN
- Duty Cycle Limit
- IDG Ranging
- CPD (Measurement and Machine Learning)
- CPD CoEx

UWB-Training

Practical part - Ranger5 Responder/Initiator (NXP Development Board)

1. Load and run NXP CW mode example (debugger and spectrum analyzer)
    1.1 Change channel 5, 9

2. Load and run NXP ranging example (debugger test and spectrum analyzer)
    2.1 Change channel 5, 9
    2.2 Change Sync Code
    2.3 Read First Path information values
    2.4 Read AoA values

3. Add crypto in NXP ranging example
    3.1 Add Encrypt/Decrypt Final Data payload

    3.2 Add incrementation of STS-Index

    3.3 e.g. Use the file UwbMwMsh_Kdf.c (FBD6) to generate the session keys and ranging keys

UWB-Training
Practical part - FBD6 CPD

1. CPD debugging (FBD6 HW)

UWB-Training
Practical part – FS19

1. Short demonstration of the UID-Ranging

UWB-Training

Practical part - FBD6 Responder/Initiator-Mode

1. Debugging Smartphone-Ranging (FBD6 HW)
   → Timing Measurements
   → Robustness tests
   → Error Analysis

2. Debugging Box2Box-Ranging (FBD6 HW)
   → Timing Measurements
   → Robustness tests
   → Error Analysis

3. Debugging UWB Subfeatures
   e.g. Loopback, CPD CoExistence, UEBTestmodes

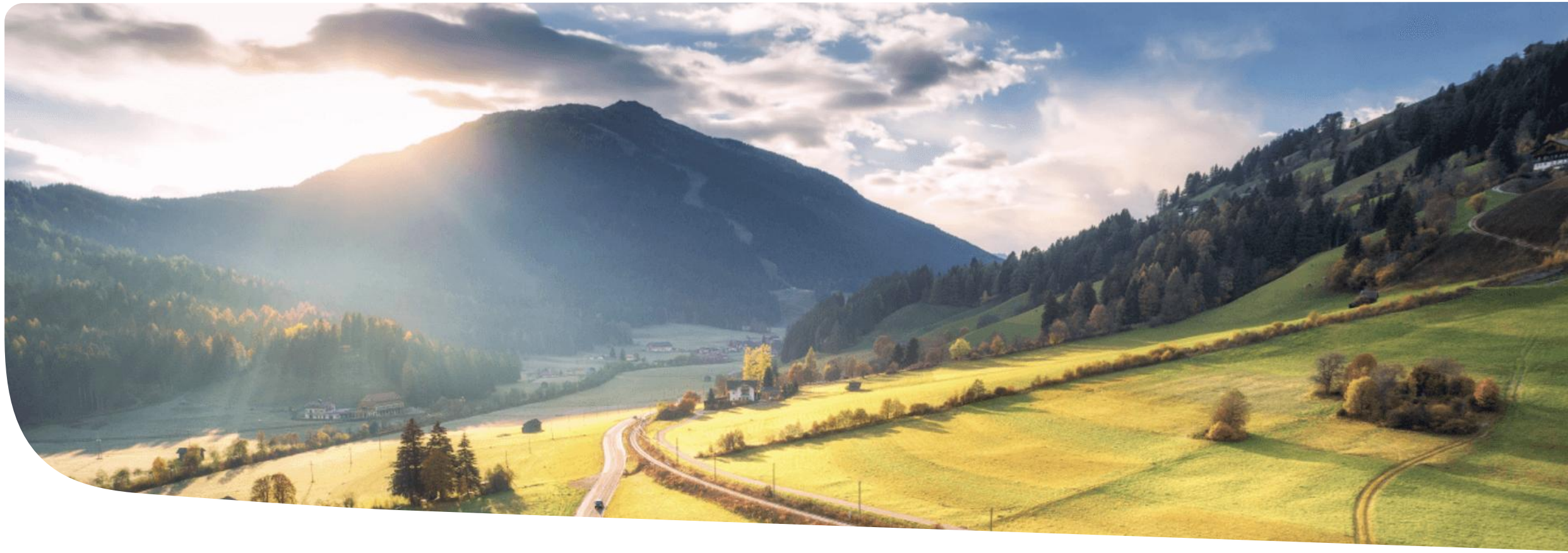
4. Test of UWB Features with Johannes Klinker

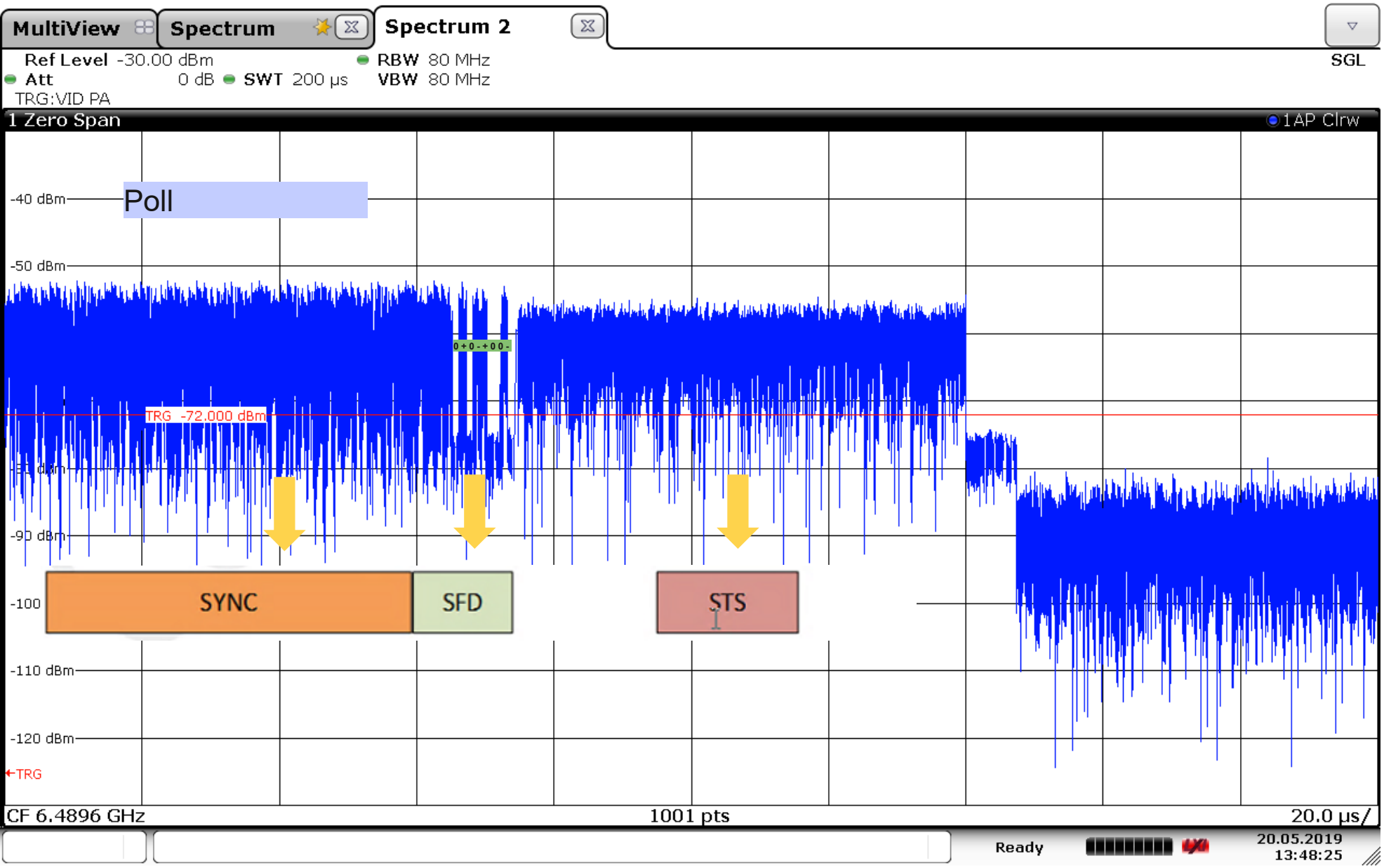
UWB-Training HVNEinarbeitungPlan.xlsx

Date	Agenda	Dauer	Name	Option	Option
08. Apr	First Day				
09. Apr	UWB Introduction	4	Dietmar	Blake	
	ToF Distance Measurement	4	Blake	Dietmar	
10. Apr	CCC	8	Dietmar	Blake	Peter D.
11. Apr	CCC	8	Dietmar	Blake	Peter D.
12. Apr	System Architecture/Overview	4	Peter D.	Dietmar	Blake
	Ranger5 Concept (API, Radio Settings, Sequences)	4	Dietmar	Blake	Frank H.
13. Apr					
14. Apr					
15. Apr	Ranger4 Concept (SMASTD/FS19)	4	Blake	Dietmar	
	Setup Tools	4	Florian J.	Blake	Frank H.
16. Apr	Short Overview - NXP Delivery	4	Dietmar	Blake	
	Short Overview - NXP Demoboard	2	Dietmar	Blake	
	Short Overview - NXP Tools (Radio Setting Generator)	1	Dietmar	Blake	
	Short Overview - NXP Tools (DSP Toolbox, Debugger Scripts)	1	Florian J.	Blake	Dietmar
17. Apr	NXP Examples - CW mode - Ranging - Crypto	8	Dietmar	Blake	
18. Apr	UWB measurements (Debugger, Spectrum analyzer, Current clamp, oscilloscope)	8	Blake	Frank	Dietmar
19. Apr	FBD6: UWB Configuration (Channel, Sync Code, Pulse Shape Combo, ...)	4	Thommy T.	Frank H.	Dietmar
	FBD6: Testmodes (CW mode, CF mode, RX mode, XTAL Pin)	2	Thommy T.	Dietmar	Frank H.
	FBD6: Loopback	2	Dietmar	Frank H.	
20. Apr					
21. Apr					
22. Apr	Box2Box (Initiator, Responder)	2	Frank H.	Dietmar	Thommy T.
	Ranging with 1 smartphone (Responder)	2	Frank H.	Dietmar	Thommy T.
	Multiranging	2	Frank H.	Dietmar	Thommy T.
	IDG Ranging	2	Thommy T.	Dietmar	Frank H.
23. Apr	UWB Window Calculation	2	Thommy T.	Dietmar	Frank H.
	Hopping Mode	2	Thommy T.	Dietmar	Frank H.
	Temperature Compensation	2	Thommy T.	Dietmar	Frank H.
	Ranging without PrePoll	2	Dietmar	Frank H.	Frank H.
24. Apr	AoA	2	Thommy T.	Dietmar	Frank H.
	Firth Path Information	2	Dietmar	Frank H.	Thommy T.
	Distance Calculation with Final Data message (FDD) over CAN	2	Dietmar	Dietmar	Thommy T.
	Duty Cycle Limit	2	Thommy T.	Frank H.	Dietmar
25. Apr	CPD (Measurement and Machine Learning)	8	Florian J.	Dietmar	
26. Apr	CPD CoEx	8	Florian J.	Dietmar	
27. Apr					
28. Apr					
29. Apr	NXP Demoboard: CW (ContWave)	4	Dietmar	Thommy T.	Frank H.
30. Apr	NXP Demoboard: TwrDs (Ranging)	8	Blake	Dietmar	
01. Mai					
02. Mai	NXP Demoboard: TwrDs (Ranging)	8	Blake	Thommy T.	Frank H.
03. Mai	NXP Demoboard: TwrDs (Ranging)	8	Blake	Thommy T.	Frank H.
04. Mai					
05. Mai					
06. Mai	NXP Demoboard: TwrDs (Ranging)	8	Blake	Thommy T.	Frank H.
07. Mai	NXP Demoboard: TwrDs (Ranging)	8	Blake	Thommy T.	Frank H.
08. Mai	NXP Demoboard: Encrypt/Decrypt & KDF	8	Patrick	Blake	
09. Mai					
10. Mai					
11. Mai					
12. Mai					
13. Mai	NXP Demoboard: Encrypt/Decrypt & KDF	8	Patrick	Blake	
14. Mai	NXP Demoboard: Encrypt/Decrypt & KDF	8	Patrick	Blake	
15. Mai	NXP Demoboard: Encrypt/Decrypt & KDF	8	Patrick	Blake	
16. Mai	FBD6 HW: CPD Debugging	8	Florian J.		
17. Mai	FBD6 HW: CPD Debugging	8	Florian J.		
18. Mai					
19. Mai					
20. Mai					
21. Mai	FBD6 HW: CPD Debugging	8	Florian J.		
22. Mai	FBD6 HW: CPD Debugging	8	Florian J.		
23. Mai	FS19 : Short demonstration of the UID-Ranging	8	Blake		
24. Mai	Test of UWB Features	8	Johannes		
25. Mai					
26. Mai					
27. Mai	Test of UWB Features	8	Johannes		
28. Mai	Test of UWB Features	8	Johannes		
29. Mai	Test of UWB Features	8	Johannes		
30. Mai	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
31. Mai	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
01. Jun					
02. Jun					
03. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
04. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
05. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
06. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
07. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
08. Jun					
09. Jun					
10. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
11. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
12. Jun	Debugging Box2Box-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
13. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
14. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
15. Jun					
16. Jun					
17. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
18. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
19. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
20. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
21. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
22. Jun					
23. Jun					
24. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
25. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
26. Jun	Debugging Smartphone-Ranging (FBD6 HW)	8	Dietmar, FrankH., Thommy		
27. Jun	Debugging UWB Features e.g. Loopback, CPD CoExistence, Testmodes	8	Dietmar, FrankH., Thommy		
28. Jun	Debugging UWB Features e.g. Loopback, CPD CoExistence, Testmodes	8	Dietmar, FrankH., Thommy		
29. Jun					
30. Jun					
01. Jul	Debugging UWB Features e.g. Loopback, CPD CoExistence, Testmodes	8	Dietmar, FrankH., Thommy		
02. Jul	Debugging UWB Features e.g. Loopback, CPD CoExistence, Testmodes	8	Dietmar, FrankH., Thommy		
03. Jul	Debugging UWB Features e.g. Loopback, CPD CoExistence, Testmodes	8	Dietmar, FrankH., Thommy		

UWB Ranging

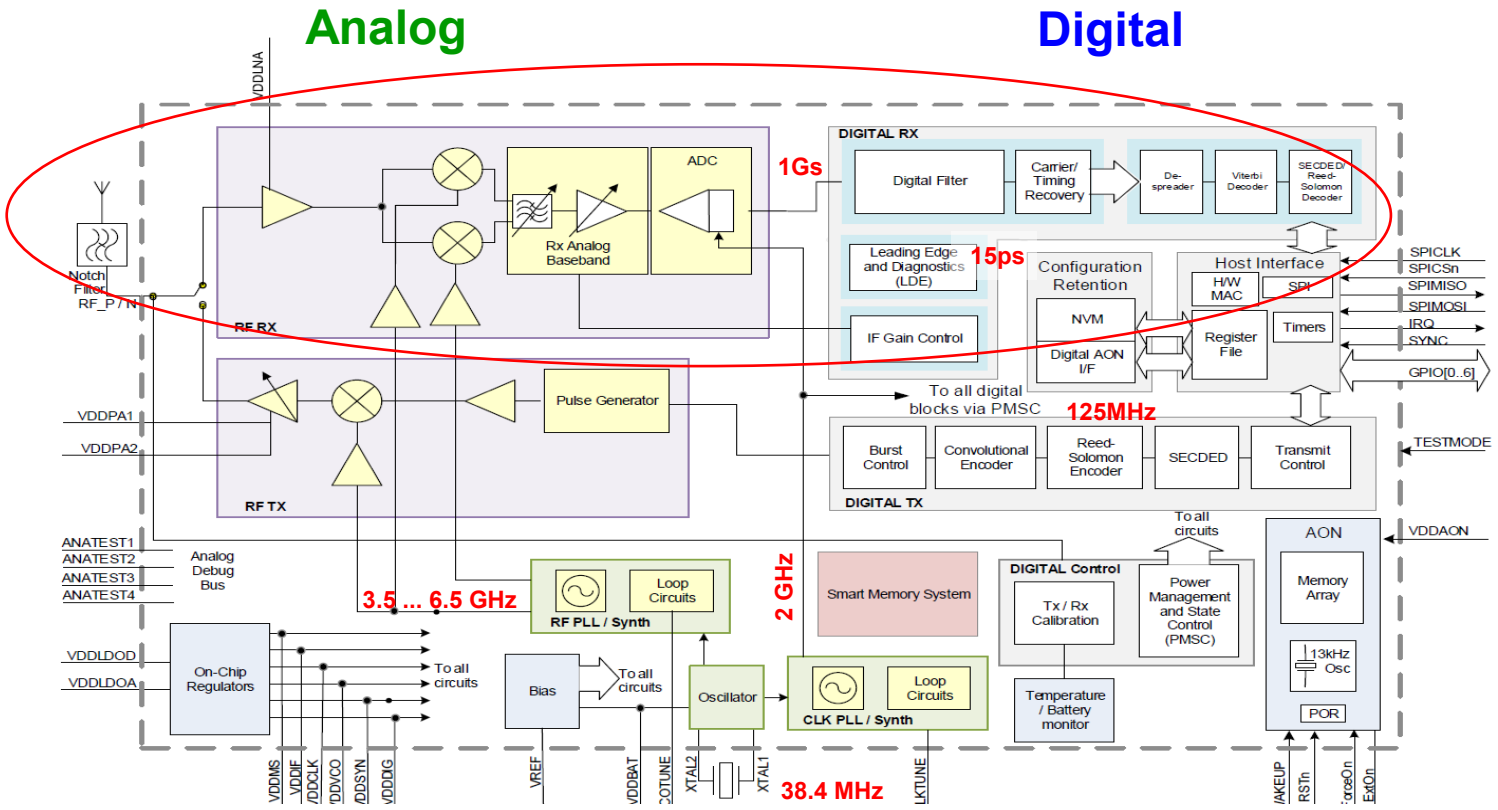
UWB Introduction



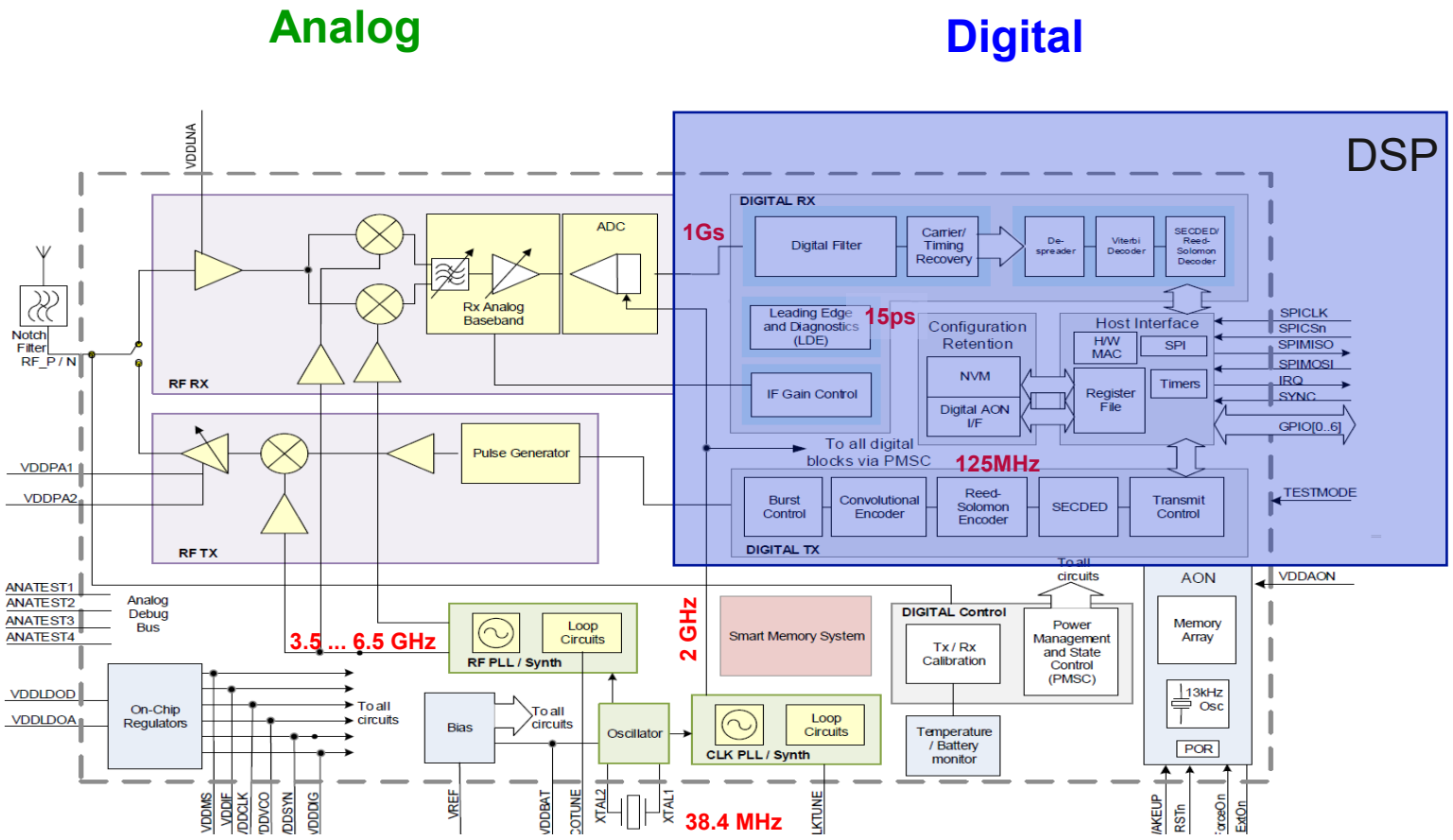


13:48:26 20.05.2019

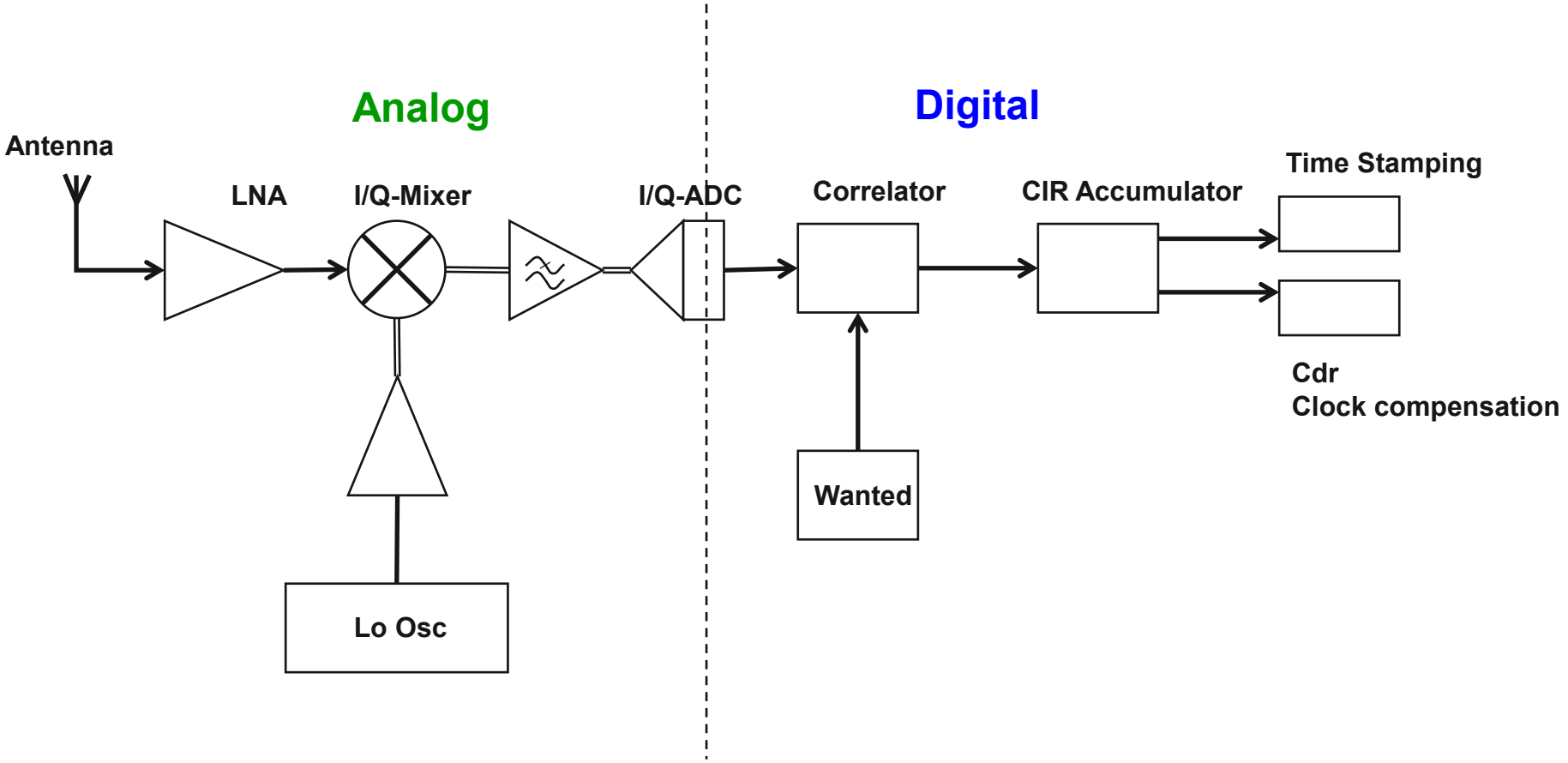
Block Schematic Ranger1 (NCJ29D0, DW1000) Receiver



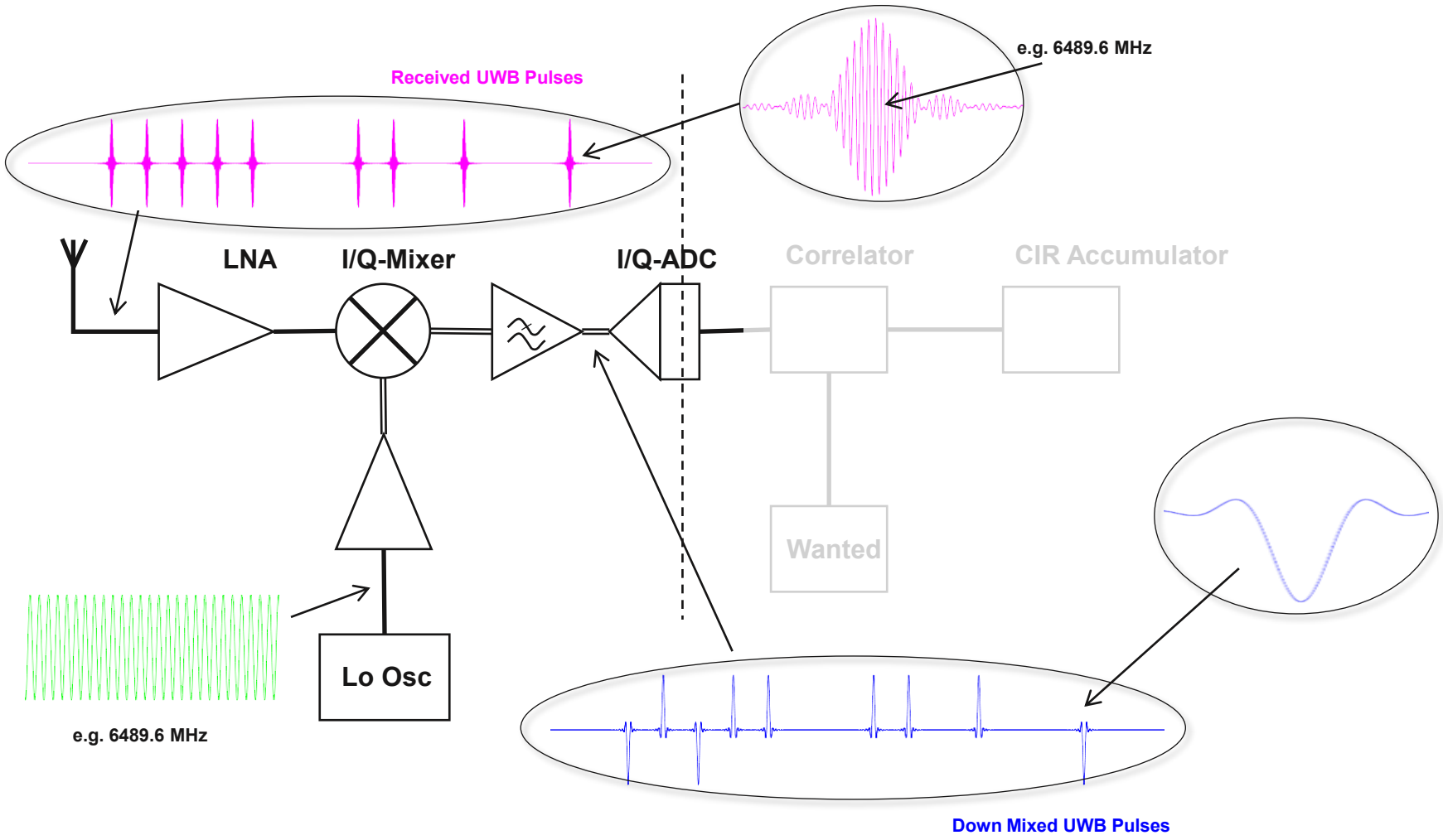
Block Schematic



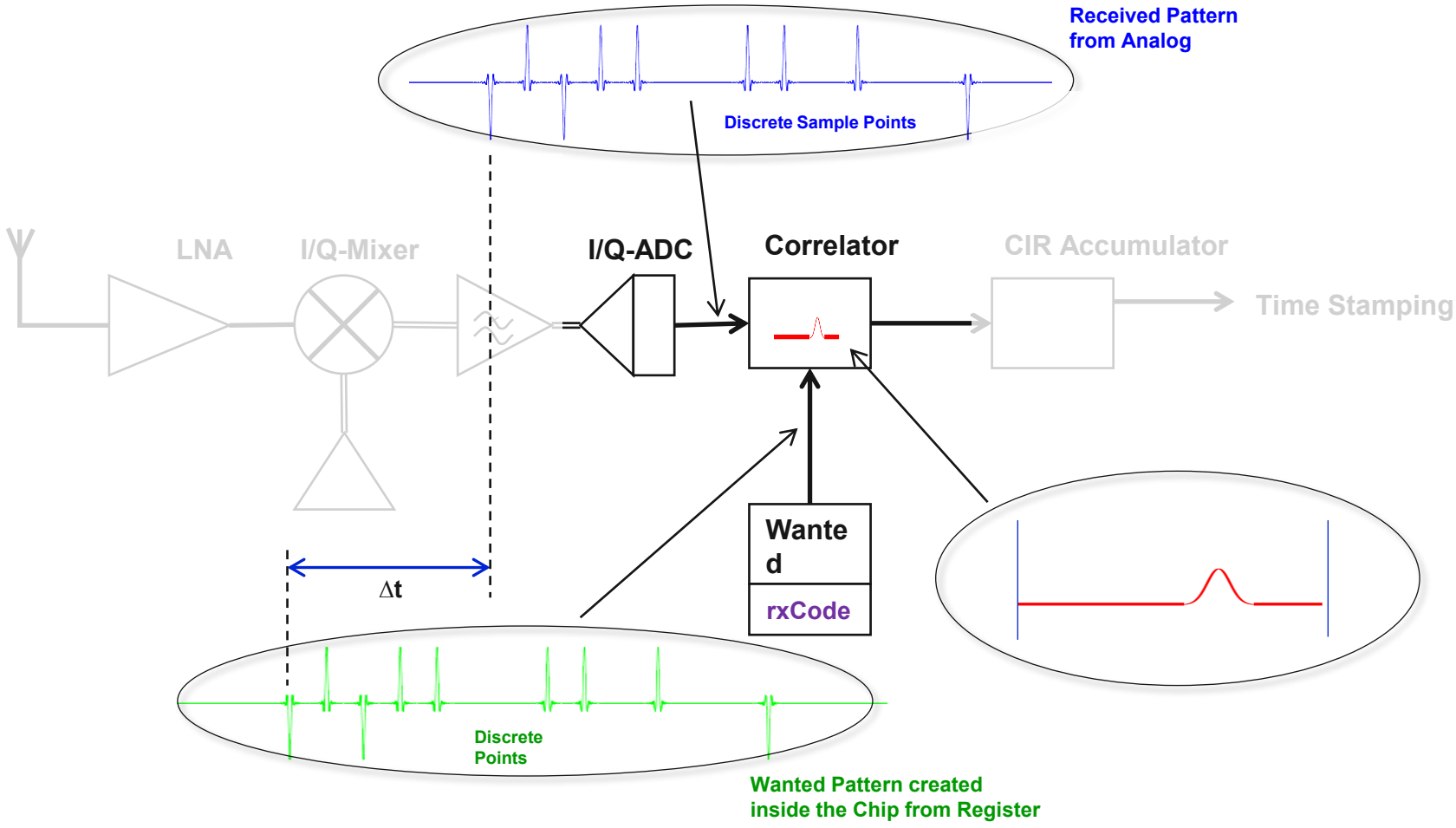
Block Schematic Receiver



Down Mixing


Received UWB Pulses
e.g. 6489.6 MHz
LNA
I/Q-Mixer
I/Q-ADC
Correlator
CIR Accumulator
Wanted
Lo Osc
e.g. 6489.6 MHz
Down Mixed UWB Pulses

Correlating



rxCode	-1 +1 -1 +1 +1 0 0 +1 +1 0 +1 0 0 -1

→ **See dwt_configure,** rxCode in SW_API_GUIDE

Correlation

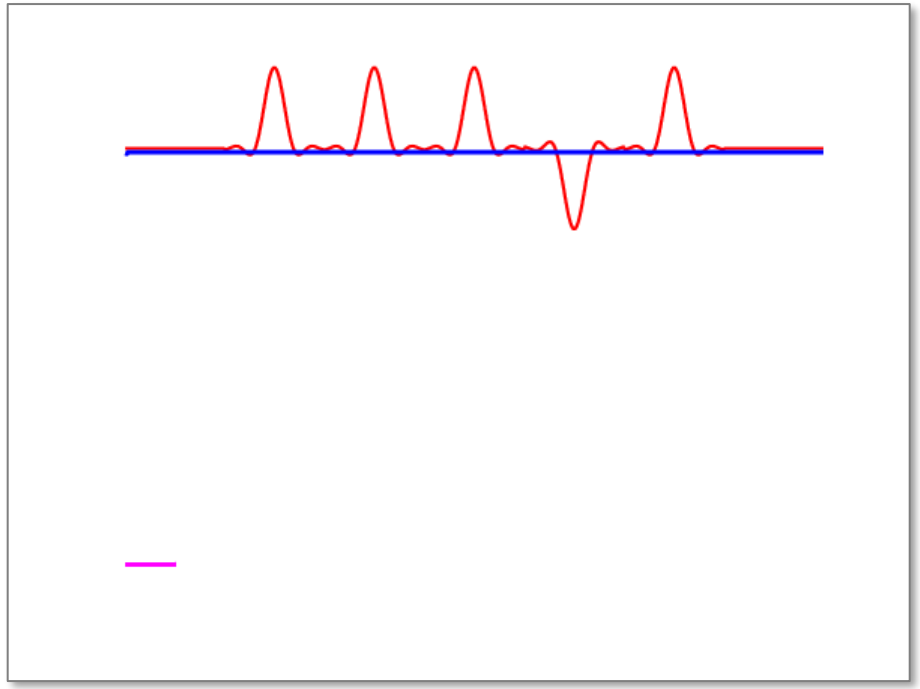
Run this Page in
Slide Show !!

It demonstrates how the UWB
Correlator is working.

Red = Received Signal
Blue = Compared Signal
Magenta = Correlation

The maximum is given, if both curves are congruent.

The small peaks shows the side slopes.
In this case with a code length of 5, the side slopes are 1/5



Correlation with Noise

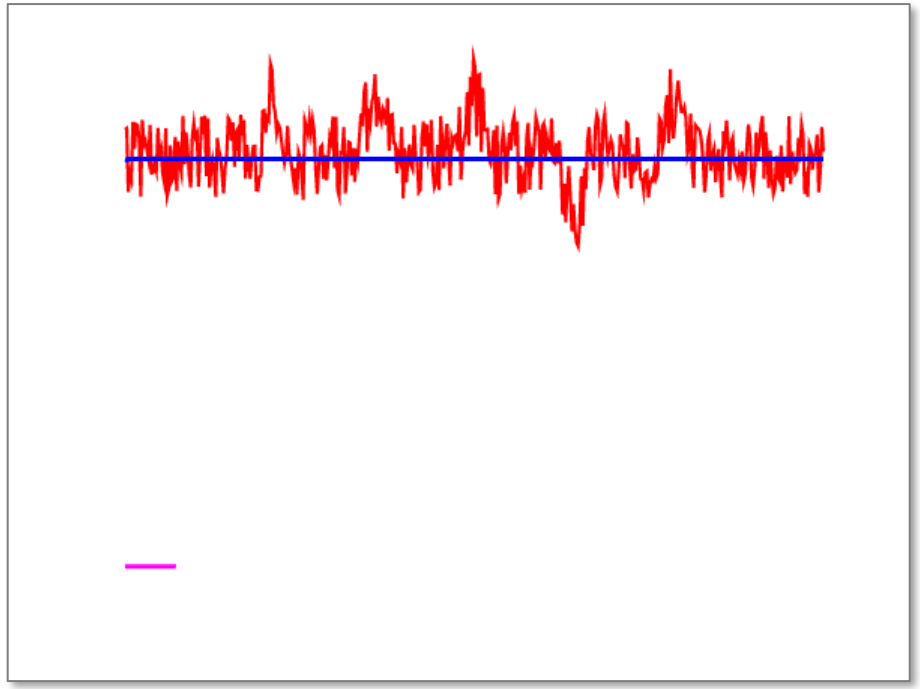
Run this Page in
Slide Show !!

It demonstrates how the UWB
Correlator is working with noise.

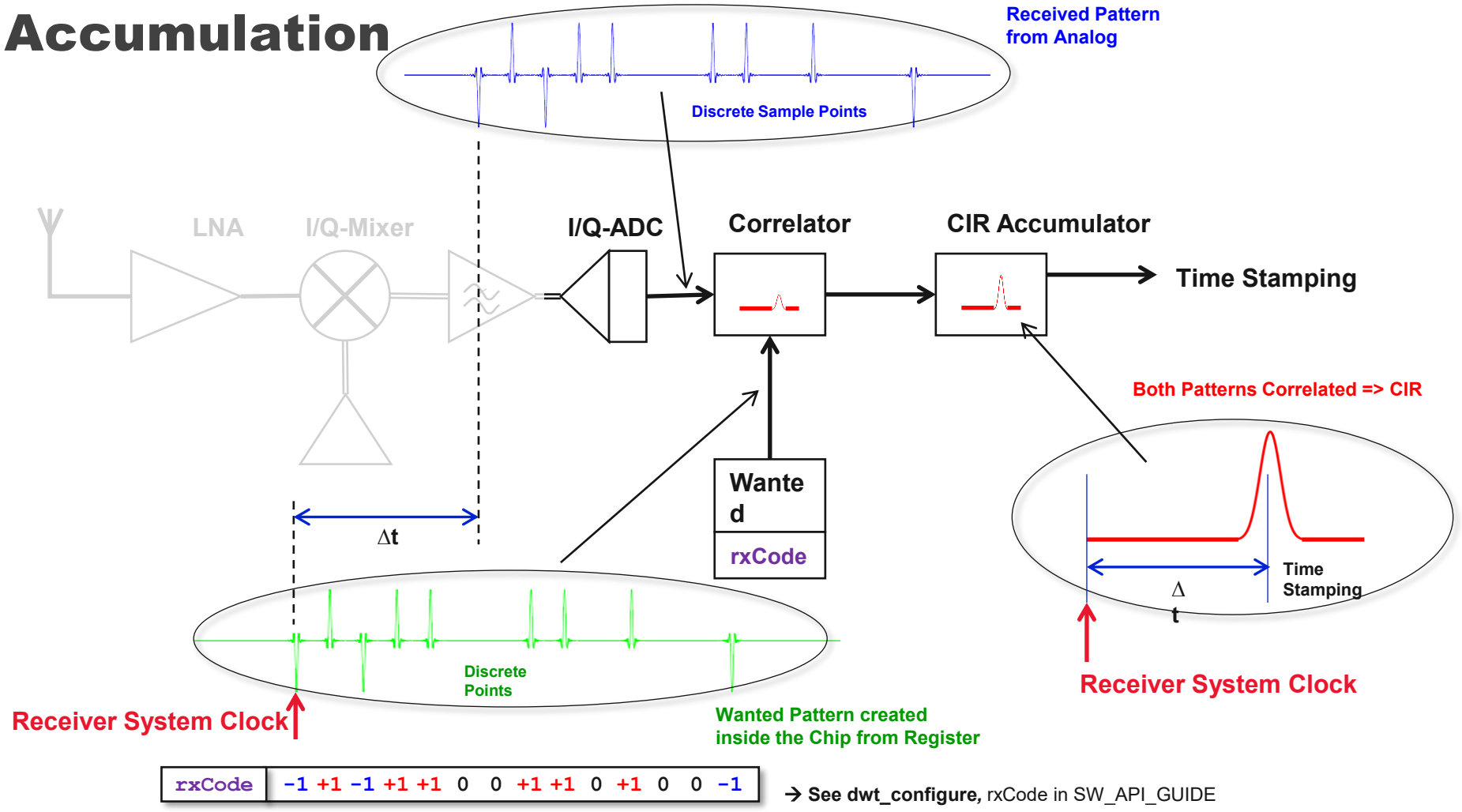
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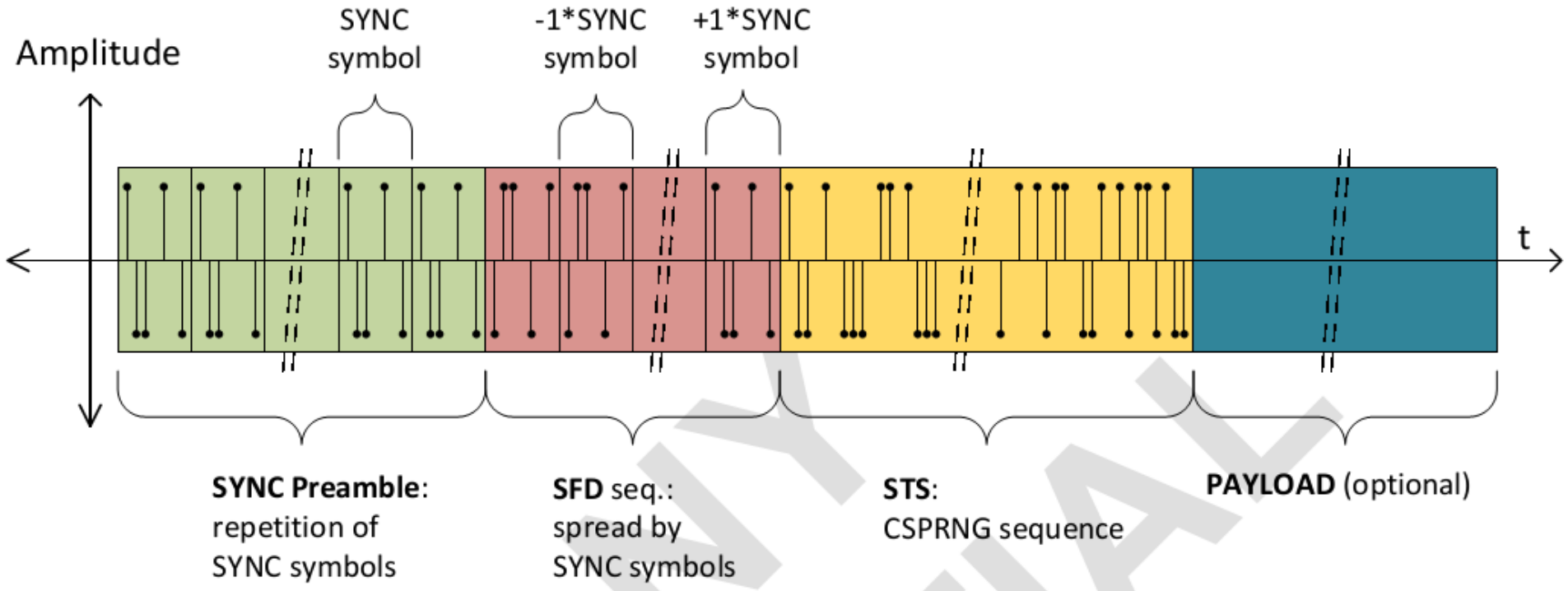
Accumulation



rxCode	-1	+1	-1	+1	+1	0	0	+1	+1	0	+1	0	0	-1

→ **See dwt_configure,** rxCode in SW_API_GUIDE

Ranging Frame



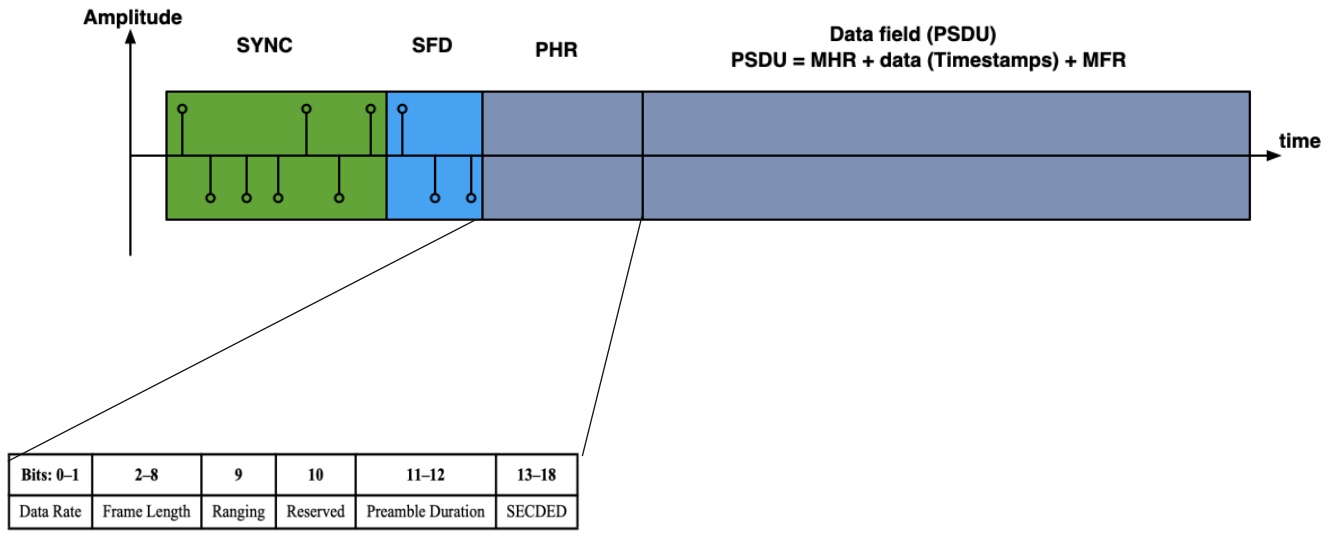
SFD start of frame delimiter:
Defined in the context of the IEEE802.15.4-2011 [1] standard. Indicates the completion of the preamble section of the frame

STS Secure Training Sequence:
The transfer of payload for authentication is not required, because the intended recipient knows upfront which STS sequence to expect and can use the STS for authentication.

Data Frames

For payload messages used:

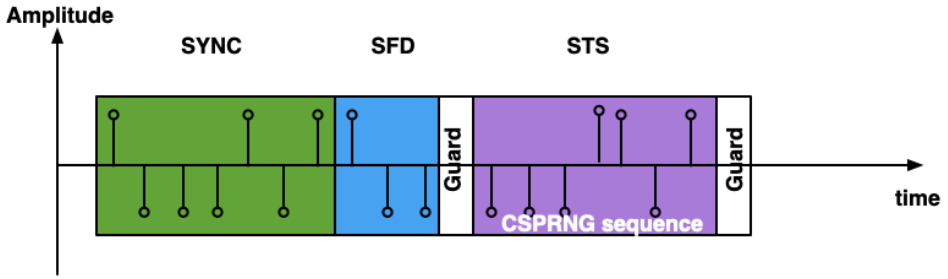
- Pre-Poll
- Final Data



Slot Index	Mnemonic	STS Packet Format
0	Pre-POLL	SP0 : Data Packet
1	POLL	SP3 : RFRAME

Ranging Frames

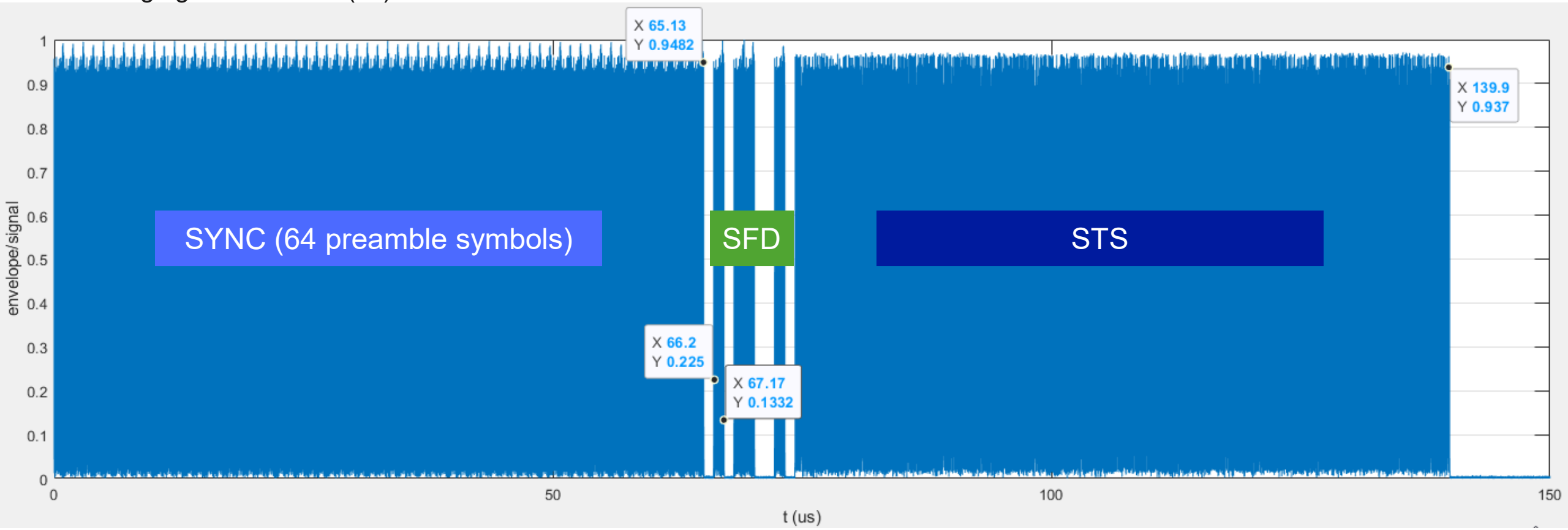
Messages:
Poll
Response
Final



Slot Index	Mnemonic	STS Packet Format
0	Pre-POLL	SP0 : Data Packet
1	POLL	SP3 : RFRAME

UWB Measurements
iPhone 11; June 23, 2020; initial tests/R&S Testkammer – IQ Recordings (code 10)

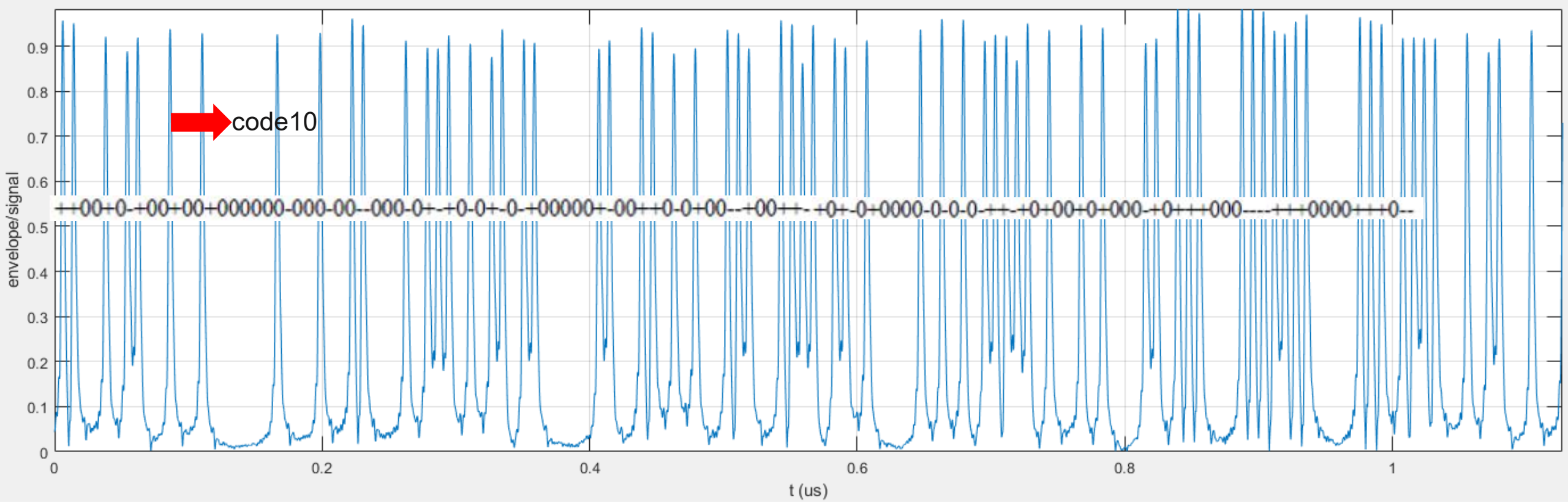
Ranging Frame -- abs(IQ)




```
[iq,samplerate] = load_iq_tar_file(filename);
t=0:1/samplerate:(length(iq)-1)/samplerate;
iq_abs_norm=abs(iq)/max(abs(iq));
t_us=10^6*t;
plot(t_us,iq_abs_norm);
xlabel('t (us)');
ylabel('envelope/signal');
```

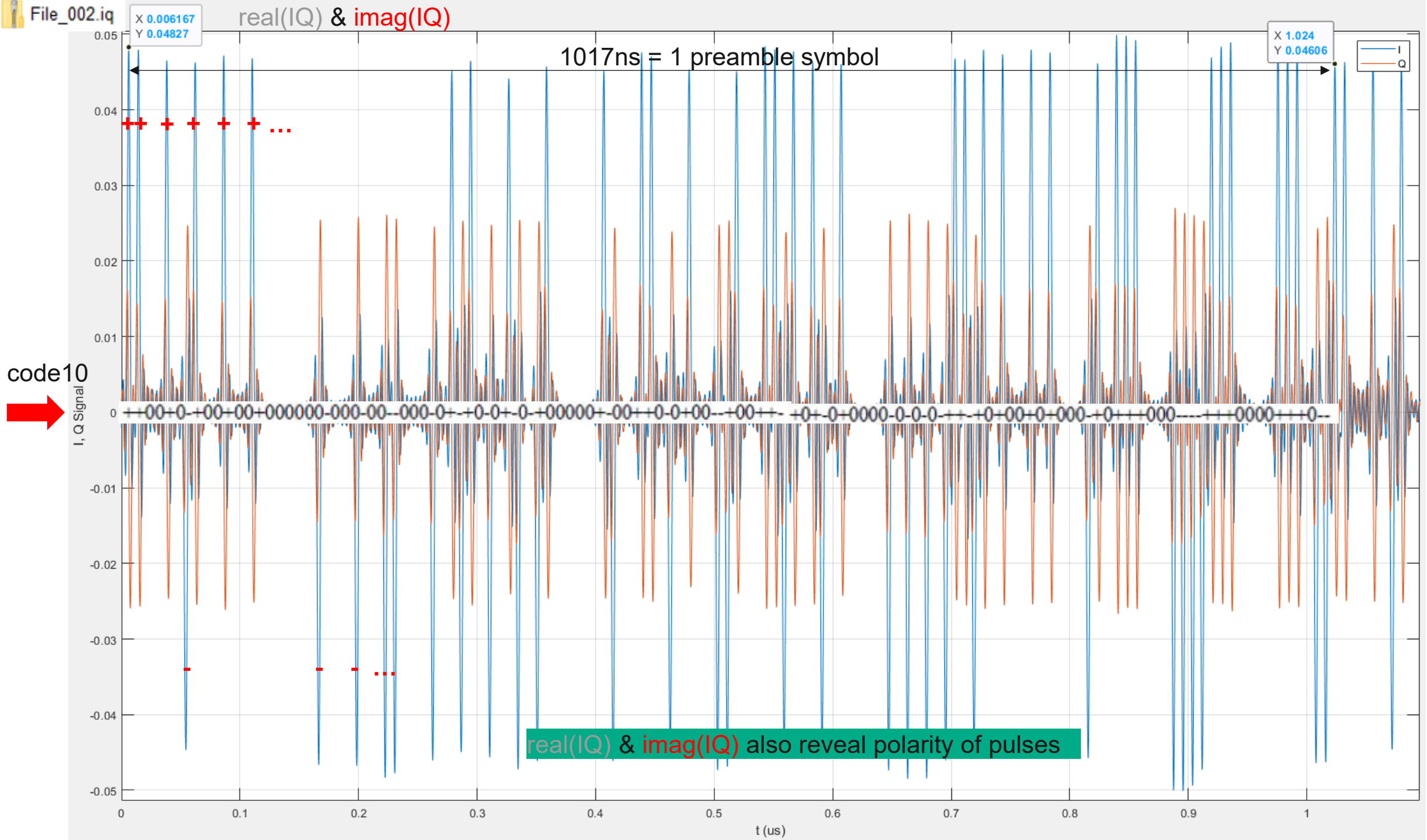
UWB Measurements
iPhone 11; June 23, 2020; initial tests/R&S Testkammer – IQ Recordings (code 10)

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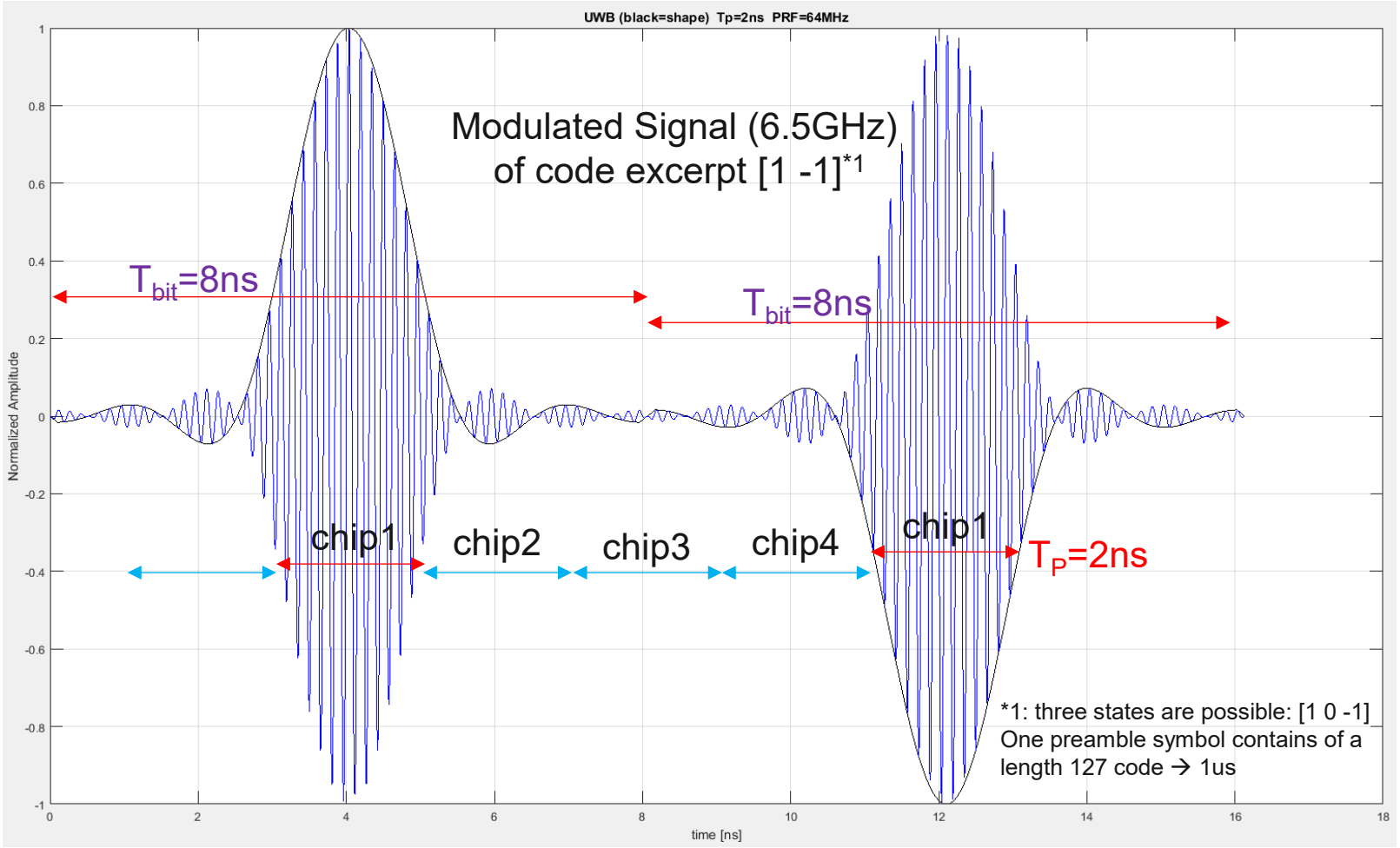
→ abs(IQ) shows pulses but no polarity


File_002.iq
X 0.006167
Y 0.04827
real(IQ) & imag(IQ)
1017ns = 1 preamble symbol
X 1.024
Y 0.04606
I
Q
code10
I, Q Signal
t (us)
real(IQ) & imag(IQ) also reveal polarity of pulses

UWB
UWB parameter comparison & optimization tools - Preamble

PRF=64MHz
Ch5:
f_c=6.4896GHz
Half of the bits (trits) of the length 127 code are „0“ (no energy)

This diagram also applies to PRF=128MHz
Difference is just that PRF128 skips „0“ bits to increase PRF



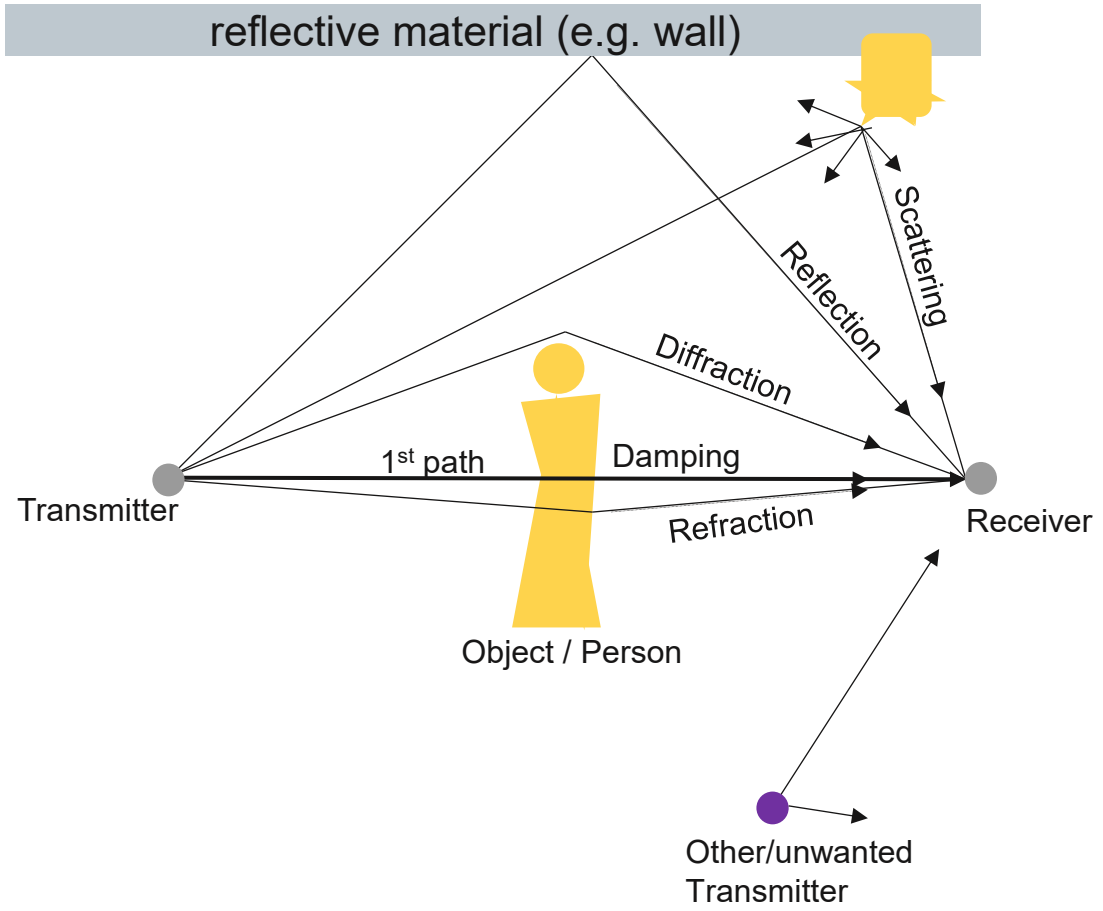
➢ Hella build Simulation tools is basis for visualization, analysis and optimization

Smartphone Localization / Hands Free Car Access
RF (propagating waves) based Localisation – Multi Path

real

Multi-Path

- **Interference & superposition at receiver**
- **Distance ≠ f(Signal Strength)**
- **Shortest path (true 1st path) signal in hands free use case typically not strongest signal**
- **Direction (=angle) measurement only useful if related to true 1st path (shortest path) → otherwise (for all other paths) detected AoA is not true direction**
- **Time of Flight (ToF) measurement only useful if related to true 1st path (shortest path) → otherwise (for all other paths) detected ToF based distance is not true distance**



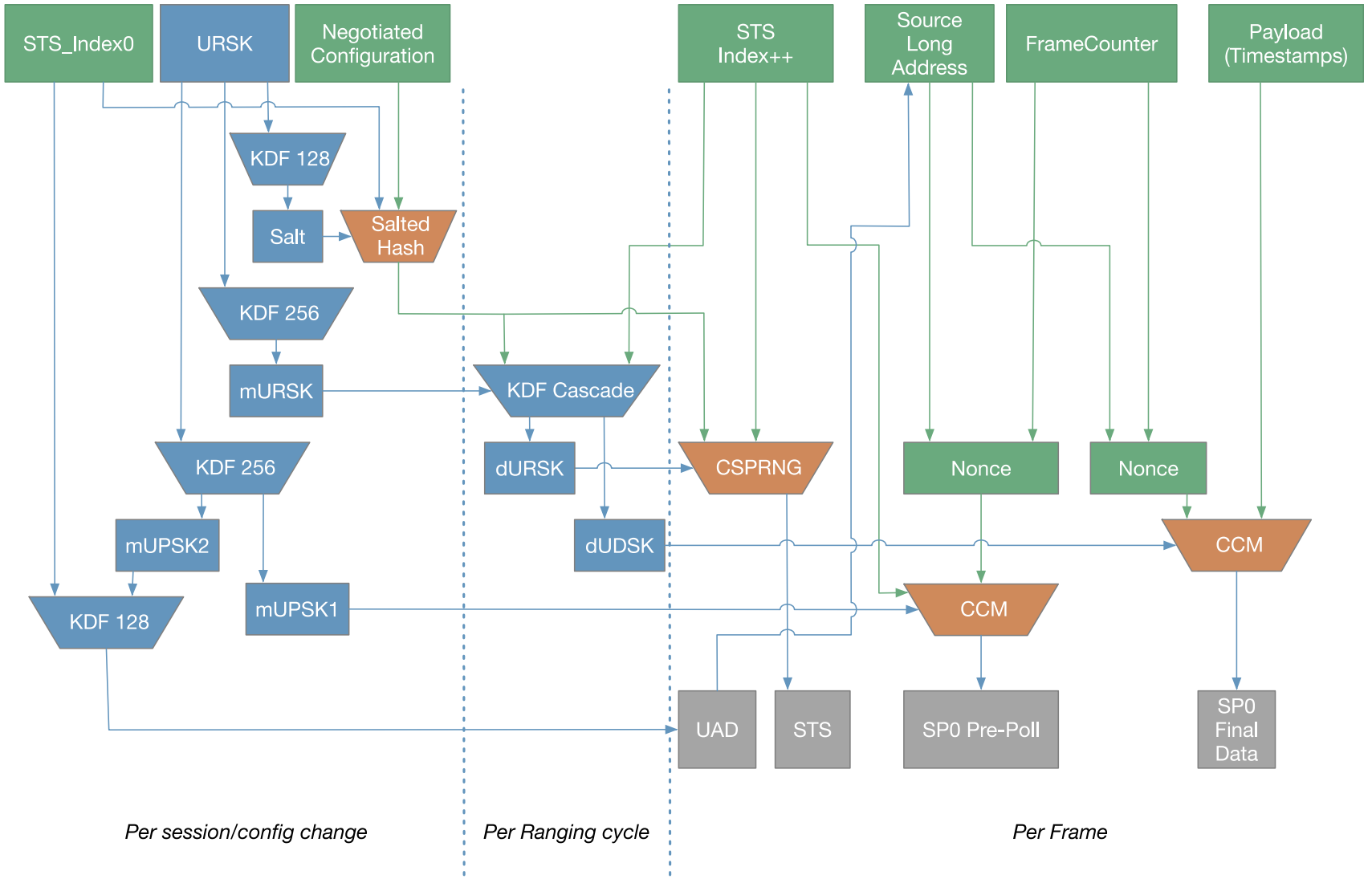
➢ Real conditions include a broad variety of superimposed impacts → 1st path detection crucial

UWB Frequency and Power

UWB Channels	UWB Center Frequency	UWB Bandwidth	Region used
0	399.36 MHz	499.2 MHz	USA
1	3494.4 MHz	499.2 MHz	USA, Europe
2	3993.6 MHz	499.2 MHz	USA, Europe, Japan, Korea
3	4492.8 MHz	499.2 MHz	USA, Europe, Japan, Korea
4	3993.6 MHz	1331.2 MHz	USA, Europe
5	6489.6 MHz	499.2 MHz	USA, Europe, China
6	6988.8 MHz	499.2 MHz	USA, Europe, China
7	6489.6 MHz	1081.6 MHz	USA
8	7488 MHz	499.2 MHz	USA, Europe, Korea, China
9	7987.2 MHz	499.2 MHz	USA, Europe, Japan, Korea, China
10	8486.4 MHz	499.2 MHz	USA, Europe, Japan, Korea, China
11	7987.2 MHz	1331.2 MHz	USA, Japan, Korea
12	8985.6 MHz	499.2 MHz	USA, Japan, Korea
13	9484.8 MHz	499.2 MHz	USA, Japan, Korea
14	9984 MHz	499.2 MHz	USA, Japan, Korea
15	9484.8 MHz	1354.97 MHz	USA, Japan, Korea

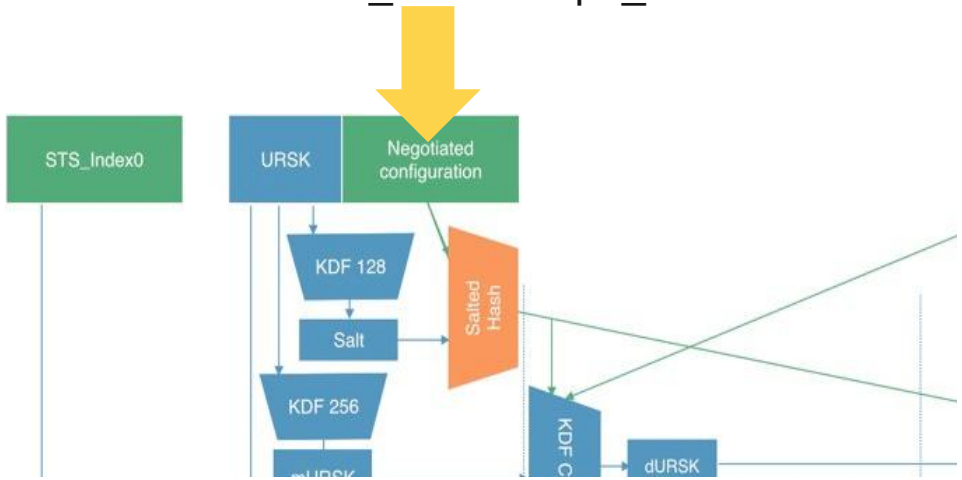
UWB Frequency	EIRP
960 to 1610 MHz	-75.3 dBm
1610 to 1990 MHz	-53.3 dBm
1990 to 3100 MHz	-51.3 dBm
3100 to 10600 MHz	-41.3 dBm
> 10600 MHz	-51.3 dBm

Ranging Frames
KDF (Key Derivation Function)



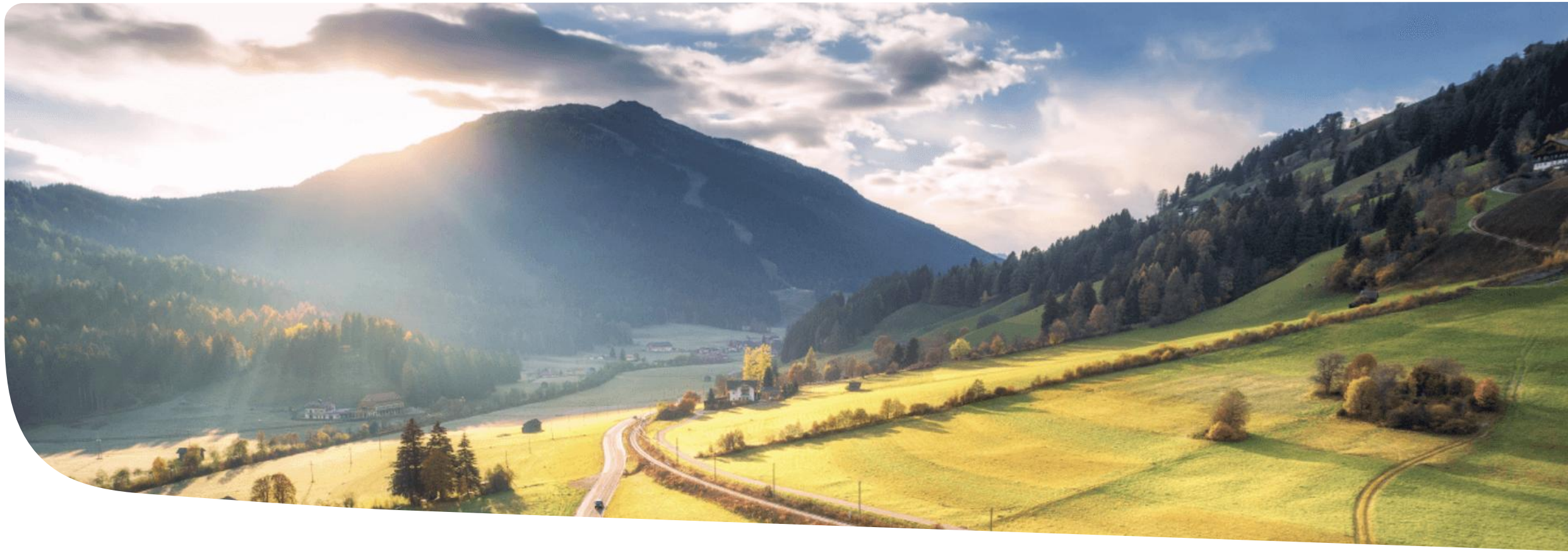
Ranging Frames
KDF (Key Derivation Function)

Selected_Protocol_Version ||
Selected_UWB_Config_Id ||
UWB_Session_Id ||
STS_Index0 ||
Number_Responder_Nodes ||
Session_RAN_Multiplier ||
Number_Slot_per_Round ||
Number_Chaps_per_Slot ||
Selected_PulseShape_Combo



UWB Ranging

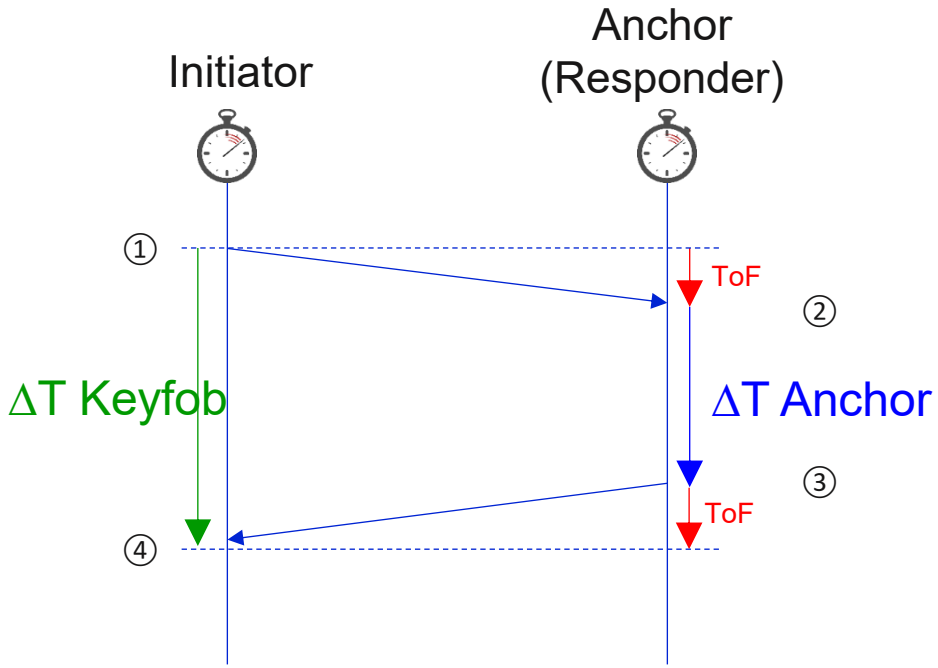
Time of Flight (ToF)
&
Distance Calculation



Single Sided - Two Way Ranging (Not used for CCC Ranging)

Consists of the following:

1. TX from Keyfob
2. RX on Anchor
3. TX from Anchor
4. RX on Keyfob



Computation on Keyfob

$$ToF = \frac{\Delta T_{KeyFob} - \Delta T_{Anchor}}{2}$$

\+ Simple

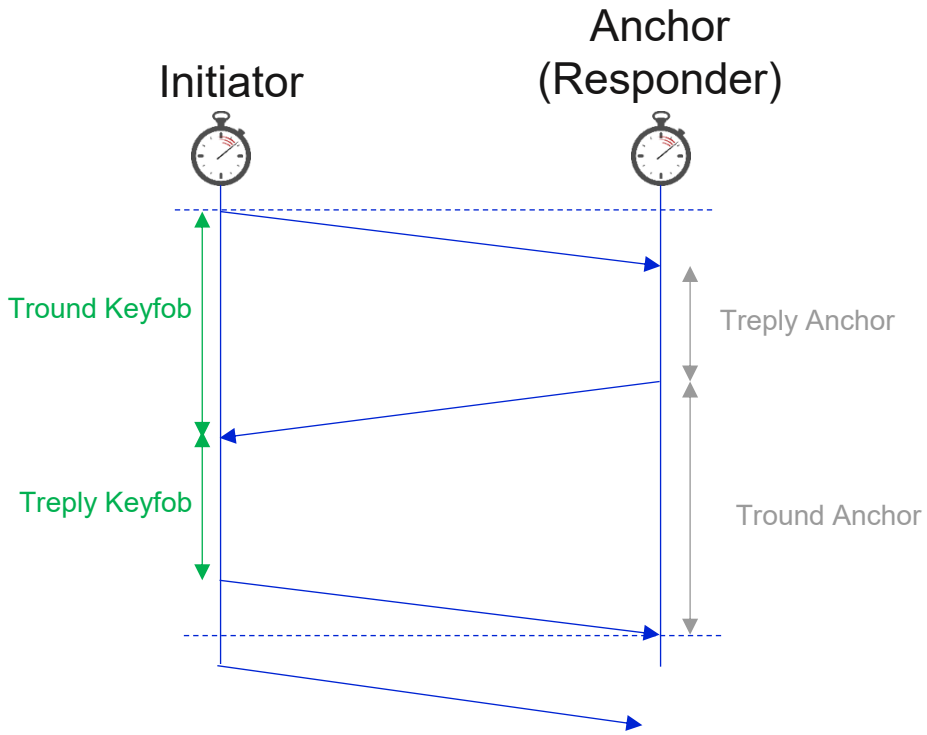
\+ Low power consumption on the Keyfob as only one TX+RX is needed

– – Clock difference leads to Big Error

Double Sided - Two Way Ranging

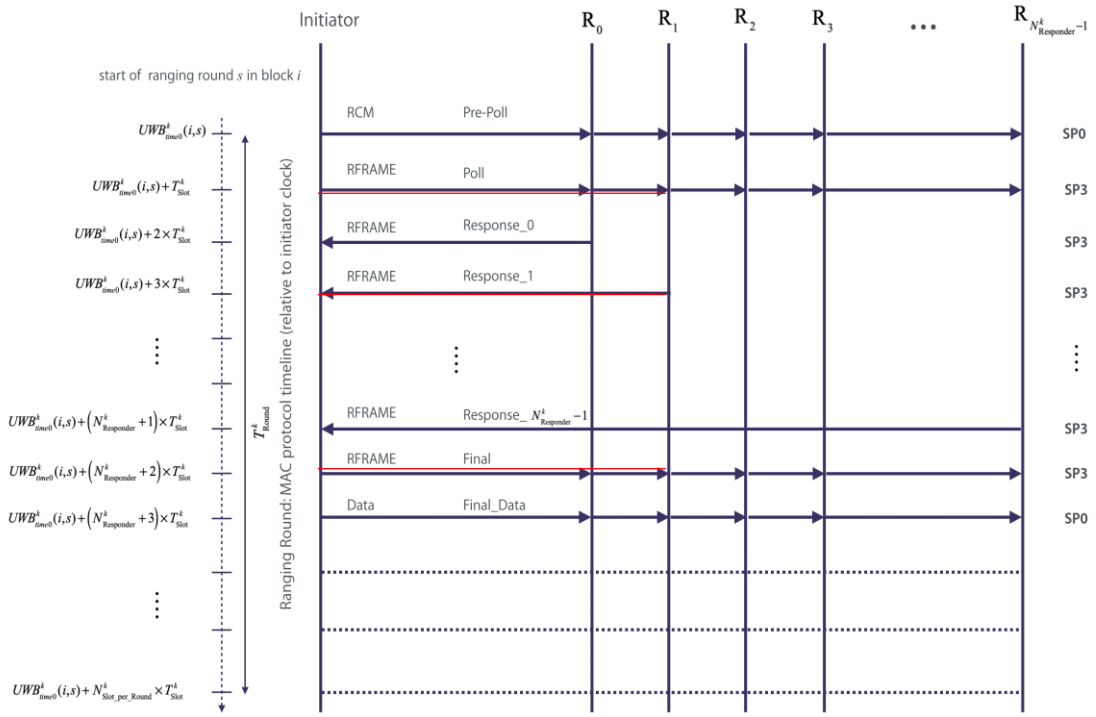
- **Consists of 3 messages:**
  - Ranging request
  - Ranging Response
  - Final message
    (with anchor timestamps)

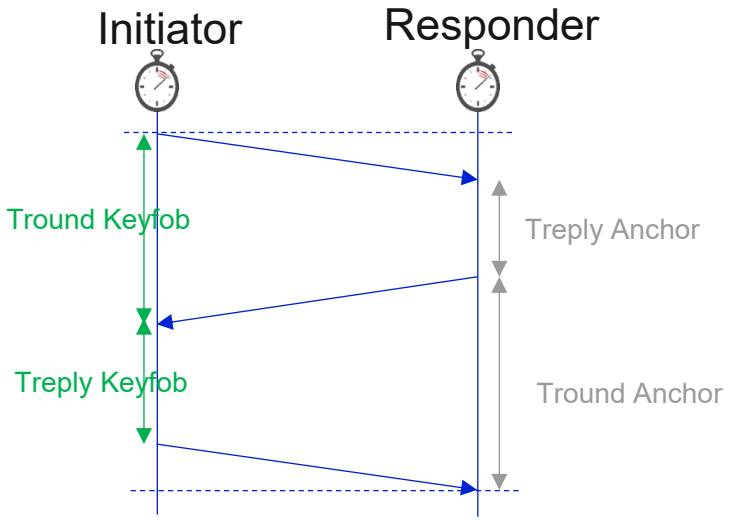
$$ToF = \frac{T_{roundA} \times T_{roundK} - T_{replyA} \times T_{replyK}}{t_{roundA} + t_{roundK} + T_{replyA} + T_{replyK}}$$



+ Automatic compensation for clock differences in the two devices
+ Time computation can be done on the car
+ More secure (preamble injection more difficult)
– need more current

CCC Ranging
e.g. Ranging Messages for Responder 1





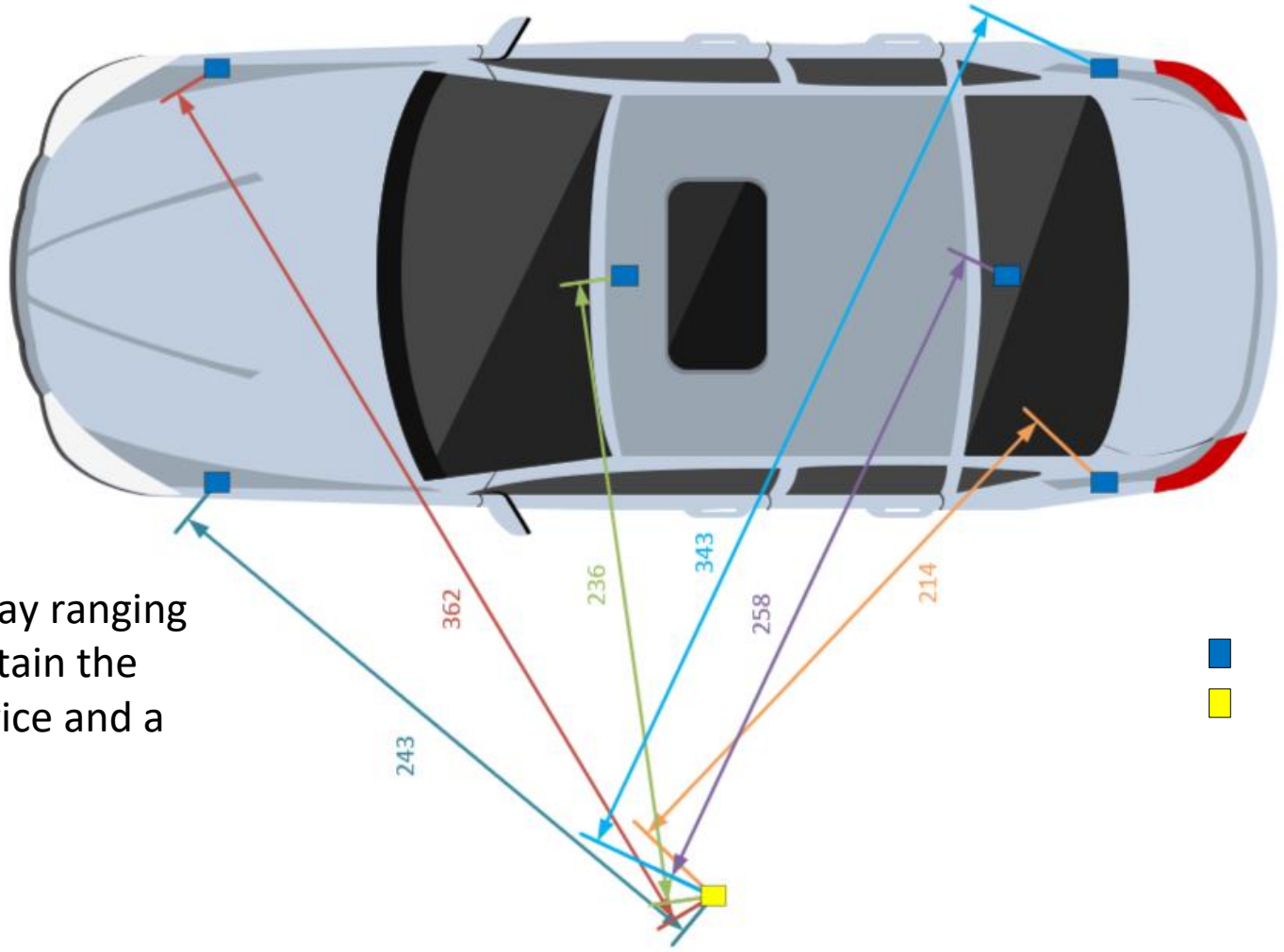
Trimension™ NCJ29D5: UWB IC for Automotive Applications

The UWB IC enables cars to know exactly where the users are. For the first time, smartphone-based car access offers the same level of convenience as state-of-the-art key fobs. Users can open and start cars, while leaving their phones in their pockets or bags, and enjoy secure remote parking via smartphone. Furthermore, the new UWB IC brings maximum level of protection against car theft through relay attacks.

Source: Trimension™ NCJ29D5 | Ultra-Wideband for Automotive IC | NXP Semiconductors

Concept Overview

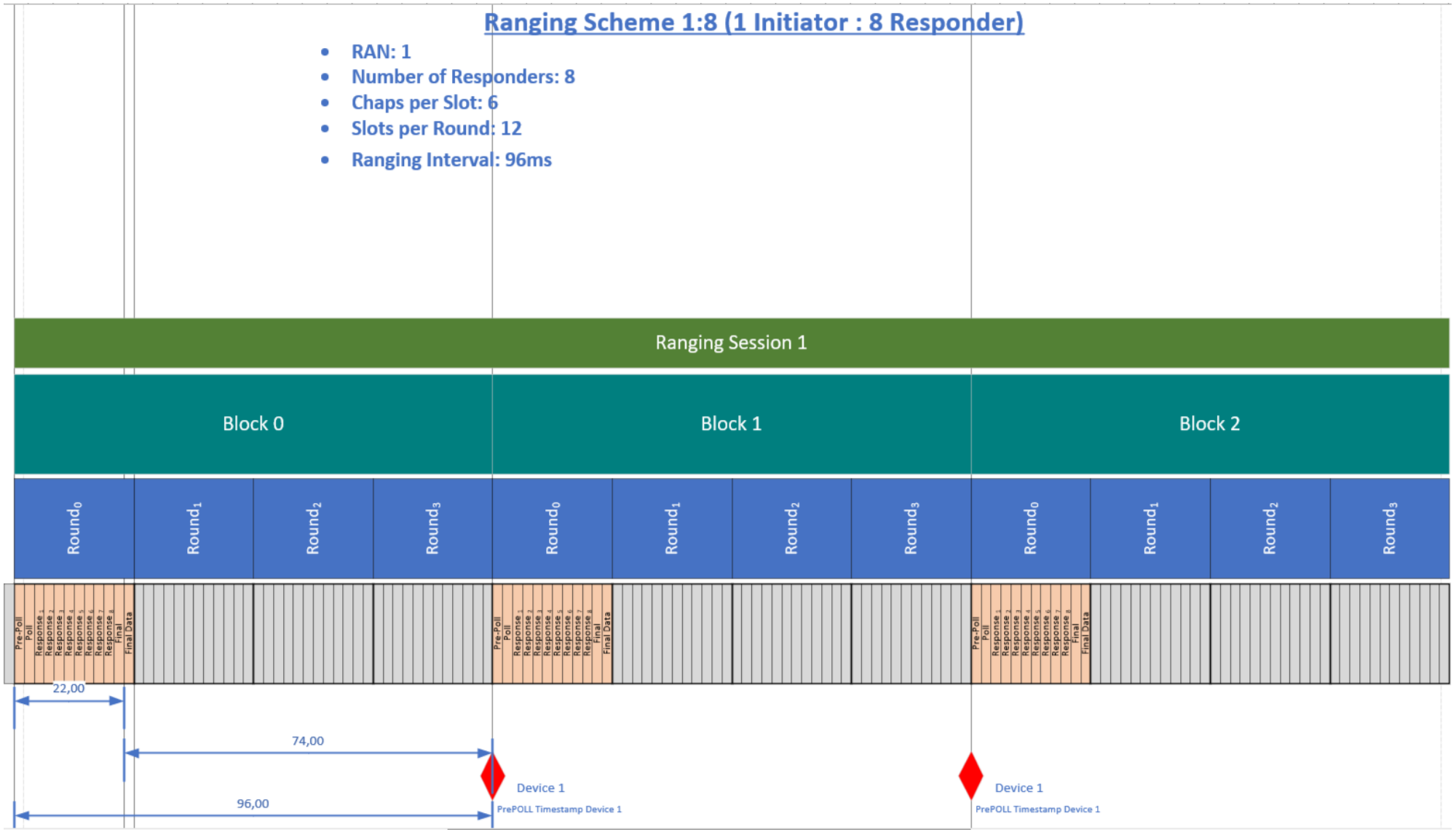
Using the double sided two-way ranging method, it is possible to ascertain the distance between a smart device and a vehicle.



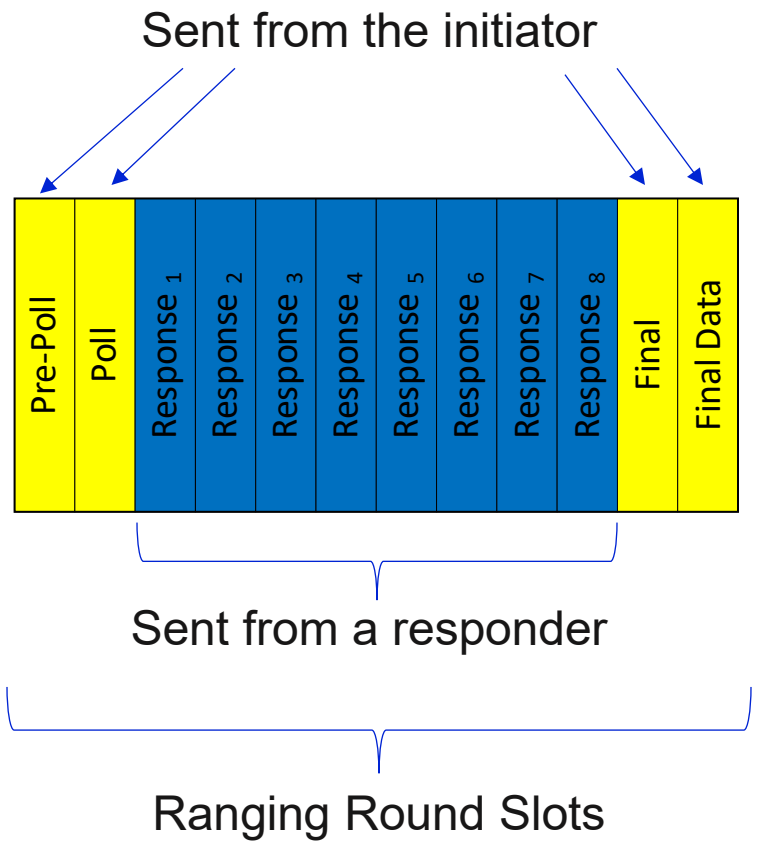
UWB Responder aka Anchor

Smart Device aka Initiator
UID, Key Fob, Smartphone, Smart Watch

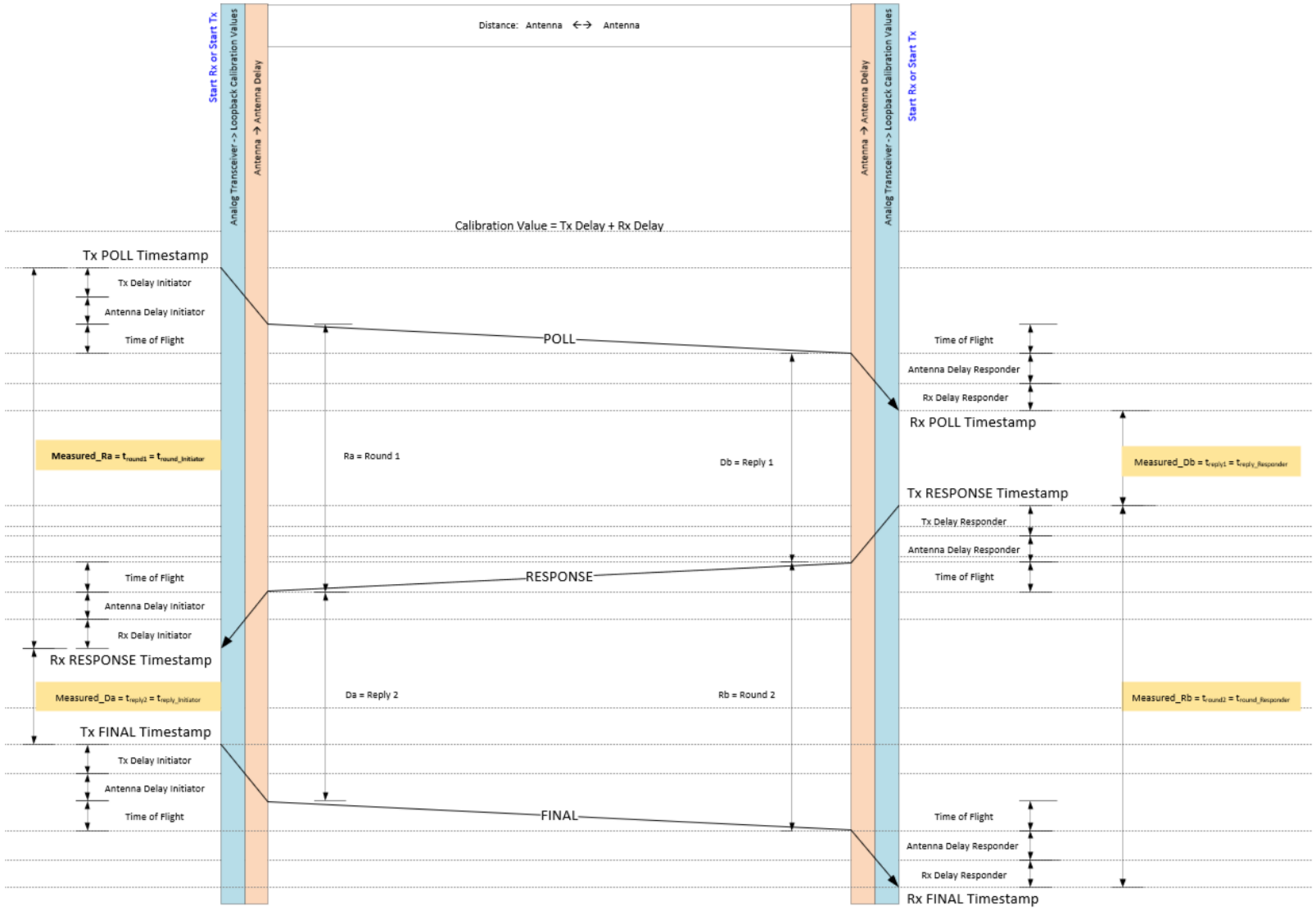
Ranging Scheme 1:8



UWB Ranging Messages



UWB Ranging: Timing and Delays



NCJ29D5: Time Stamps

The NCJ29D5 measures time using a time stamp timer.

	Integer Part	Fractional Part			
Time Stamp Timer	8.0128ns	15.65ps			
Tick = Fractional Part	15.65ps				
	m/s	**cm/s**	**cm/ns**	**cm/ps**	**cm/tick**
Speed of Light	299792458	29979245800	29.9792458	0.02997925	0.469175
distance = tof x 0.469175cm pro tick					
maximum Time Stamp Timer Value	0xFFFF FFFF = 4,294,967,295		4,294,967,295 x 8.0128ns =		34.41471394s

Distance Calculation

INITIATOR

FINAL DATA Payload Data:

Ranging_Timestamp_Final_Tx = Tx FINAL Timestamp + Tx Delay + Antenna Delay - (Tx POLL Timestamp + Tx Delay + Antenna Delay)
= Tx FINAL Timestamp + Tx Delay + Antenna Delay - Tx POLL Timestamp - Tx Delay - Antenna Delay)
= Tx FINAL Timestamp – Tx POLL Timestamp

Ranging_Timestamp_Responder = Rx RESPONSE Timestamp – Rx Delay - Antenna Delay - (Tx POLL Timestamp + Tx Delay + Antenna Delay)
= Rx RESPONSE Timestamp – Rx Delay - Antenna Delay - Tx POLL Timestamp - Tx Delay - Antenna Delay
= Rx RESPONSE Timestamp – Tx POLL Timestamp – Rx Delay - Tx Delay - (2 x Antenna Delay)
= Rx RESPONSE Timestamp – Tx POLL Timestamp - Loopback Calibration Value - (2 x Antenna Delay)

RESPONDER

Ra = Round1 = Ranging_Timestamp_Responder
Da = Reply2 = Ranging_Timestamp_Final_Tx – Ra

Db = Reply1 = (Tx Response Timestamp + Tx Delay + Antenna Delay) – (Rx POLL Timestamp - Rx Delay - Antenna Delay)
Db = Reply1 = Tx Response Timestamp + Tx Delay + Antenna Delay – Rx POLL Timestamp + Rx Delay + Antenna Delay
Db = Reply1 = Tx Response Timestamp - Rx POLL Timestamp + Tx Delay + Rx Delay + (2 x Antenna Delay)
Db = Reply1 = Tx Response Timestamp - Rx POLL Timestamp + Loopback Calibration + (2 x Antenna Delay)

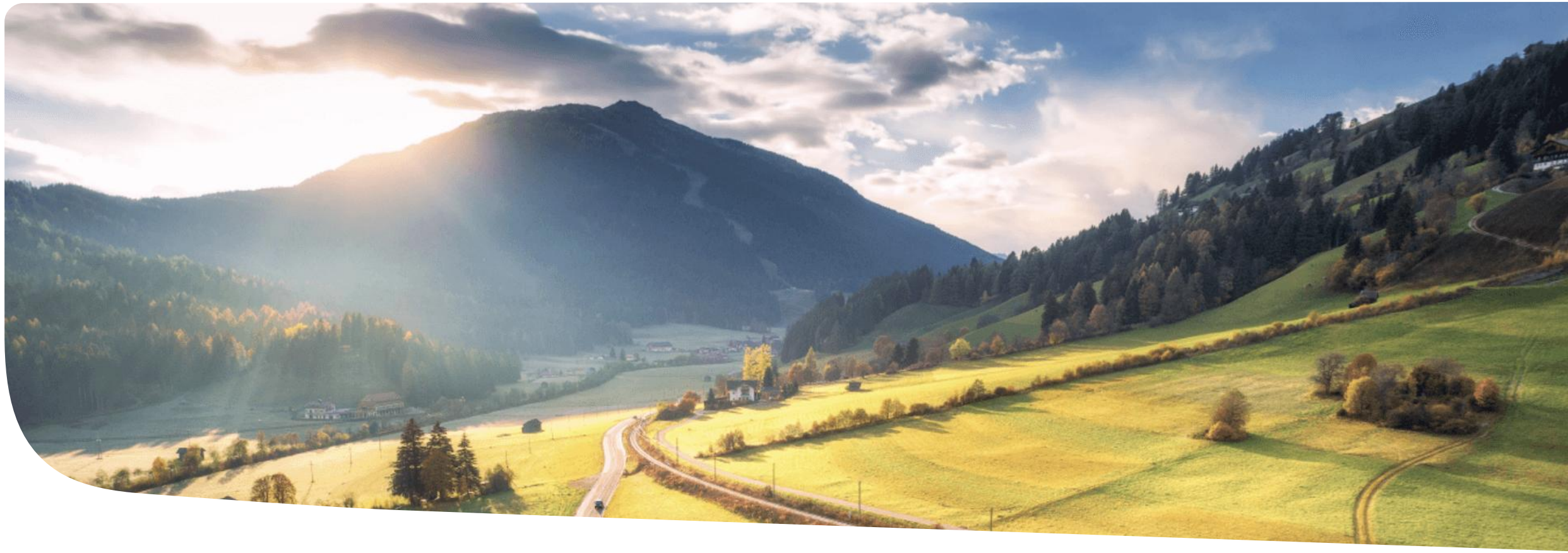
Rb = Round2 = (Rx Final Timestamp - Rx Delay - Antenna Delay) - (Tx Response Timestamp + Tx Delay + Antenna Delay)
Rb = Round2 = Rx Final Timestamp - Rx Delay - Antenna Delay - Tx Response Timestamp - Tx Delay - Antenna Delay
Rb = Round2 = Rx Final Timestamp - Tx Response Timestamp - Rx Delay - Tx Delay - (2 x Antenna Delay)
Rb = Round2 = Rx Final Timestamp - Tx Response Timestamp - Loopback Calibration - (2 x Antenna Delay)

tof = ((Ra x Rb) – (Da x Db)) / (Ra + Rb + Da + Db)
tof = ((Round 1 x Round2) – (Reply2 x Reply1)) / (Round 1 + Round2 + Reply2 + Reply1)

d = tof x (469175 / 1000000)

UWB Ranging

CCC



CCC

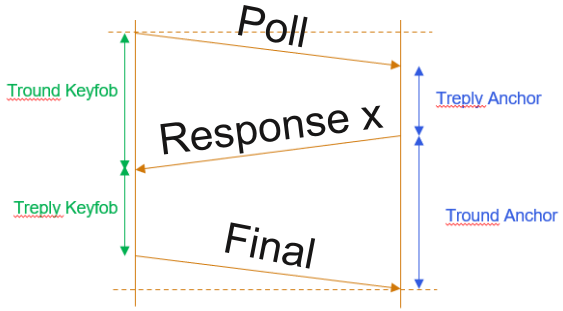


20.1 UWB MAC Architecture – RAN Concept
20.2 MAC Time Grid
20.5.1 *Ranging Exchange Sequence*
20.5.2 Ranging *Session Setup*
20.6 STS Index Incrementation
21.3 UWB Frame Format
21.4.1 *Standard compliant SYNC*
21.5 Pulse Shape Combinations
22 UWB SECURITY
….

Ranging Protocol

\41_SW_UWB\50_Docu\82_UWB\FBD6_UWB_Ranging_Protocol.xlsx

CCC Ranging
e.g. Anchor 5 (Responder 5)



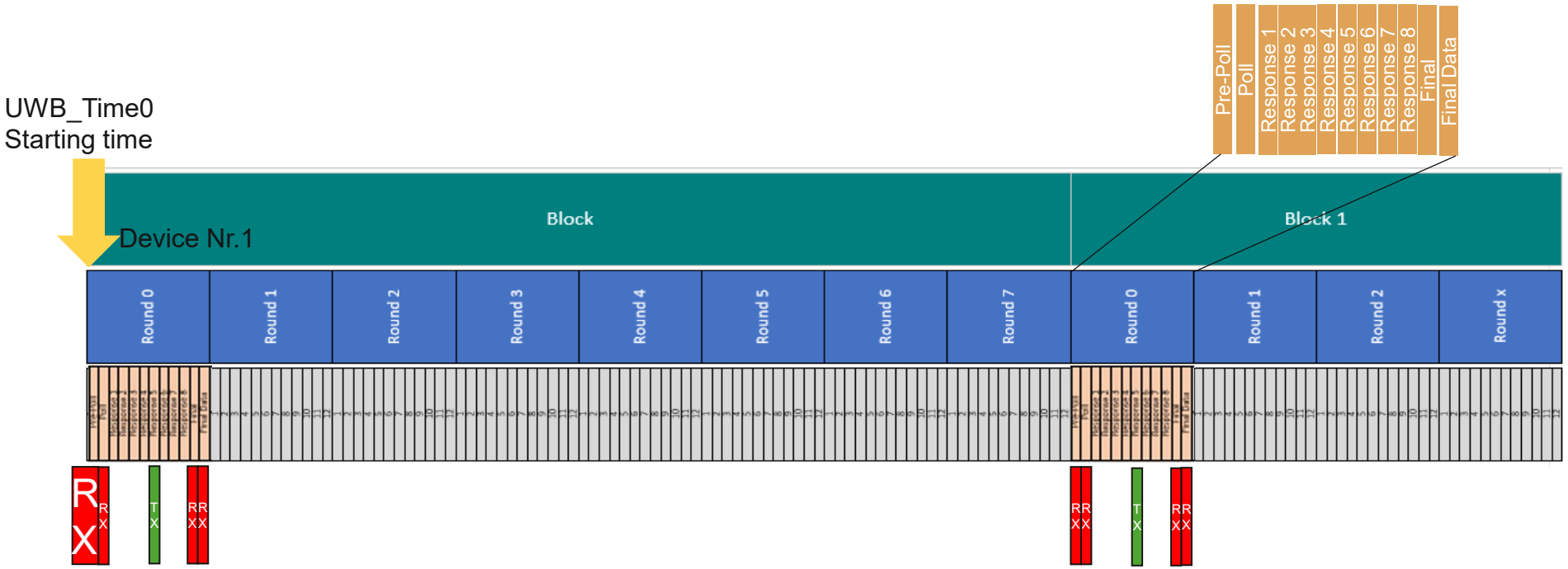
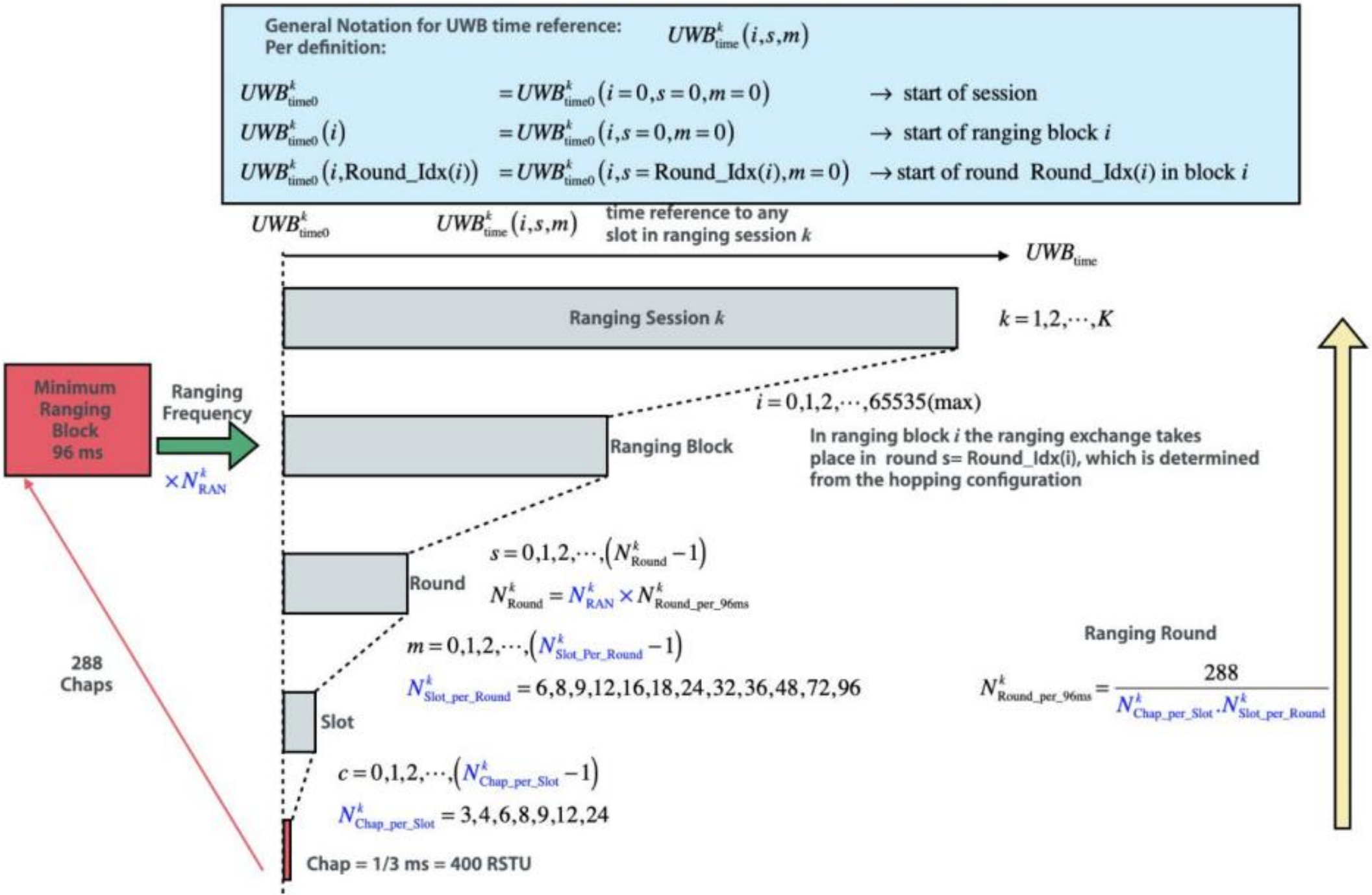


Figure 20-3: Overview of MAC Grid (See Table G- 1).



Pulse Shape

Von: Shubhamsha Karkada <shubhamsha.karkada@nxp.com>
Gesendet: Donnerstag, **10. September 2020** 17:44
An: Hobbie, Renke <Renke.Hobbie@hella.com>; Robert Spreitzer <robert.spreitzer@nxp.com>; Sridhar Sanka <sridhar.sanka@nxp.com>; Frühling, Dietmar <Dietmar.Fruehling@hella.com>; Hendrik Seidel <hendrik.seidel@nxp.com>; Stefan Lemsitzer <stefan.lemsitzer@nxp.com>; Daniel Tomas <daniel.tomas@nxp.com>
Cc: Galipo, Blake <Blake.Galipo@hella.com>; Weghaus, Ludger <Ludger.Weghaus@hella.com>; Brian Yoo <brian.yoo@nxp.com>; Euler, Manuel <Manuel.Euler@hella.com>; Bohner, Florian <Florian.Bohner@hella.com>; Schwarte, Thorsten <Thorsten.Schwarte@hella.com>; Bachmann, Jens <Jens.Bachmann@hella.com>; Kowolik, Nils <Nils.Kowolik@hella.com>; Li, Fiona <Fiona.Li@hella.com>; Mirko Schimkat <mirko.schimkat@nxp.com>; Sundaresan Swaminathan <sundaresan.swaminathan@nxp.com>; hc77.kim@samsung.com
Betreff: RE: AW: RE:(2) AW: RE: RE: SAMSUNG HELLA - Short Aligment on Next Step

Hello Renke,

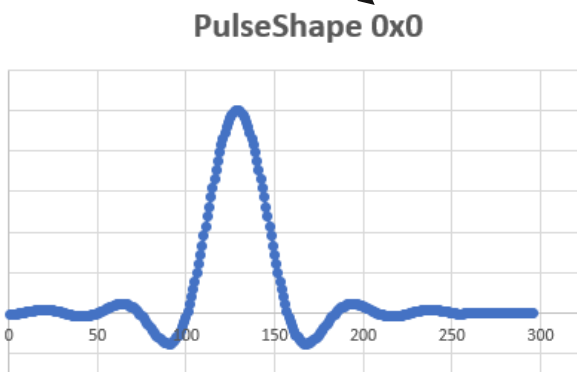
In SR100T, only **minimum-phase pulse, i.e. pulse shape 0x11 is supported**. Selected_PulseShape_Combo is used only for key derivation.

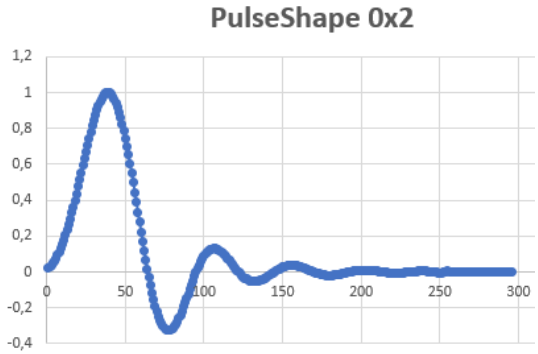
Pulse shape 0x00 will be supported in future.

Regards
Shubhamsha



?





21.5.2.1 Precursor-Free Pulse – PulseShape 0x2

PulseShape 0x2 is a special case within PulseShape 0x1, by applying minimum-phase conversion of PulseShape 0x0 using method described in [34]. A Python implementation of the linear-to-minimum phase conversion is included in Appendix G, and FIR samples for a span of 8 chips at oversampling of 32x and β of 0.45 for PulseShape 0x2 are again included in Appendix H, both provided as informative references. It is recommended to use PulseShape 0x2 whenever possible if transmitter chooses to implement precursor-free pulse shape.

Pulse Shape	Description	bitfield
Symmetrical Root-Raised-Cosine Pulse	symmetrical Root-Raised-Cosine Pulse, referred to as PulseShape 0x0, the RF envelope (or equivalent baseband pulse amplitude) is expressed by [31]	0x0
Precursor-Free Pulse	group of pulse shapes that meet the time-domain mask of Fig. 89 in [33].	0x1
	a special case within PulseShape 0x1, by applying minimum-phase conversion of PulseShape 0x0 using method described in [34].	0x2

Table 21-3: Summary of pulse shapes

Table 21-3 summarizes the different pulse shape allowed in the specification, and it is recommended to use PulseShape 0x2.

21.5.1 *Symmetrical Root-Raised-Cosine Pulse – PulseShape 0x0*

For the symmetrical Root-Raised-Cosine Pulse, referred to as PulseShape 0x0, the RF envelope (or equivalent base-band pulse amplitude) is expressed by [31],

$$r(t) = \frac{4\beta}{\pi\sqrt{T_p}} \frac{cos\left((1+\beta)\pi t/T_p\right) + \frac{sin\left((1-\beta)\pi t/T_p\right)}{4\beta(t/T_p)}}{1-(4\beta t/T_p)^2}$$

where:

β = roll-off factor with a value of 0.45 ±0.05

T_p = Pulse duration, which is inverse of the chip frequency (1/499.2MHz)

Finite impulse response (FIR) samples of PulseShape 0x0 using β of 0.45 with 32x oversampling per chip and total span of 8 chips are included in Appendix H as informative reference.

21.5.2 *Precursor-Free Pulse – PulseShape 0x1*

In general, "precursor-free" refers to the pulse's property to monotonically raise its amplitude to the first peak that also represents its main lobe. PulseShape 0x1 represents a group of pulse shapes that meet the time-domain mask of Fig. 89 in [33].

21.5.3 *PulseShape Combinations*

Compliant receivers shall be able to receive and suitably process both symmetrical and precursor-free pulses. It is up to the implementer whether the same receiver configuration is used for PulseShape 0x2 and PulseShape 0x1, or whether reception of PulseShape 0x2 is explicitly optimized via a separate receiver configuration. A list of possible transmitter pulse shape combinations (PulseShape_Combo) for initiator and responder is shown in Table 21-3. The responder's and initiator's receive configurations shall be set according to the PulseShape selections made by the initiators and responder's transmitters, respectively.

PulseShape_Combo	Initiator	Responder
	Transmit PulseShape	Transmit PulseShape
0x00	0x0	0x0
0x01	0x0	0x1
0x02	0x0	0x2
0x10	0x1	0x0
0x11	0x1	0x1
0x12	0x1	0x2
0x20	0x2	0x0
0x21	0x2	0x1
0x22	0x2	0x2

Table 21-4: PulseShape_Combo

Re: Tx-power; status; next steps

MC Michael Coady <mcoady@apple.com>
An Weghaus, Ludger
Cc Hobbie, Renke; Curkin, Roman; Frühling, Dietmar; Li, Fiona

Aktionselemente

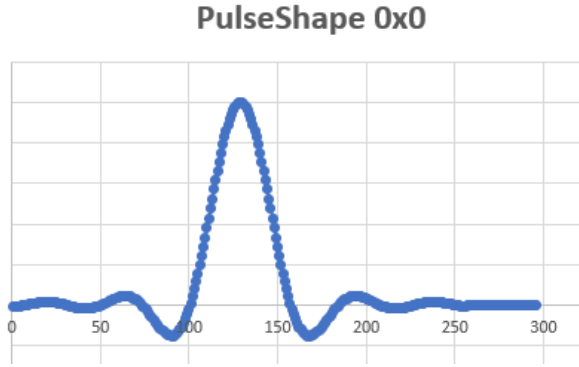
Hello Ludger,

1-
Latest upload in the link to CCC below , which has major update to Pulse Shape definitions.
We will support both Linear-SRRC and precursor-free pulse shape down the road, but only support L-SRRC in the current FW.

Using the IEEE figure as an illustration, we are transmitting the pulse on the right, and our receiver is expecting this pulse shape as well.

https://causeway.carconnectivity.org/wg/DKTG/document/8540

2- Can you remind me which UWB silicon provider you are using?



Regards,
Mike

Michael Coady
□
408-219-9941
mcoady@apple.com

On Aug 19, 2020, at 10:04 AM, Weghaus, Ludger <Ludger.Weghaus@hella.com> wrote:

Hi Mike

Two brief questions:

1. In the IEEE802.15.4 there are 2 pulse shapes ("compliant pulse example") being shown as shown in the picture below. Cou
this be adjusted)? This question is intended so we set our parameters fitting best to the iPhone to have best ranging perfo
the transmitted pulse shape and the reference pulse r(t) on the right is for internal cross correlation calculations but I am
implementation. If I wanted to measure this I'd need to open/break the iPhone which I'd prefer not to. So any informatio

<image001.jpg>

Ranging Protocol (Payload Messages)

FBD6_UWB_Ranging_Protocol.xlsx
C: » Projekte » BMW_FBD6 » 50_FBD6 » 41_SW_UWB » 50_Docu » 82_UWB

Pre-Poll:

After to Decryption				
Pre-Poll: Byte number	**Meaning**		**Pre-Poll: Byte number**	**Meaning**
dataPayload[0] = 0x49	Frame Control		dataPayload[23] = 0xBA	UWB Session ID
dataPayload[1] = 0x2B	Frame Control		dataPayload[24] = 0x73	UWB Session ID
dataPayload[2] = 0x3E	Destination address		dataPayload[25] = 0xC3	UWB Session ID
dataPayload[3] = 0xDD	Destination address		dataPayload[26] = 0xA8	UWB Session ID
dataPayload[4] = 0x16	Security Control		dataPayload[27] = 0x01	Poll STS Index
dataPayload[5] = 0x0	Frame Counter		dataPayload[28] = 0x00	Poll STS Index
dataPayload[6] = 0x0	Frame Counter		dataPayload[29] = 0x00	Poll STS Index
dataPayload[7] = 0x0	Frame Counter		dataPayload[30] = 0x00	Poll STS Index
dataPayload[8] = 0x0	Frame Counter		dataPayload[31] = 0x00	Ranging Block
dataPayload[9] = 0x68	Key Source (Low) (Source Short Address)		dataPayload[32] = 0x00	Ranging Block
dataPayload[10] = 0x11	Key Source (Low) (Source Short Address)		dataPayload[33] = 0x00	Hop Flag
dataPayload[11] = 0x87	Key Source (High) (Source PAN ID)		dataPayload[34] = 0x00	Round Index
dataPayload[12] = 0xBC	Key Source (High) (Source PAN ID)		dataPayload[35] = 0x00	Round Index
dataPayload[13] = 0xAA	Key Index		dataPayload[36] = 0xFA	MIC
dataPayload[14] = 0x5	Vendor Specific Header IE: Length		dataPayload[37] = 0x93	MIC
dataPayload[15] = 0x0	Vendor Specific Header IE: ID		dataPayload[38] = 0x86	MIC
dataPayload[16] = 0x0E	Vendor OUI		dataPayload[39] = 0x74	MIC
dataPayload[17] = 0x8B	Vendor OUI		dataPayload[40] = 0xE3	MIC
dataPayload[18] = 0x60	Vendor OUI		dataPayload[41] = 0xA4	MIC
dataPayload[19] = 0x1	UWB MAC Message ID		dataPayload[42] = 0x5B	MIC
dataPayload[20] = 0x0D	UWB MAC message payload length		dataPayload[43] = 0xD7	MIC
dataPayload[21] = 0x80	Header Termination IE		dataPayload[44] = 0xED	FCS (CRC)
dataPayload[22] = 0x3F	Header Termination IE		dataPayload[45] = 0xF4	FCS (CRC)

Final Data:

After to Decryption				
Final Data: Byte number	**Meaning**		**Final Data: Byte number**	**Meaning**
dataPayload[0] = 0x49	Frame Control		dataPayload[23] = 0x01	UWB Session ID
dataPayload[1] = 0x2B	Frame Control		dataPayload[24] = 0x00	UWB Session ID
dataPayload[2] = 0x39	Destination address		dataPayload[25] = 0x00	UWB Session ID
dataPayload[3] = 0xAD	Destination address		dataPayload[26] = 0x00	UWB Session ID
dataPayload[4] = 0x16	Security Control		dataPayload[27] = 0x00	Ranging Block
dataPayload[5] = 0x01	Frame Counter		dataPayload[28] = 0x00	Ranging Block
dataPayload[6] = 0x0	Frame Counter		dataPayload[29] = 0x00	Hop Flag
dataPayload[7] = 0x0	Frame Counter		dataPayload[30] = 0x00	Round Index
dataPayload[8] = 0x0	Frame Counter		dataPayload[31] = 0x00	Round Index
dataPayload[9] = 0x1	Key Source (Low) (Source Short Address)		dataPayload[32] = 0x0A	Final STS Index
dataPayload[10] = 0x0	Key Source (Low) (Source Short Address)		dataPayload[33] = 0x00	Final STS Index
dataPayload[11] = 0x0	Key Source (High) (Source PAN ID)		dataPayload[34] = 0x00	Final STS Index
dataPayload[12] = 0x0	Key Source (High) (Source PAN ID)		dataPayload[35] = 0x00	Final STS Index
dataPayload[13] = 0xAA	Key Index		dataPayload[36] = 0x00	Ranging Timestamp Final Tx
dataPayload[14] = 0x5	Vendor Specific Header IE: Length		dataPayload[37] = 0x00	Ranging Timestamp Final Tx
dataPayload[15] = 0x0	Vendor Specific Header IE: ID		dataPayload[38] = 0x8E	Ranging Timestamp Final Tx
dataPayload[16] = 0x69	Vendor OUI		dataPayload[39] = 0x44	Ranging Timestamp Final Tx
dataPayload[17] = 0xDF	Vendor OUI		dataPayload[40] = 0x08	Number of Ranging Responders
dataPayload[18] = 0x4	Vendor OUI		dataPayload[41] = 0x00	Responder Index
dataPayload[19] = 0x2	UWB MAC Message ID		dataPayload[42] = 0x6A	Ranging Timestamp Responder #
dataPayload[20] = 0x4A	UWB MAC message payload length		dataPayload[43] = 0x83	Ranging Timestamp Responder #
dataPayload[21] = 0x80	Header Termination IE		dataPayload[44] = 0xA4	Ranging Timestamp Responder #
dataPayload[22] = 0x3F	Header Termination IE		dataPayload[45] = 0x07	Ranging Timestamp Responder #
			dataPayload[46] = 0x00	Ranging Timestamp Uncertainty Responder #
			...	
			dataPayload[95] = 0x00	Ranging Timestamp Uncertainty Responder #
			dataPayload[96] = 0x02	Ranging Status Responder #
			dataPayload[97] = 0xA4	MIC
			dataPayload[98] = 0xB0	MIC
			dataPayload[99] = 0x0A	MIC
			dataPayload[100] = 0x8B	MIC
			dataPayload[101] = 0x92	MIC
			dataPayload[102] = 0x35	MIC
			dataPayload[103] = 0x1A	MIC
			dataPayload[104] = 0xA1	MIC
			dataPayload[105] = 0x8C	FCS (CRC)
			dataPayload[106] = 0x91	FCS (CRC)

7. MAC frame formats

7.1 Device extended address

A device's extended address shall be a 64-bit extended universal identifier (EUI-64), as defined by IEEE Std 802-2014 and assigned by the IEEE Registration Authority.[12]

The EUI-64 shall be sent in the canonical form defined in IEEE Std 802-2014.

7.2 General MAC frame format

The general MAC frame for Frame Type values other than fragment and extended, as defined in Table 7-1, shall be formatted as illustrated in Figure 7-1.

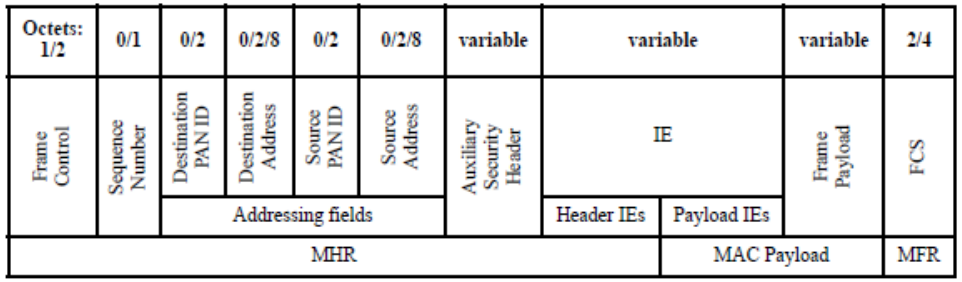


Figure 7-1—General MAC frame format

The fields of the MHR appear in a fixed order; however, some fields may not be included in all frames.

If the Frame Type field indicates a Fragment packet or Frak then the frame format shall be as formatted as defined in 23.3.3 and 23.3.6.2, respectively.

If the Frame Type field indicates an Extended frame, then the frame format shall be formatted as illustrated in 7.3.6.

7.2.1 Frame Control field

The Frame Control field for frames other than the Multipurpose frame, Fragment frame and Extended frame shall be formatted as illustrated in Figure 7-2. The Frame Control fields for the Multipurpose frame and Extended frame are specified in 7.3.5 and 7.3.6, respectively.

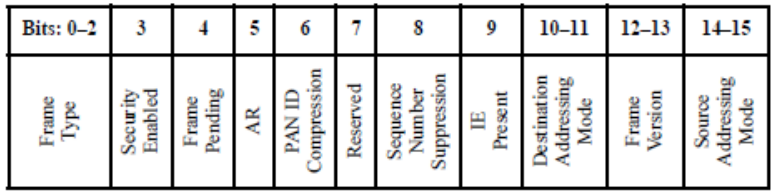


Figure 7-2—Format of the Frame Control field

The Fragment frame type does not have a Frame Control field.

7.2.1.1 Frame Type field

The Frame Type field shall be set as defined in Table 7-1.

Table 7-1—Values of the Frame Type field

Frame type value b2 b1 b0	Description
000	Beacon
001	Data
010	Acknowledgment
011	MAC command
100	Reserved
101	Multipurpose
110	Fragment or Frak[a]
111	Extended

[a]The Fragment and Frak formats are defined in 23.3.3 and 23.3.6.2, respectively.

Frame formats for each of the Frame Type field values are specified in 7.3.

7.2.1.2 Security Enabled field

The Security Enabled field shall be set to one if the frame is protected by the MAC sublayer and shall be set to zero otherwise. The Auxiliary Security Header field of the MHR shall be present only if the Security Enabled field is set to one.

7.2.1.3 Frame Pending field

The Frame Pending field shall be set to one if the device sending the frame has more data for the recipient, as described in 6.7.3. This field shall be set to zero otherwise.

The Frame Pending field shall be used only in Beacon frames or frames transmitted either during the CAP by devices operating on a beacon-enabled PAN or at any time by devices operating on a nonbeacon-enabled PAN.

When operating in low-energy (LE) CSL mode, the frame pending bit may be set to one to indicate that the transmitting device has pending frames to send to the same recipient and expects the recipient to keep the radio on until the frame pending bit is reset to zero.

When operating in TSCH mode, the frame pending bit can be set to one to indicate that the recipient should stay on in the next timeslot and on the same channel if there is no link scheduled.

At all other times, it shall be set to zero on transmission and ignored on reception.

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Ranging Protocol (Payload Messages)

Description	Frame Control		Sequence Number	Destination PAN ID		Destination address								Source PAN ID		Source address								Auxiliary Security Header										Vendor Specific Header							Header Termination IE		Frame Payload and MIC														FCS (CRC)	
																								Security Control	FRAME Counter	FRAME Counter	FRAME Counter	FRAME Counter	Key Source (Source Short Address)	Key Source (Source Short Address)	Key Source (Source PAN ID)	Key Source (Source PAN ID)	Key Index	Key Identifier	Key Identifier	Key Identifier	Key Identifier	Vendor Specific Header	Vendor Specific Header	Vendor Specific Header	Vendor Specific Header	Vendor Specific Header	Vendor Specific Header	Vendor Specific Header																					
																								Key Source Field																																					
																												Key Source Low		Key Source High																															
Message Evaluation	49	2B				39	AD	---	---	---	---	---	---											10	C8	71	1	0	1	0	0	0	AA					5	0	69	DF	4	1	0D	80	3F	1	0	0	0	C1	AA	22	0	E4	B8	0	0	0	42	57
Byte Number	0	1				2	3																	4	5	6	7	8	9	10	11	12	13					14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
RCM Message	49	2B				39	AD																	10	C8	71	1	0	1	0	0	0	AA					5	0	69	DF	4	1	0D	80	3F	1	0	0	0	C1	AA	22	0	E4	B8	0	0	0	42	57

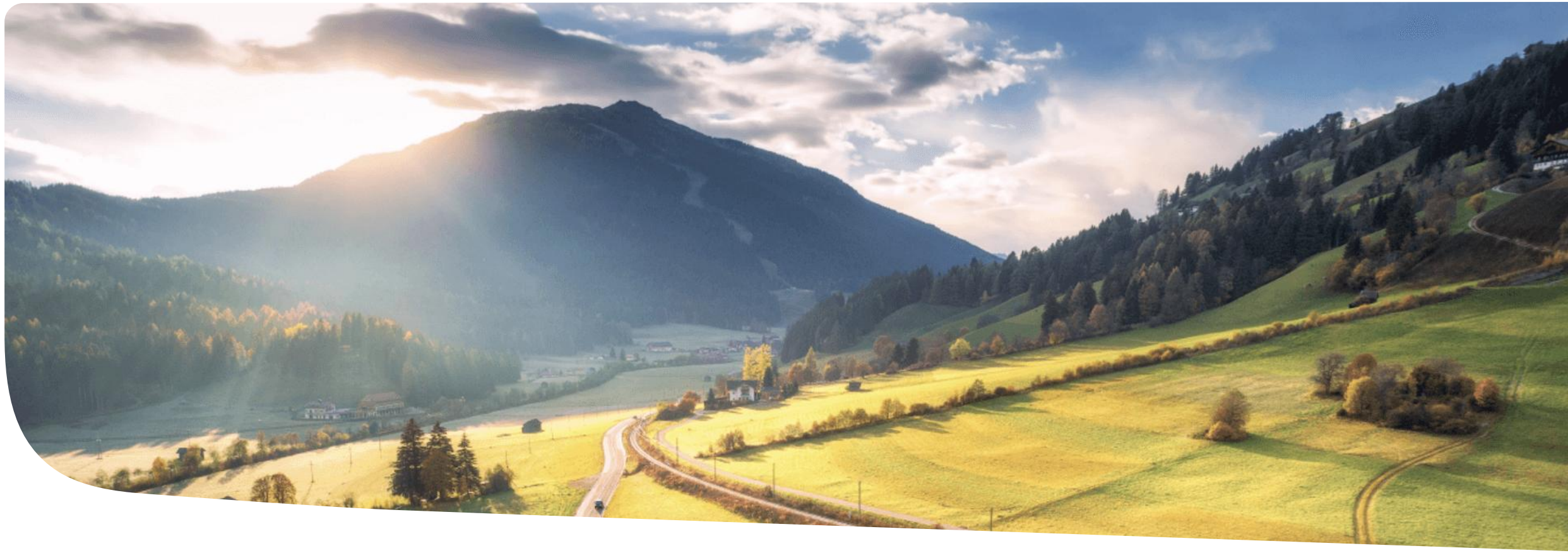
Bit Number	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Frame Control	Reserved	PAN ID Compression	AR ACK Required	Frame Pending	Security Enabled	Frame Type			Source Addressing Mode		Frame Version		Destination Addressing Mode		IE Present	Sequence Number Suppression
	0	1	0	0	1	0	0	1	0	0	1	0	1	0	1	1
Evaluation Result			ACK Not Required	Not Pending	Enabled	Data			PAN ID and address fields are not present		IEEE Std 802.15.4-2006		Address field contains a short address (16 bit)		IE Present	Supressed

Bit Number	7	6	5	4	3	2	1	0
Security Control	Reserved	ASN in NONCE	Frame Counter Suppression	Key Identifier Mode		Security Level		
	0	0	0	1	0	0	0	0

Bit Number	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	0	1	2	0	1
Vendor Specific Header	Length							IE ID								Type	Vendor OUI			Vendor Info	UWB MAC message payload length
	0	0	0	0	1	0	1	0	0	0	0	0	0	0	0	0	4	DF	69	oll UWB MAC message	0D = 13

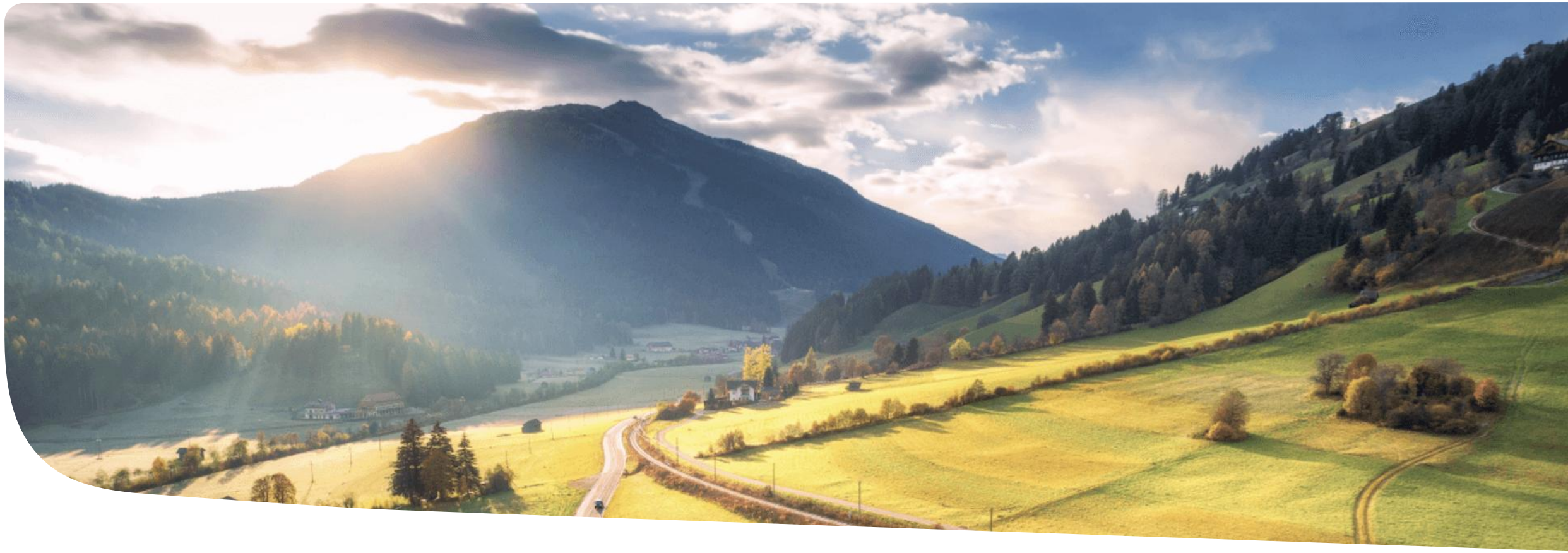
UWB Ranging

System Architecture/Overview

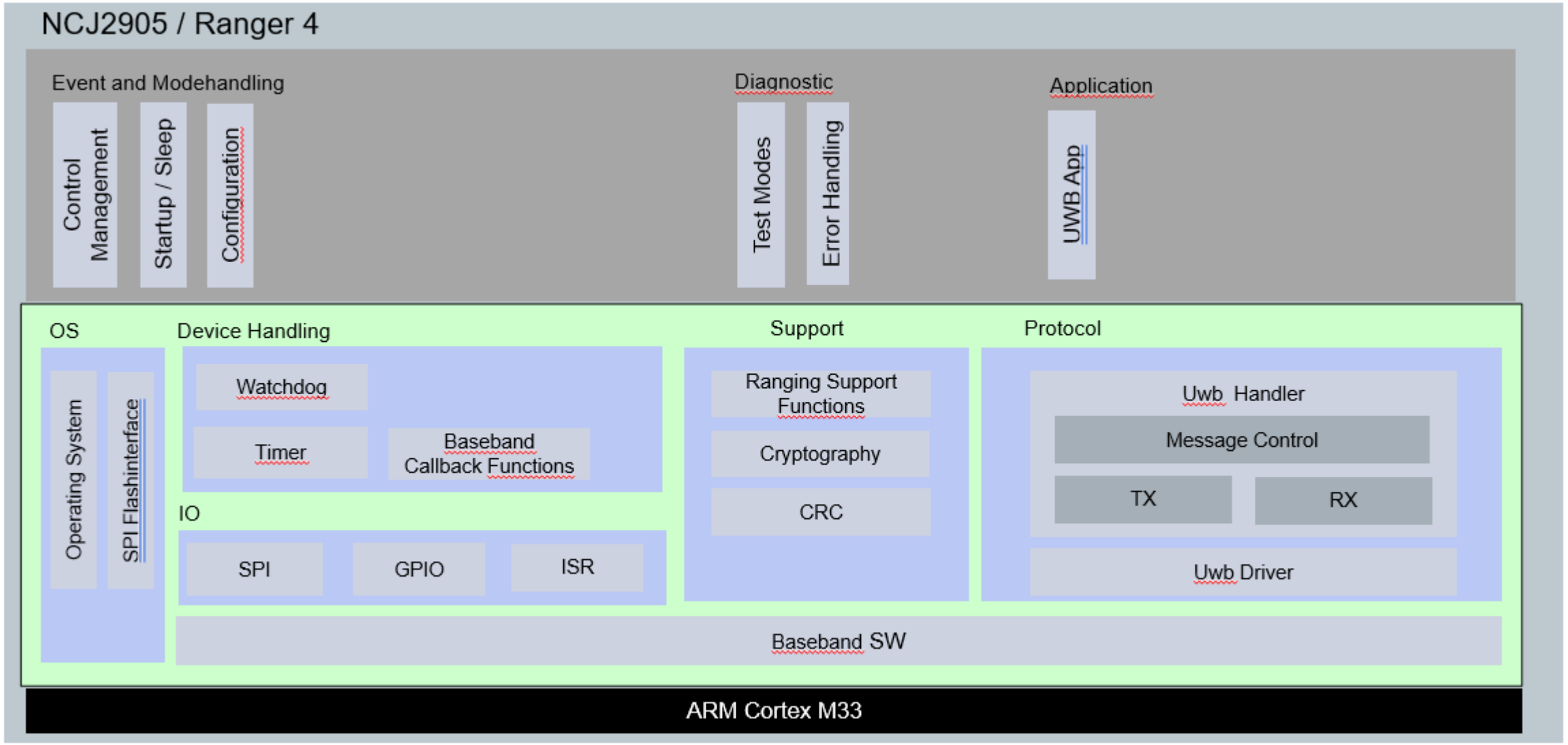


UWB Ranging

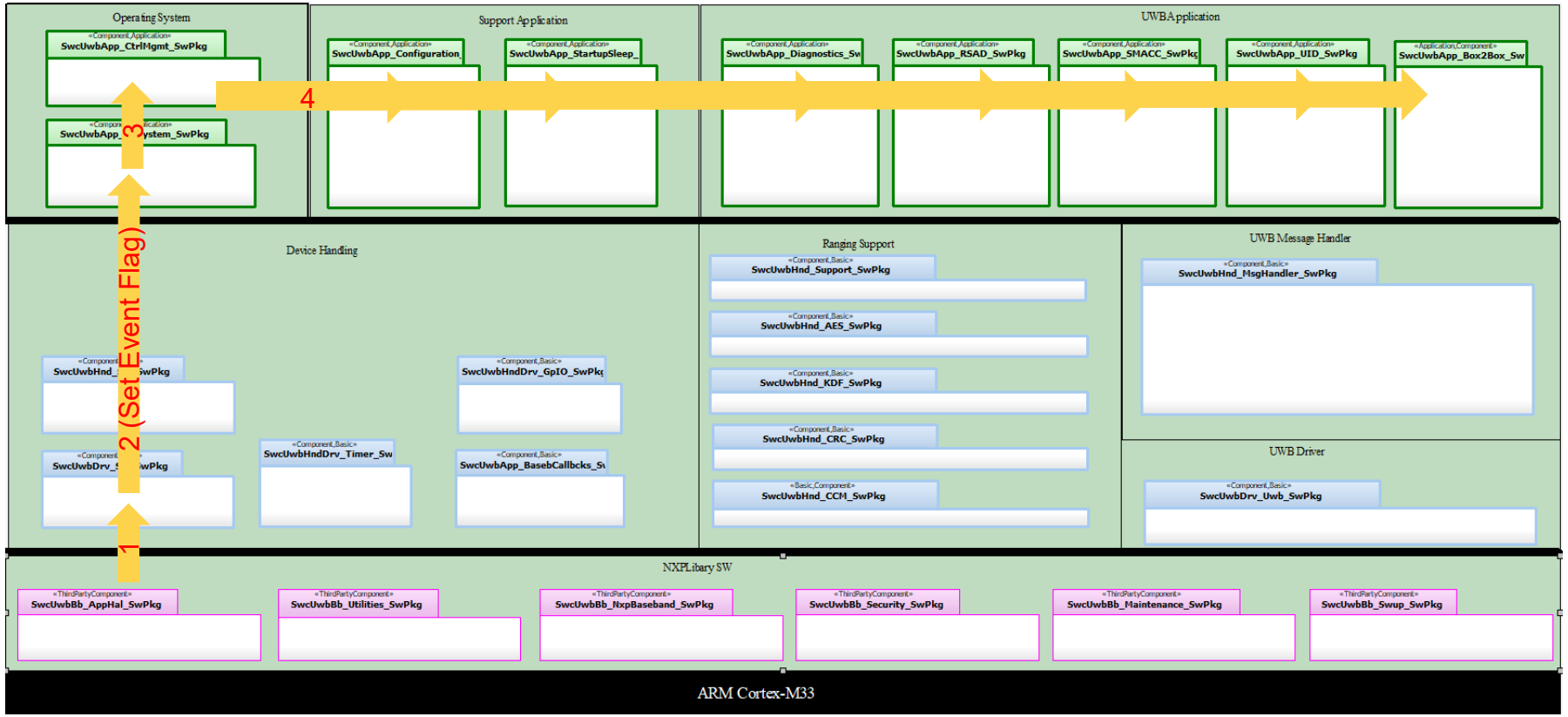
SW Architecture



Ranger4 SW Architecture (SMASTD/Ford)

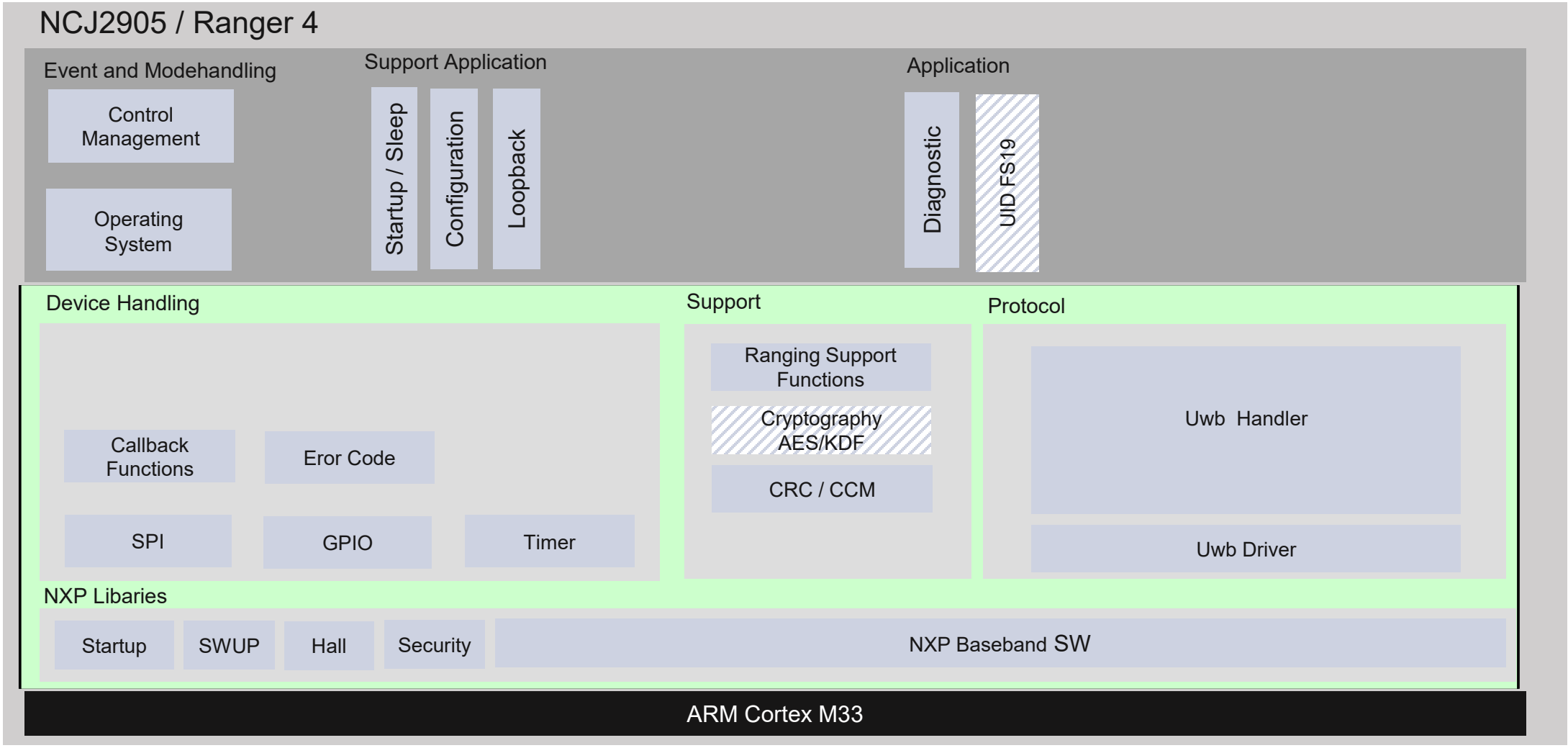


SW Architecture
e.g. new SPI Command

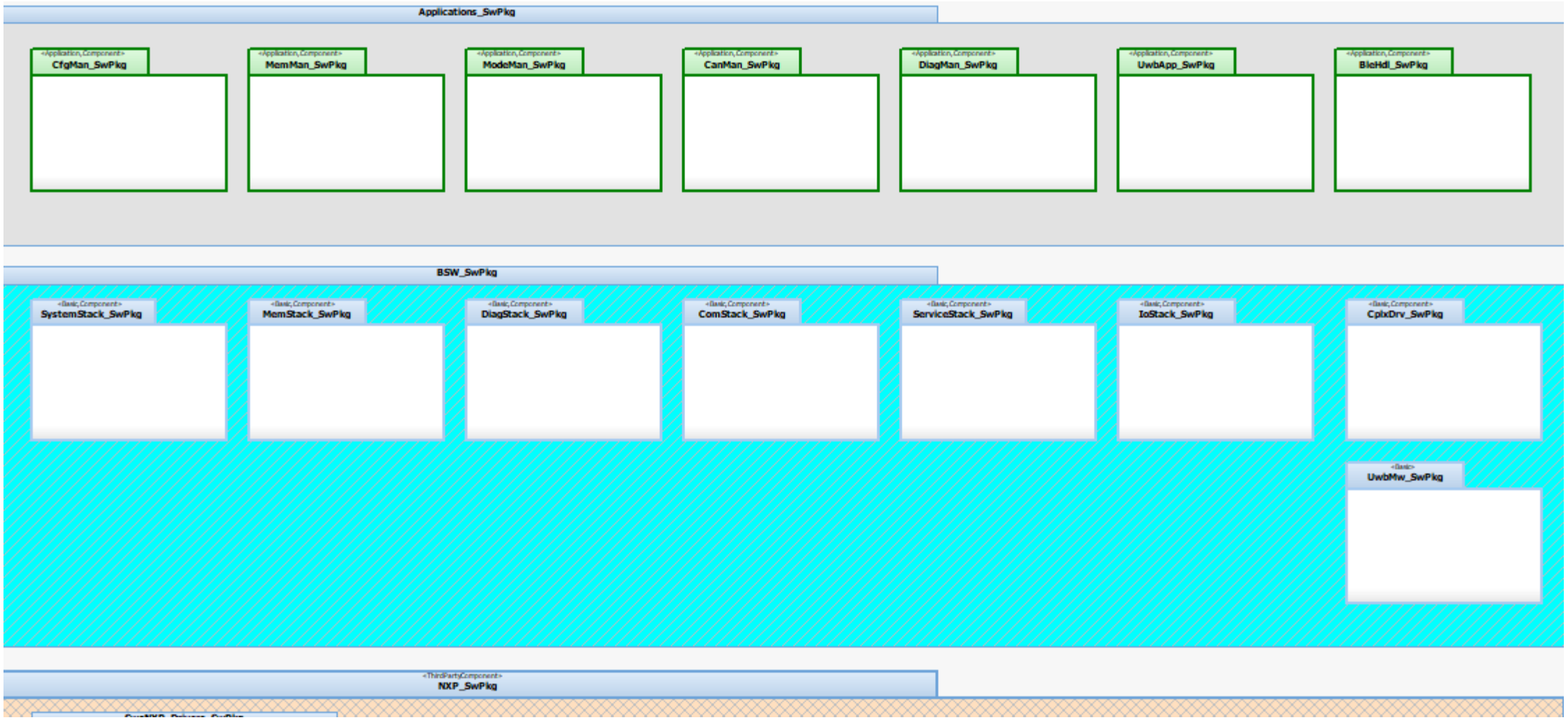


Ranger4 - SW Architecture Concept FS19

Changes necessary

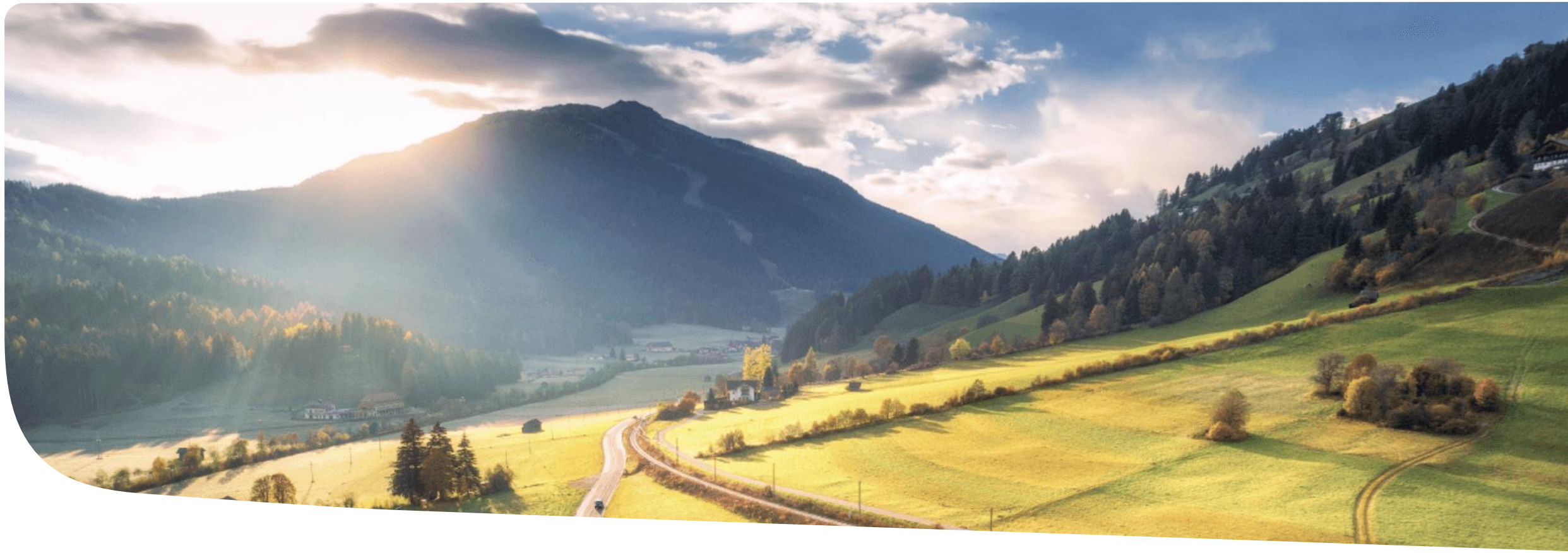


Ranger5 Architecture (FBD6)

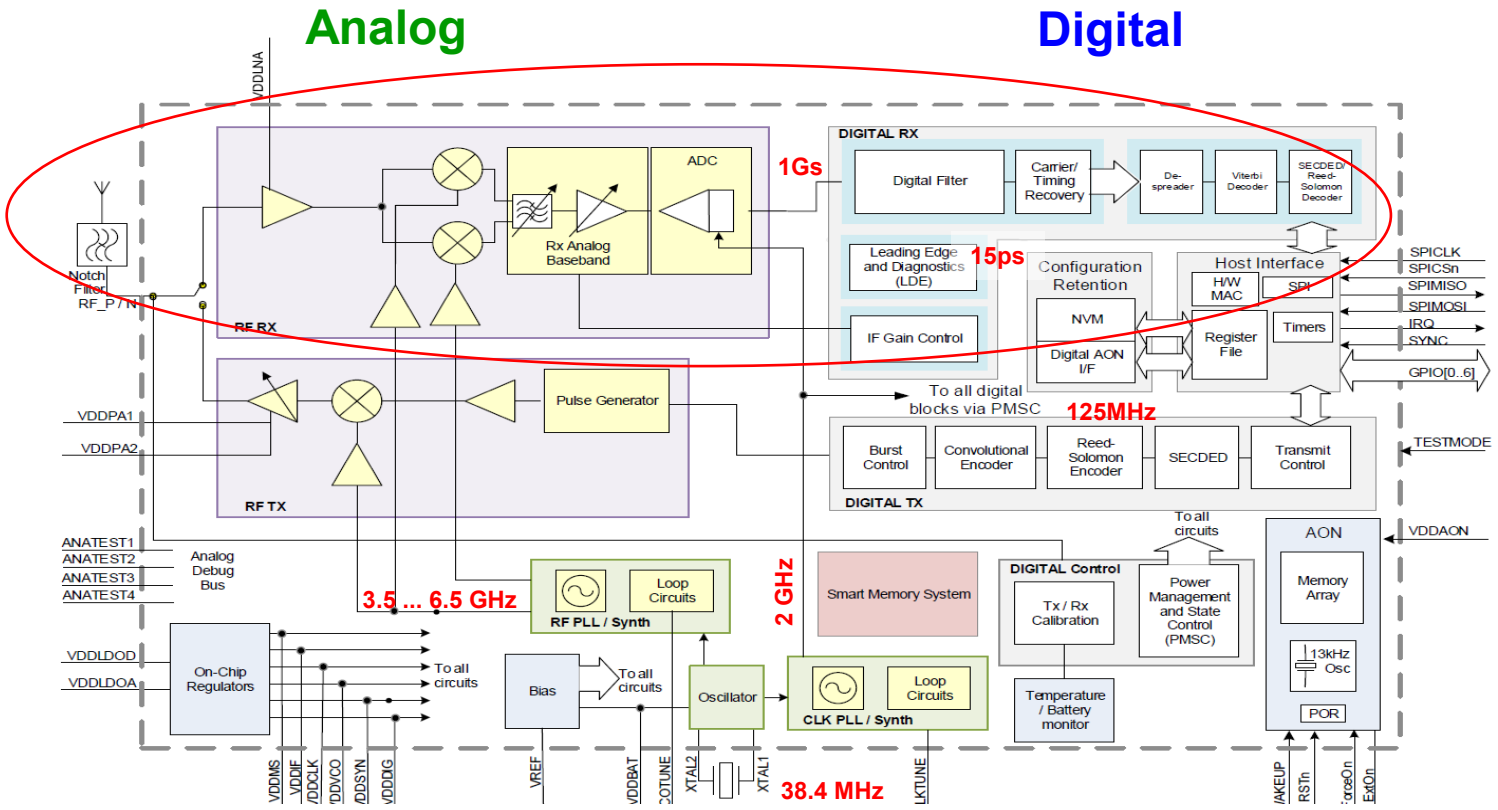


UWB Ranging

Ranger 4



Block Schematic Ranger1 (NCJ29D0, DW1000) Receiver



Ranger 4 Architecture

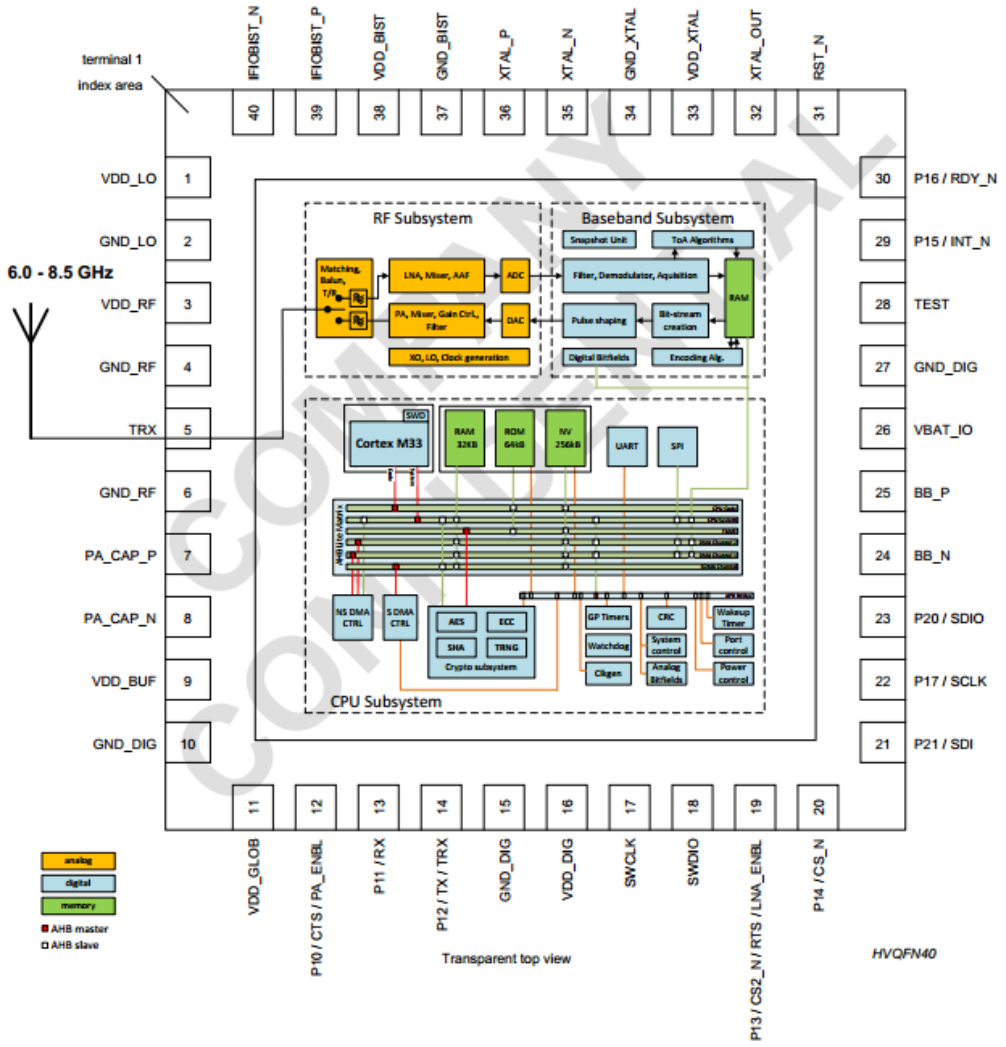


Figure 3. Block diagram

UWB for Hands Free Entry/Go
System Overview / Interfaces

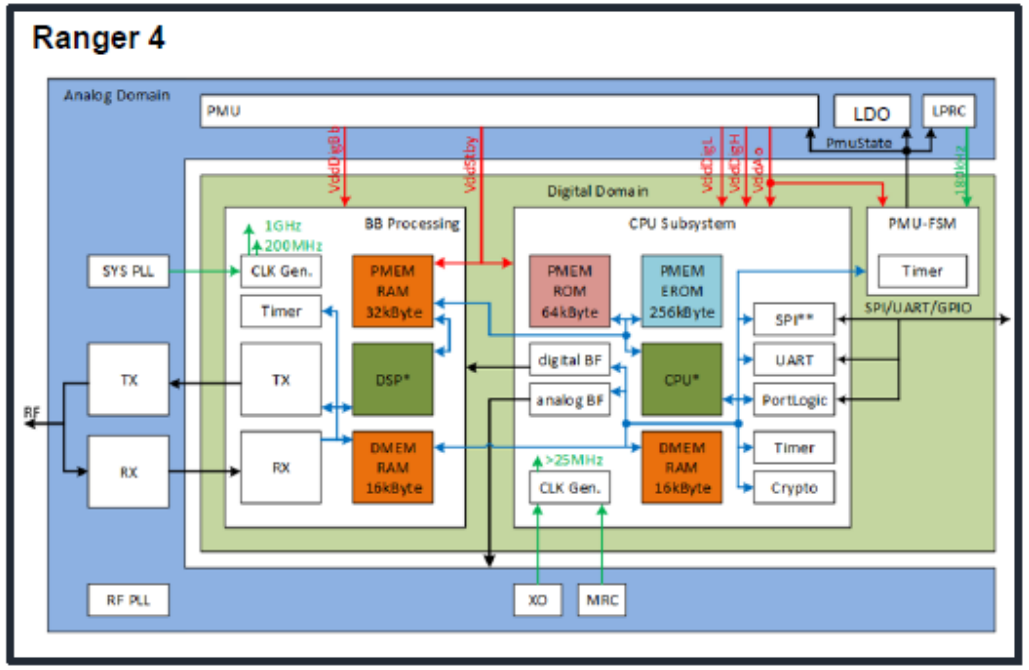
Ranger 4 – Ranging and IoT Interoperability

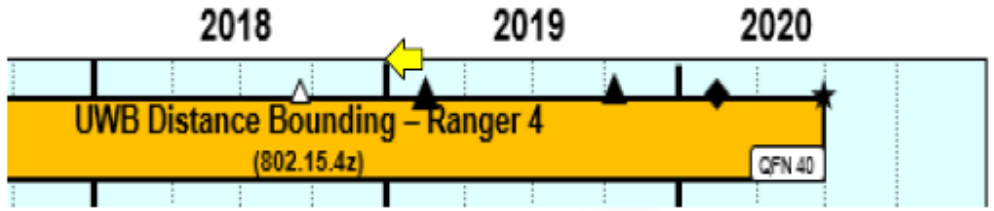
Key Feature

- High band operation (6.5 ~ 8.0 GHz)
- Lowest BOM and integrated Balun
- IEEE 802.15.4a compliant for IoT communication
- IEEE 802.15.4z for Secure Ranging (STS)
- Short Frame capable for lowest Rx energy (RSD Application)
- ARM M33 core with 256KB Flash (128KB User Code)
- QFN 40 (6x6mm)

Development Status

- Eng. Samples available and functional validation in progress
  - STS & Payload reception OK
  - Ranging test planned
  - Helios IntOp planned
- Frimware implementation started (Ranging Control Interface)
- Customer sampling (Q1/2019) pull-in considered

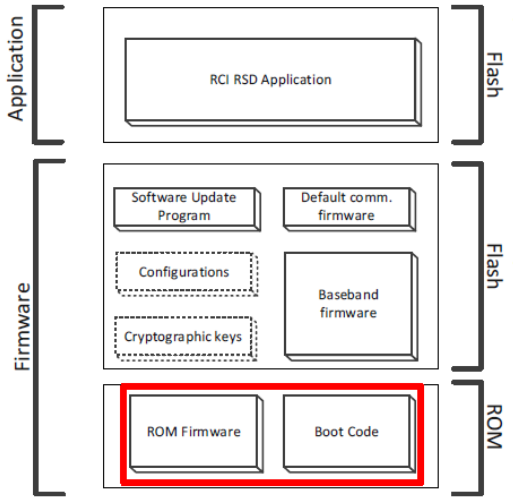




1.

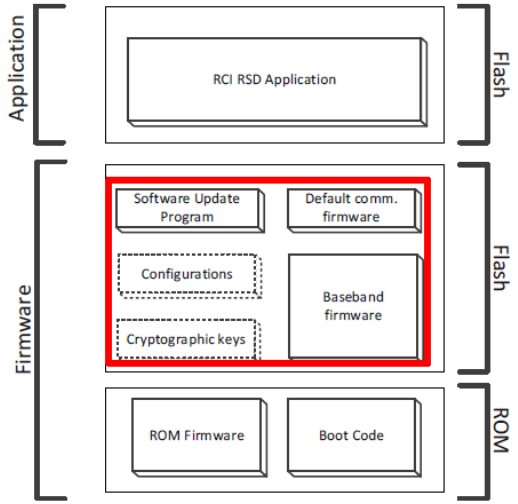
CONFIDENTIAL

Ranger 4 – Software Overview (1)



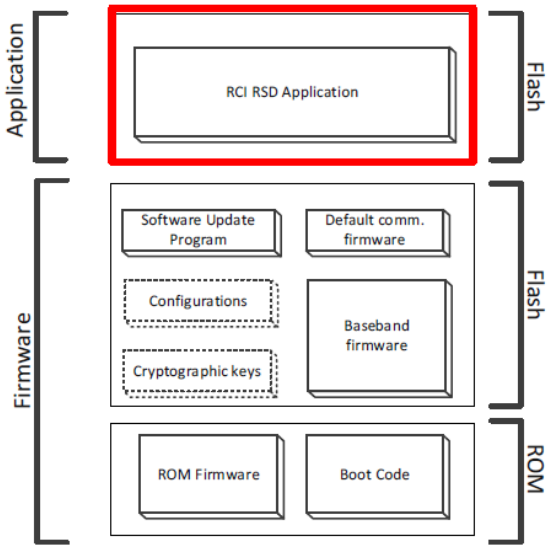
- ROM Firmware
  - Security and cryptographic functions
  - Cryptographic key management
  - Flash drivers
  - Power management
  - HAL (Hardware Abstraction Layer)
- Boot code
  - Performs device startup
  - Applies device configurations

Ranger 4 – Software Overview (2)



- Flash Firmware
  - Software updater program
    - Securely updates software components
    - Securely applies patches to ROM
  - Storage: Cryptographic Keys
    - Cannot be read out
    - Stored securely in Flash
  - Storage: Configurations
    - Radio configuration
    - Device configurations
  - Baseband firmware
    - Controls the UWB radio
  - Default communication firmware
    - Communication firmware for software update over wired interface

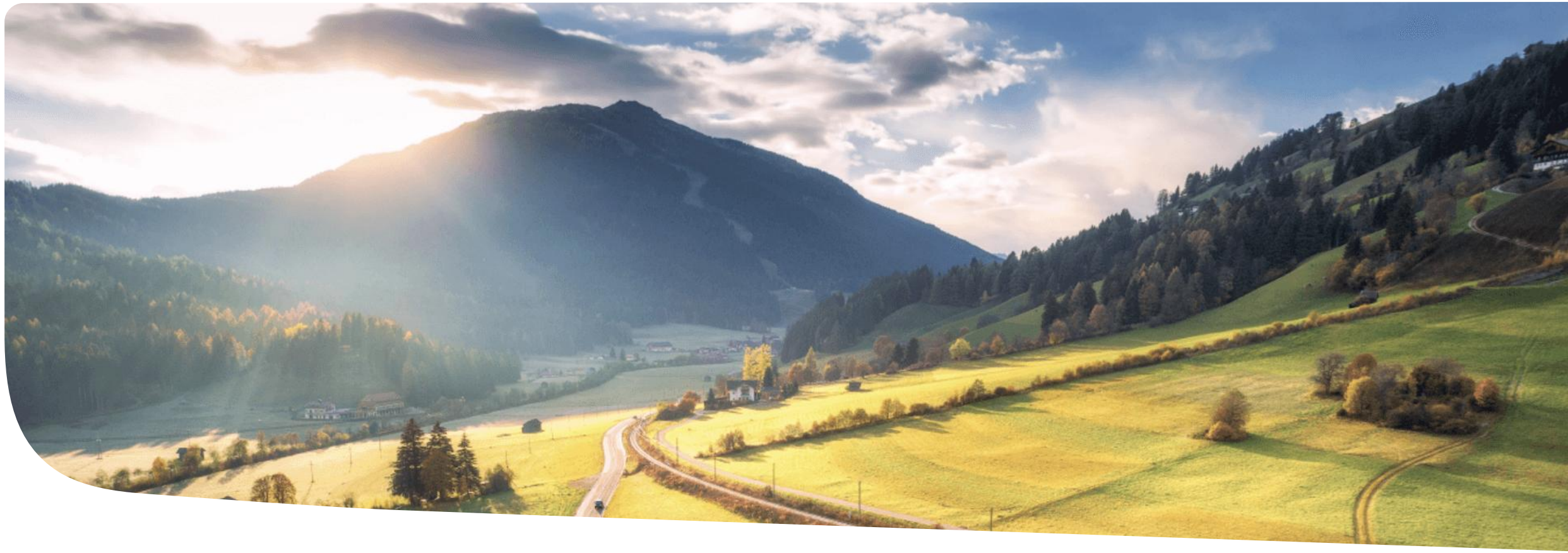
Ranger 4 – Software Overview (3)



- Application (Draft definition)
  - RCI RSD Application
    - Performs distance measurement for RSD
    - Configured and controlled by RCI commands
    - RCI supported by communication libraries for wired interfaces
      - SPI
      - LIN
  - Can be replaced by any customer application.

Ranger5

SW Architecture



UWB/NCJ29D6 – Block diagram

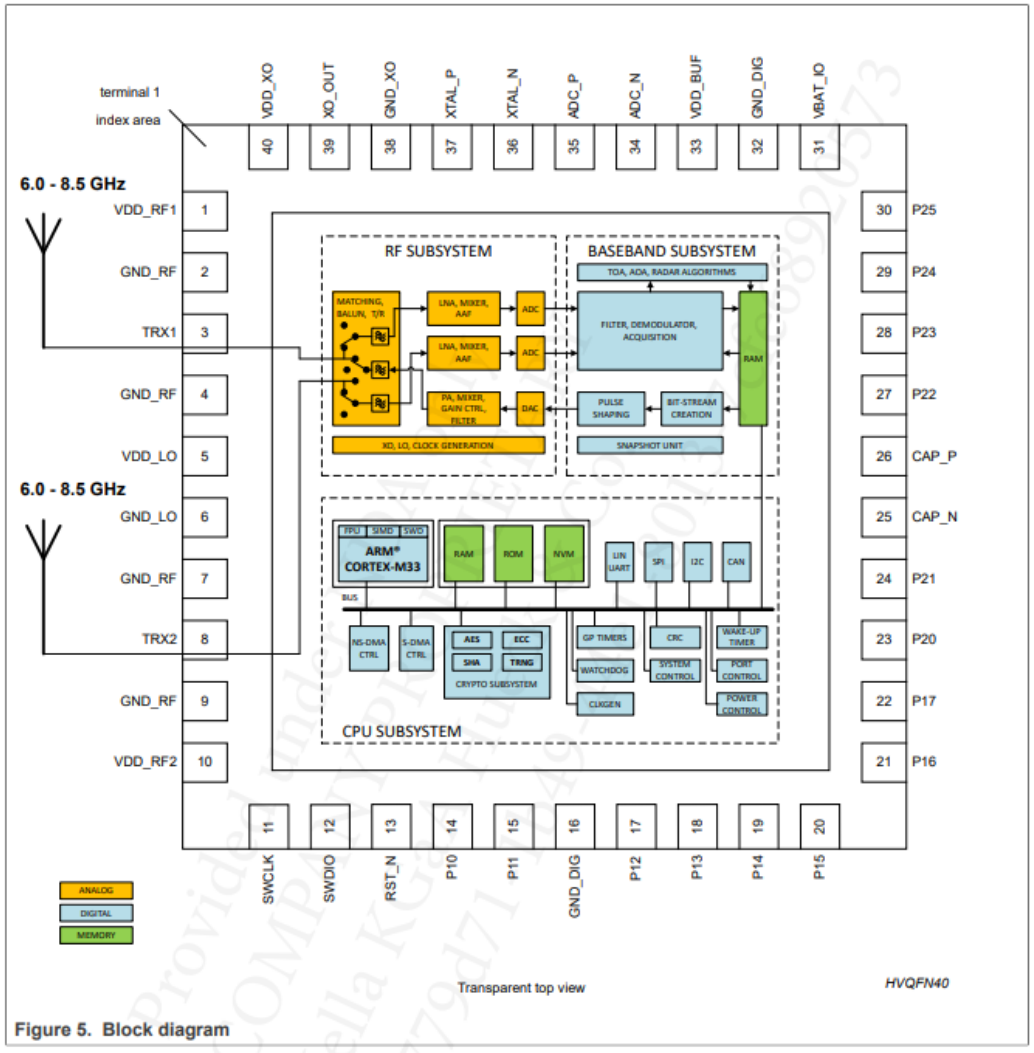


Figure 5. Block diagram

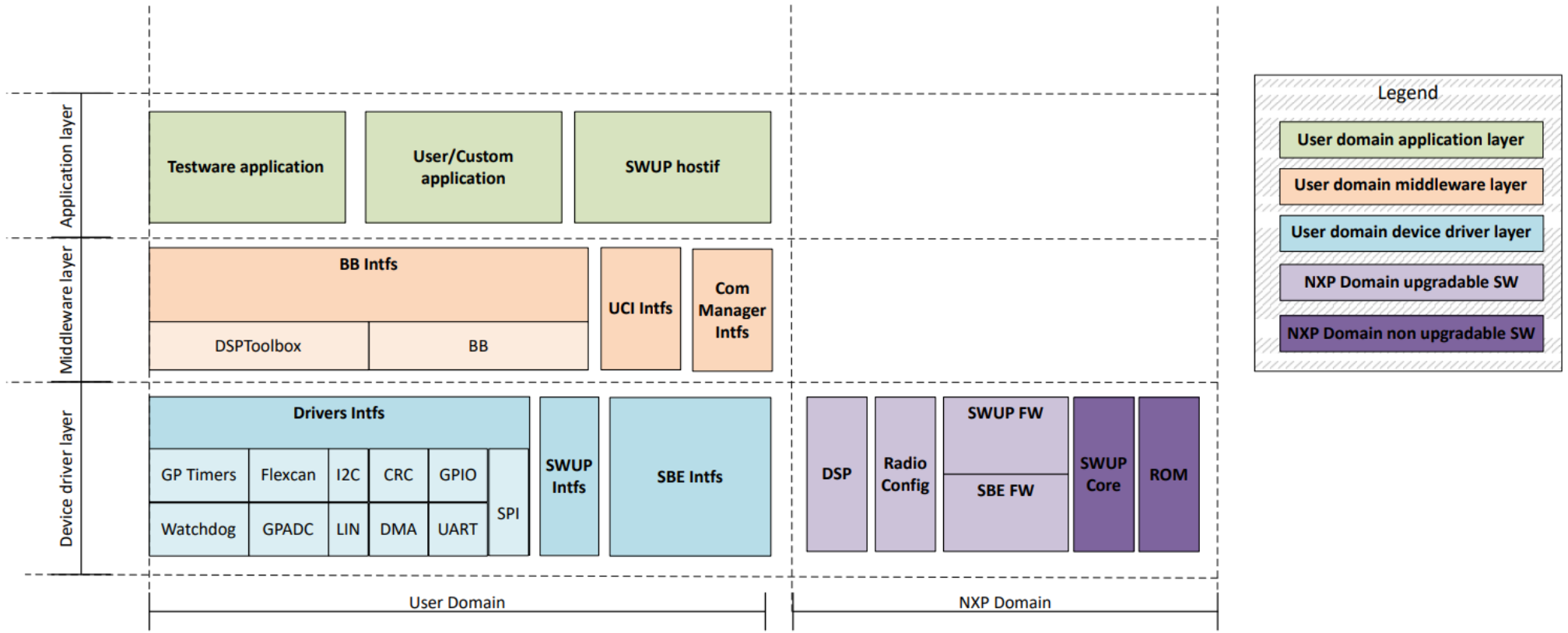
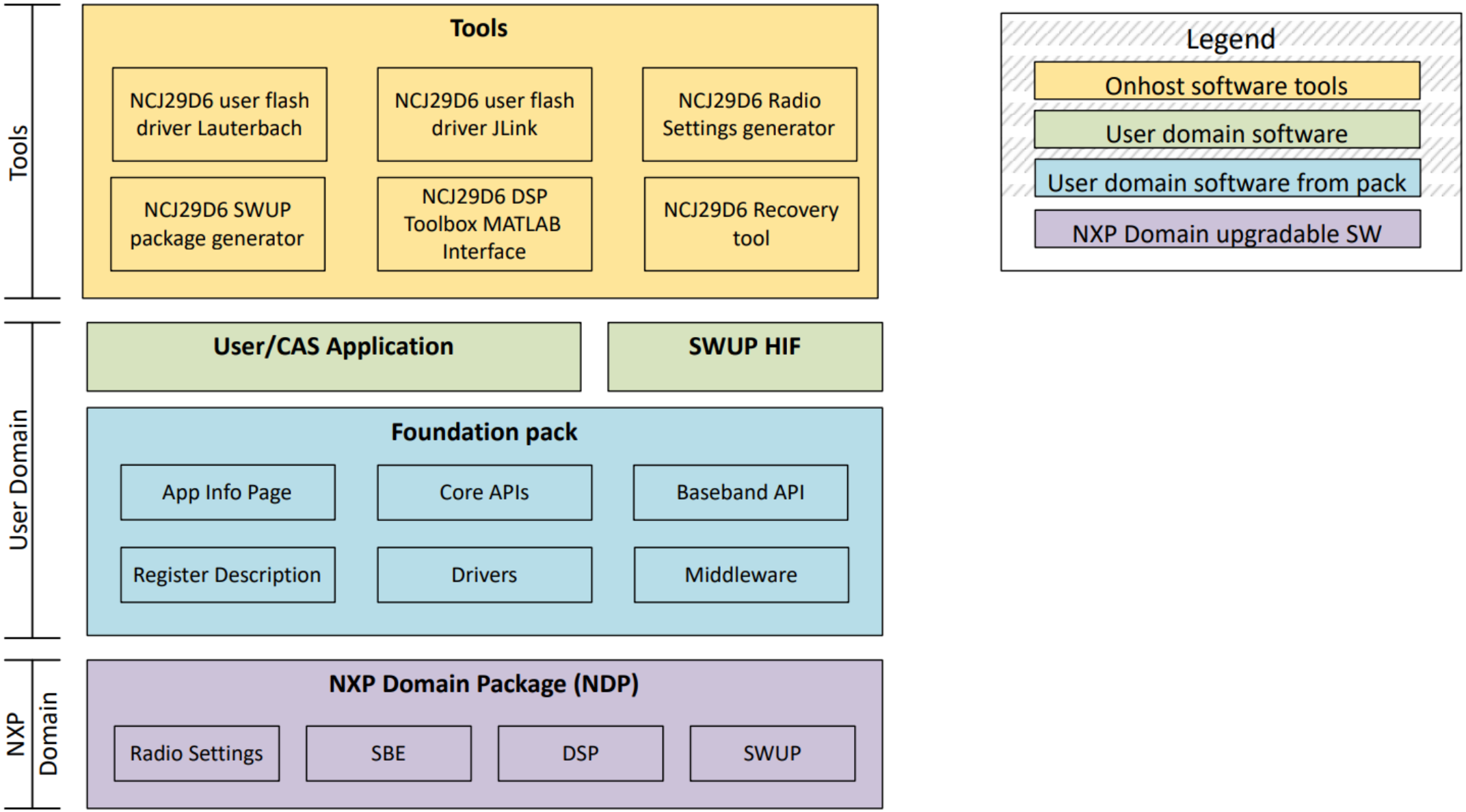


Figure 12. Overview of software architecture

NXP Overview



NXP Domain

The NXP Domain is under control of NXP. It contains elementary blocks to run the device like the following elements:

- **DSP Image:** Holding the DSP image for the baseband subsystem. This image is loaded during run time to the DSP memory.
- **Radio settings:** The radio settings are needed by the baseband subsystem to know how the UWB frame shall look like.
- **Secure Boot Extensions (SBE):** Functionality to control critical hardware blocks.
- **Software update program (SWUP):** The Software Update Program to update the device.

The customer can update the elements in the NXP domain via SWUP packages provided by NXP.

Radio settings

Slot	Scenario	Description
RX		
0	Rx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Rx_Sc406	High Performance (min phase 1 GHz)
2	Rx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Rx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Rx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Rx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Rx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Rx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Rx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Rx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)
TX		
0	Tx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Tx_Sc406	High Performance (min phase 1 GHz)
2	Tx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Tx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Tx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Tx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Tx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Tx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Tx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Tx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)

User Domain

The user domain is under the control of the customer. It's the customers responsibility that this domain holds the needed elements. The user domain can be split into the following elements:

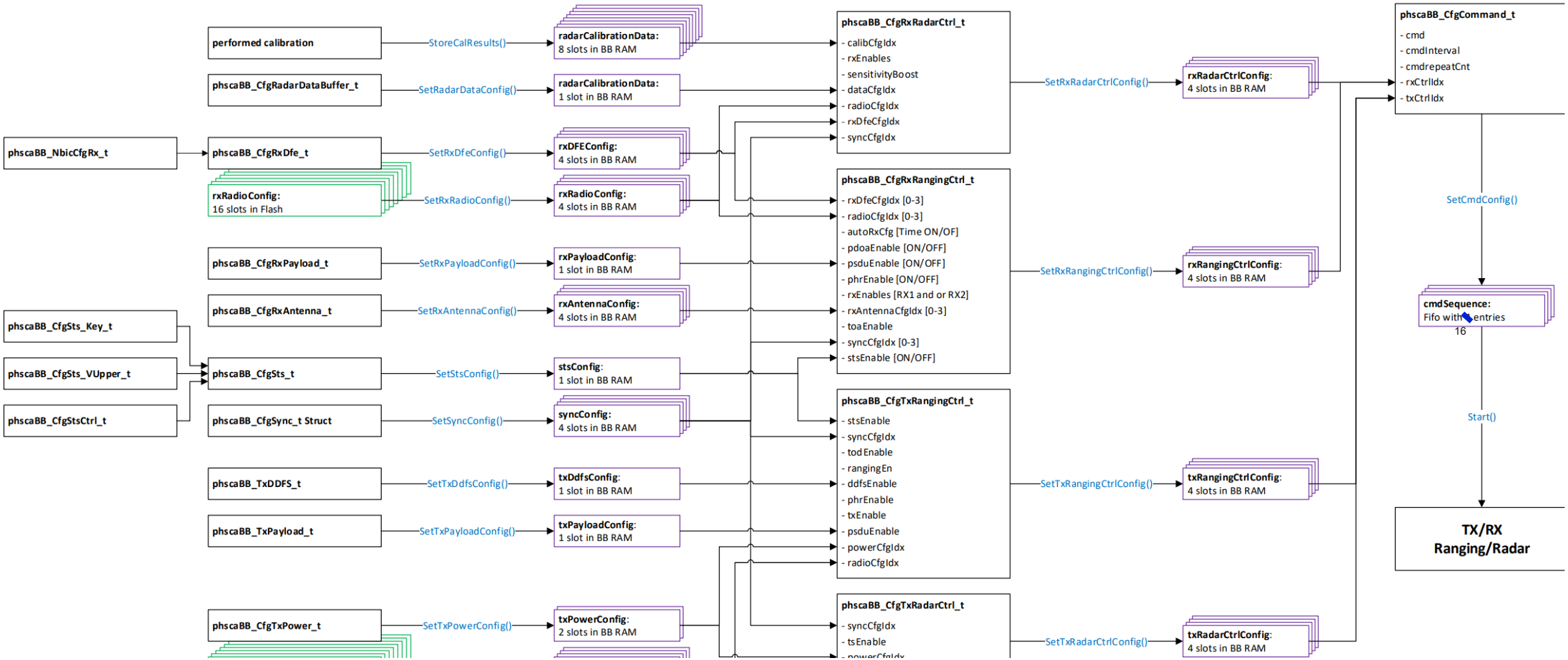
- **Foundation pack:** The foundation pack is Keil µVision IDE software development pack for NCJ29D6, It holds all elements which are required to run an application on the NCJ29D6. This package is mandatory and must be used by the customer. If the customer doesn't want to use foundation pack components drivers and middleware they are free to write their own drivers.
- **User flash drivers:**
- **User application:** The customer must provide an application, running the NCJ29D6.
- **SWUP custom host Interface:** The SWUP custom host interfaces handles the communication between an interface (e.g. SPI 6 wire/ CAN) and the SWUP. This interface application is a separate application next to the user application and is started by the SWUP.

Tools

The NCJ29D6 support tools which run on Windows environment or on user domain area of the device NCJ29D6:

- **NCJ29D6 MATLAB® DSP Toolbox:** The DSP toolbox MATLAB® interface component provides interfaces to DSP toolbox configuration structures and functions in MATLAB®.
- **NCJ29D6 Radio Settings Generator:** The radio settings are needed by the baseband subsystem to know how the UWB frame shall look like.
- **NCJ29D6 Software Update Package Generator:** The Software Update Package Generator allows its user to create packages that can be applied to NCJ29D6 device using the Software Update Program (SWUP).
- **NCJ29D6 SWUP Recovery Tools:** The SWUP Recovery tools contain the scripts and environment to restore the device from recovery mode. It also includes default SWUP Host Interface binary.
- **NCJ29D6 User Flash Driver Lauterbach:** The NCJ29D6 user flash driver Lauterbach permits loading binary data into the user flash area of the NCJ29D6 using the Lauterbach debug probe.
- **NCJ29D6 User Flash Driver SEGGER J-Link:** The NCJ29D6 user flash driver SEGGER JLink permits loading any data into the user flash area of the NCJ29D6 using the SEGGER J-Link debug probe.

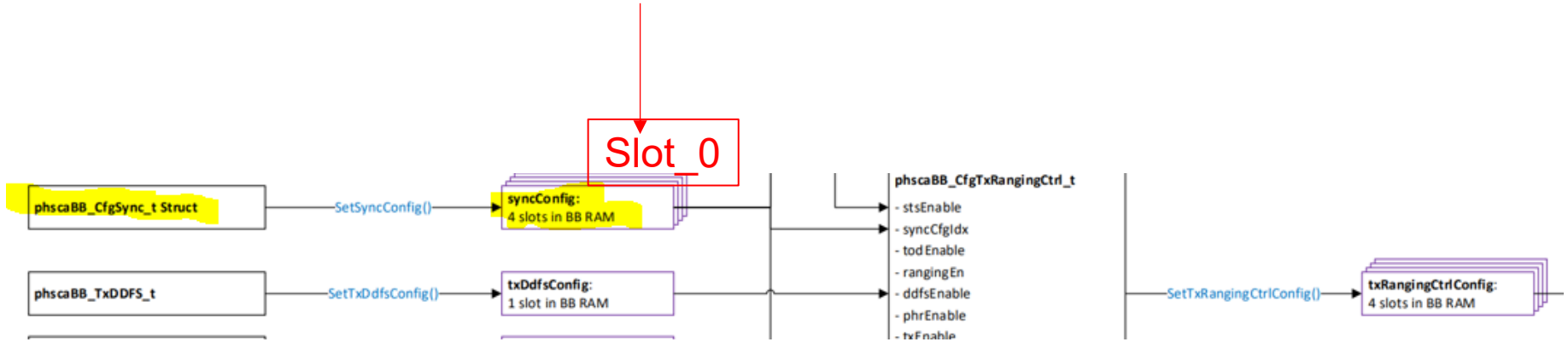
Configuration Concept

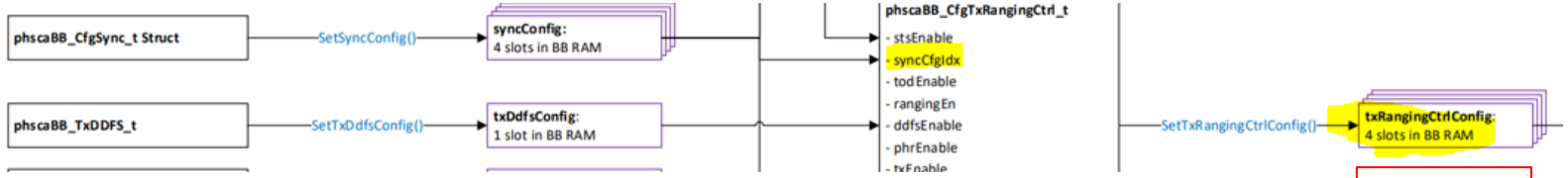


Example Configuration Concept

Build Output | UwbMwConf_Control.c | phscaBB.h

```
341         case EN_RANGING_BB_CFG:
342     #ifndef UNDEF_PULSE_SHAPE
343             /* Get Device Ranging Configuration */
344             /* Misra: 2.2 Qac: 2983 Cmt:This variable is only used to stock the return of the function.Intended , no impact on software*/
345             lReturnValue_t = MemMan_Uwb_GetDataUwbDeviceRangingCfg(aSessionIdShort_u8, &lUwbDeviceRangingCfg_pst);
346             lBbReturnValue_en = (((PHSCA_API_SUCCESS == lBbReturnValue_en) && (lUwbDeviceRangingCfg_pst != NULL)) ? UwbMwConf_LoadPulseShape(lUwbDeviceRangin
347
348             if((void *)NULL != lUwbDeviceRangingCfg_pst)
349             {
350                 /* Configure the Sync setting slots in Baseband */
351                 lSyncCodeCfg_st.syncId = lUwbDeviceRangingCfg_pst->SyncCodeIndex_u8;
352                 lBbReturnValue_en = ((PHSCA_API_SUCCESS == lBbReturnValue_en) ? phscaBB_SetSyncConfig(0u, &lSyncCodeCfg_st) : lBbReturnValue_en);
353
354                 if(TRUE == lUwbDeviceRangingCfg_pst->IsIDGRanging_bln)
355                 {
356                     MemManNvData_p_t_FBD6_IDG_UWB_Poll_Timeout_BeforeSync_Get(&lTimeoutBeforeSyncMilliSec_u8);
```

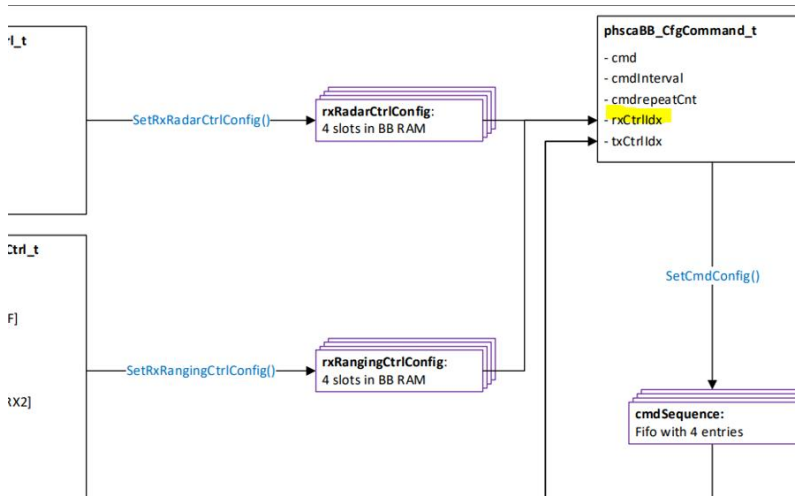




Slot_2

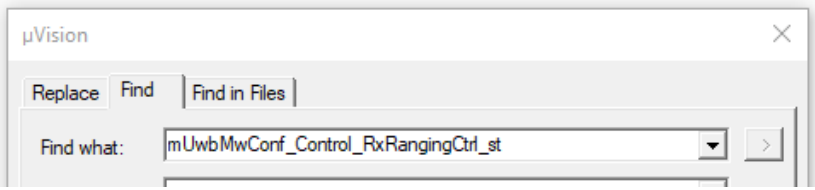
```
phscaBB_CfgRxRangingCtrl_t mUwbMwConf_Control_RxRangingCtrl_st[7] = {
…..
 [3] =
  {
    .radioCfgIdx          = PHSCAUWB_RX_SC100,
    .syncCfgIdx           = 0x2u,   /* W_Task 1743907, use CfgIdx 0 */
    .rxDfeCfgIdx          = PHSCA_UWB_FETCH_FROM_SETTING,              // Fetch from radio setting
    .rxAntennaCfgIdx      = PHSCA_UWB_FETCH_FROM_SETTING,              // Fetch from radio setting
    .sfdCfgIdx            = 0x0u,   /* W_Task 1743907, use CfgIdx 0 */
    .stsEnable            = 0x00u,
    .phrEnable            = 0x01u,
    .psduEnable           = 0x01u,
    .toaEnable            = 0x01u,
    .pdoaEnable           = PHSCA_UWB_DISABLE,
    .rxSyncDisable        = 0x00u,
    .autoRxOffTimeCfg     = 0x00u,              // No automatic restart
    .autoRxOnTimeCfg      = (2000u*1280u)/1248u,   // ~200us timeout
    .rxEnables            = PHSCAUWB_RX_POS_3     // Enable RX 1 and Rx 2
  },
```

```
/* Configure the Rx Ranging control slots in Baseband */
phscaBB_SetRxRangingCtrlConfig(2u, &mUwbMwConf_Control_RxRangingCtrl_st[2]) : lBbReturnValue_en);
```

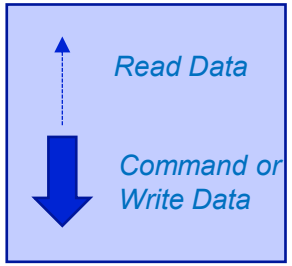



```
case EN_SEQ_RESPONDER_CCC_RANGING:
    if(lDutyCycleReturn != E_NOT_OK)
    {
      #ifndef UNDEF_PULSE_SHAPE
          lReturnValue_t = ((E_OK == lReturnValue_t) ? UwbMwConf_ConfigureBaseBand(aDeviceConfigId_u8, EN_COEXISTENCE_BB_CFG): lReturnValue_t);
    #endif
  /* Set the frequency params */
  /* Misra: 2.2 Qac: 2982 Cmt:Variable is only used to stock the return of the function.No impact on software*/
  /* Misra: 4.1 Qac: 2812 Cmt:Error acknlowdged.No impact on software*/
  lFreqId_u8 = UwbMwConf_SetFrequency(lUwbDeviceRangingCfg_pst->FrequencyChannelIndex_u8, EN_SEQ_RESPONDER_CCC_RANGING);
  lTxFrameCounter_u8++;
  /* Configuration for Rx Pre-Poll message */
   lFifo_st[0].cmd =  PHSCA_BB_CMD_RANGING_RX;                          /* Command to be executed */
   lFifo_st[0].rxCtrlIdx    = 0x02u;                                   /* Index to select Rx control configuration */
   lFifo_st[0].txCtrlIdx    = 0x00u;                                   /* Index to select Tx control configuration */
   lFifo_st[0].cmdRepeatCnt = 0x01u;                                   /* Number of repetitions of the commands */
   lFifo_st[0].cmdRepeatInterval = 0u; /* Interval between commands  */
   /* Misra: 2.2 Qac: 2985 Cmt:Warning acknlowdged.C_UWB_MSH_RX_WINDOW_OFFSET may change value*/
   lFifo_st[0].cmdSequenceInterval = (uint32_t)(((uint32_t)lSlotLength_u32-C_UWB_MSH_RX_WINDOW_OFFSET)*1248u)/10u;                /* Interval between sequences */
   lFifo_st[0].freqCtrlIdx = lFreqId_u8;

  /* Configuration for Rx Poll message */
   lFifo_st[1].cmd =  PHSCA_BB_CMD_RANGING_RX;
   lFifo_st[1].rxCtrlIdx    = 0x01u;
   lFifo_st[1].txCtrlIdx    = 0x00u;
   lFifo_st[1].cmdRepeatCnt = 0x01u;
   lFifo_st[1].cmdRepeatInterval =0u;
```



Message & Sequence Handling (MSH Layer)



App Man

Responder SMA

CPD

Initiator Box2Box

Testmodes (Diagnostic)

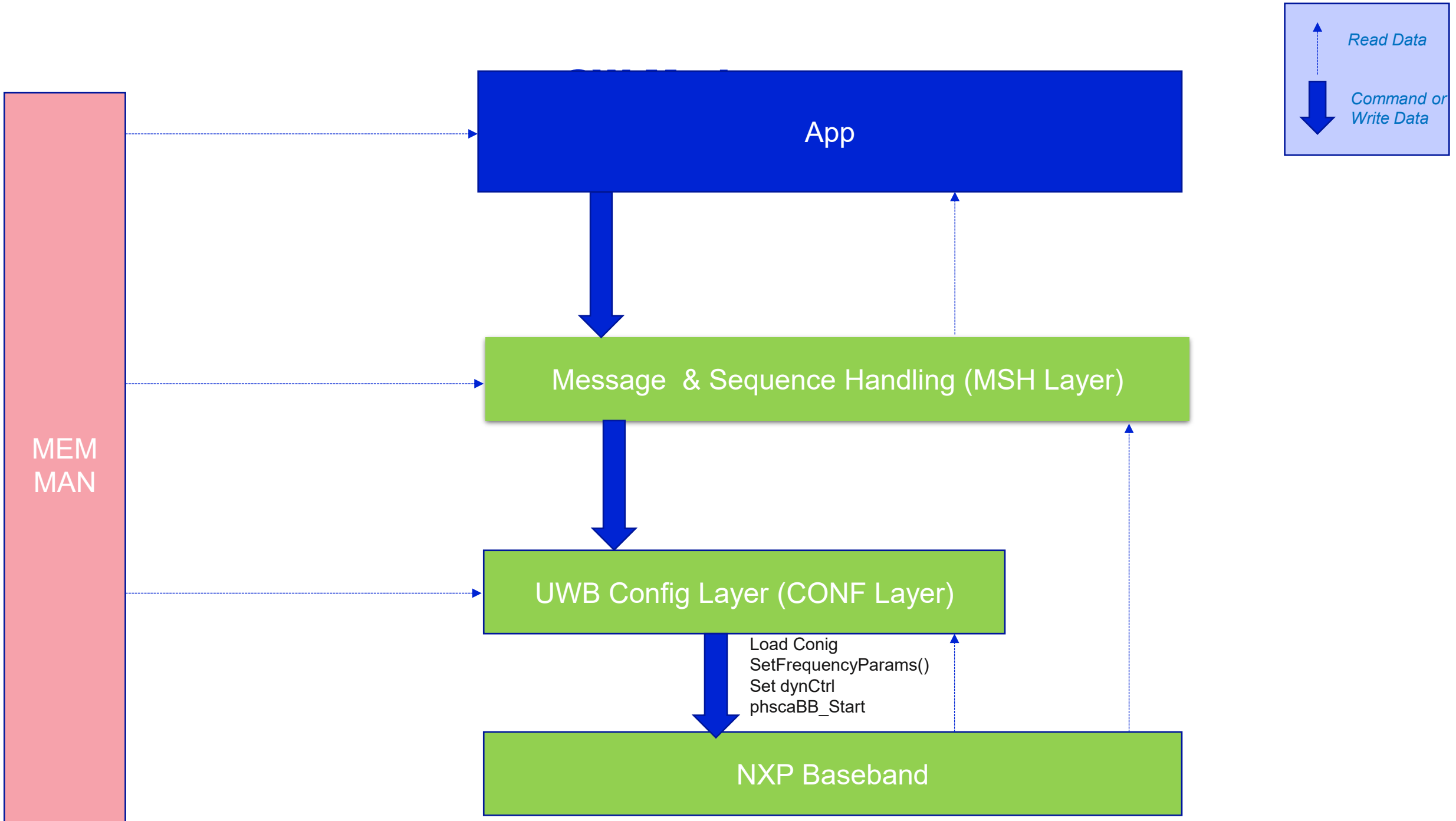
MshStart(Device Config ID, Sequence_ID, Start Time in µs)

ezogen auf die aktuelle Reference-Zeitstempel)

MshStart(Device Config ID, Sequence_ID, Start Time in µs))

MshStart(Device Config ID, Sequence_ID, Start Ti
MshReadVersion(tbd)
MshXtalPin(ON/OFF)

Message & Sequence Handling (MSH Layer)
(e.g. Timing CPD&Ranging)


Read Data
Command or
Write Data
App
MEM
MAN
Message & Sequence Handling (MSH Layer)
UWB Config Layer (CONF Layer)
Load Conig
SetFrequencyParams()
Set dynCtrl
phscaBB_Start
NXP Baseband

UwbConfiguration

```
Build Output | Build Output | Build Output | SAB_Main.c

    if (SYSSTATEM_Mode_LowLevel == aMode_en)
    {
      /* Misra: 10.3 Qac: 1297 Cmt: Type of this variable is considered with no risk.*/
    /*      (void)UwbMwConf_Init(3u);*/ /* Select the XO as clock source of 55.2Mhz for Baseband operation */
    }
    else if (SYSSTATEM_Mode_Normal == aMode_en)
    {
      CanManMain_Init();
      MemManMain_Init();
      ModeManMain_Init();
      CfgManMain_Init();
      DiagManMain_Init();
      BleHdlMain_Init();
      UwbApp_Cpd_Init();
      UwbApp_Responder_Init();
      TSyncMain_Init();
      AppMan_Init();
      CryptoMain_Init();
      AppManFBDOnlyMode_Init();
      /* Misra: 10.3 Qac: 1297 Cmt: Type of this variable is considered with no risk.*/
      (void)UwbMwConf_Init(3u); /* Select the XO as clock source of 55.2Mhz for Baseband operation */
    }
    else if (SYSSTATEM_Mode_LowPowerStop == aMode_en)
    {
      /*  */
    }
    else if (SYSSTATEM_Mode_LowPowerRun == aMode_en)
```

```
2339 Std_ReturnType UwbMwConf_Init(phscaBB_XoClock_Selection_t aXoClockSel_st)
2340 {
2341     /*#[ operation UwbMwConf_Init(phscaBB_XoClock_Selection_t) */
2342
2343     Std_ReturnType lErrorFlag_t = E_OK;
2344     Std_ReturnType lReturnValue_t;
2345
2346     lReturnValue_t = UwbMwConf_PowerUpLoadDsp(aXoClockSel_st);
2347
2348     lReturnValue_t |= UwbMwConf_ConfigureBaseBand(0u, EN_DEFAULT_BB_CFG);
2349
2350     /*commended to avoid problems with cpd coexistence*/
2351     /*lReturnValue_t |= UwbMwConf_ConfigureTempMeasurement();*/
2352
2353     lReturnValue_t |= UwbMwConf_ReadTemperature();
2354
2355     if (E_OK != lReturnValue_t)
2356     {
2357       lErrorFlag_t = E_NOT_OK;
2358     }
2359
2360     return lErrorFlag_t;
2361     /*#]*/
2362 }
```

```
static Std_ReturnType UwbMwConf_PowerUpLoadDsp(const phscaBB_XoClock_Selection_t aXoClockSel_st)
{
    /*#[ operation UwbMwConf_PowerUpLoadDsp(phscaBB_XoClock_Selection_t) */
    Std_ReturnType lErrorFlag_u8 = E_OK;
    phscaApi_Status_t lBbReturnValue_en = PHSCA_API_SUCCESS;
    phscaApi_Status_t lApiReturnValue_en;

     /* Enable memory blocks */
    /* Misra: 1.1 Qac: 0604 Cmt:  No impact, current project use c99 .*/
    phscaApi_Dsp_MemCtrl_t dspMemCtrl =
    {
        /* Misra: 1.1 Qac: 1053 Cmt:  C99 is available in FBD6 project. Therefore Designators are allowed for better readability. */
        /* Misra: 10.3 Qac: 1297 Cmt: Type of this variable is considered with no risk.*/
        .rangingMemCtrl = PHSCAAPI_DSP_MEM_ACTIVE, //0x02u,
        /* Misra: 10.3 Qac: 1297 Cmt: Type of this variable is considered with no risk.*/
        .radarMemCtrl   = PHSCAAPI_DSP_MEM_ACTIVE, //0x02u,
    };

    lApiReturnValue_en = phscaApi_Dsp_MemStateCtrl(&dspMemCtrl);

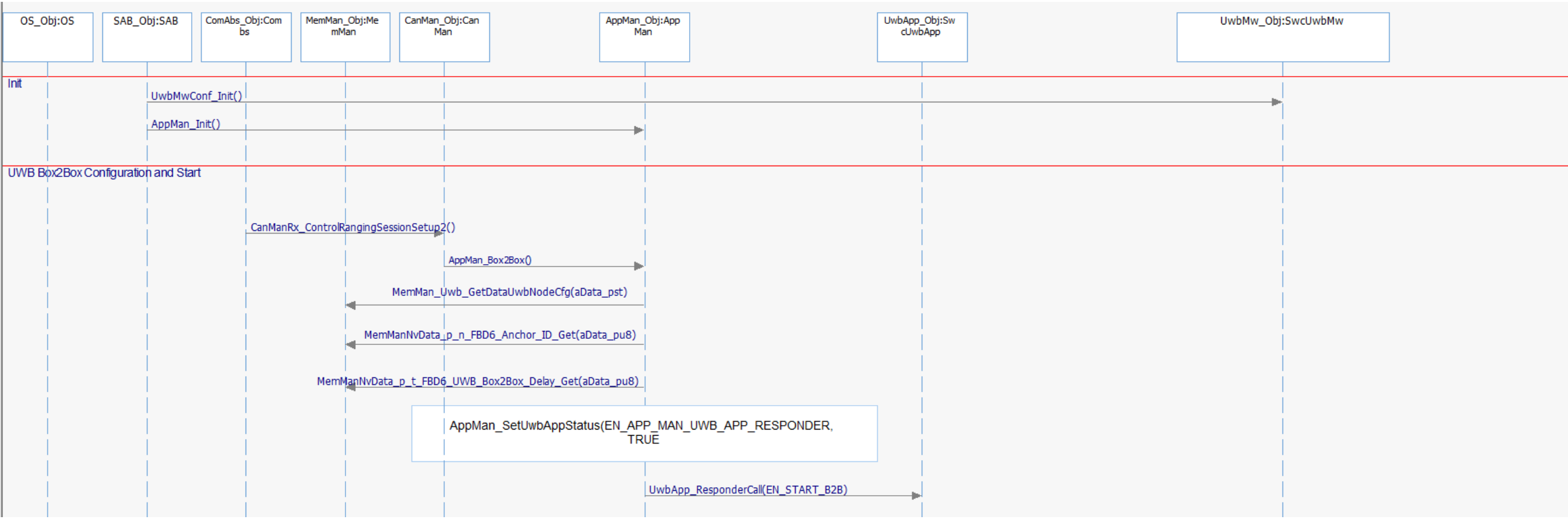
    /* Load P/X/Y memory with CoolFlux Ranging binaries. */
    if (PHSCA_API_SUCCESS == lApiReturnValue_en)
    {
        lApiReturnValue_en = phscaApi_Dsp_LoadImage(PHSCADSP_BASEBAND_IMAGE);
    }

    if(PHSCA_API_SUCCESS != lApiReturnValue_en)
    {
        lErrorFlag_u8 = E_NOT_OK;
    }

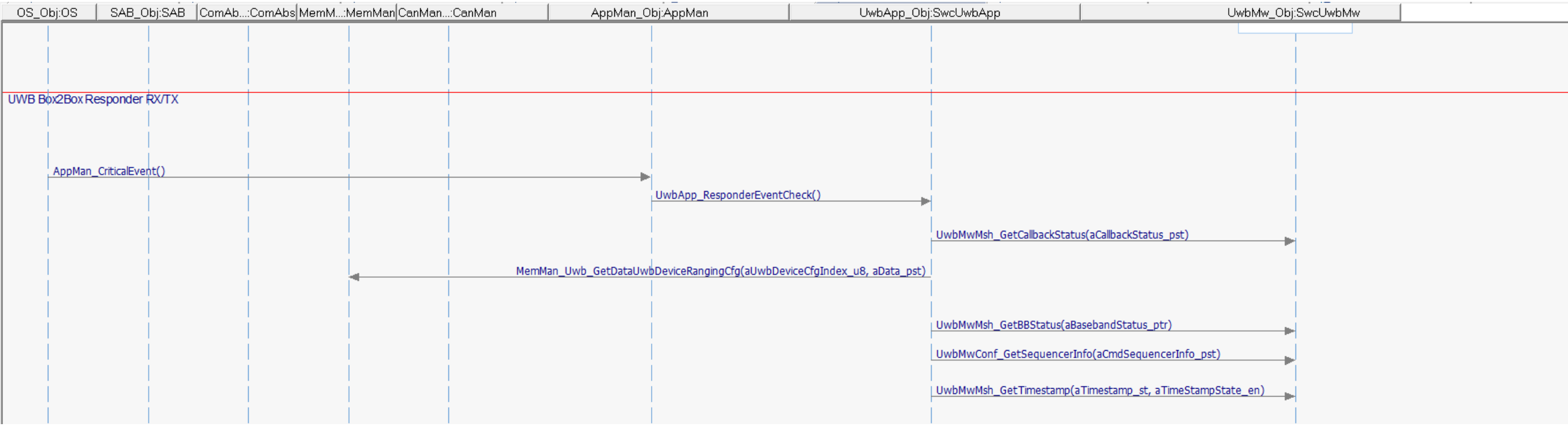
    /* initialize the Baseband library. */
    if (PHSCA_API_SUCCESS == lApiReturnValue_en)
    {
        lBbReturnValue_en = phscaBB_Init(PHSCA_BB_XO_CLOCK_INPUT_SOURCE_55P2Mhz);//PHSCA_BB_XO_CLOCK_INPUT_SOURCE_55P2Mhz);
    }

    /* Powerup the BB powerdomain. */
    if (PHSCA_API_SUCCESS == lApiReturnValue_en)
    {
        lApiReturnValue_en = phscaApi_Dsp_BasebandUp();
    }
```

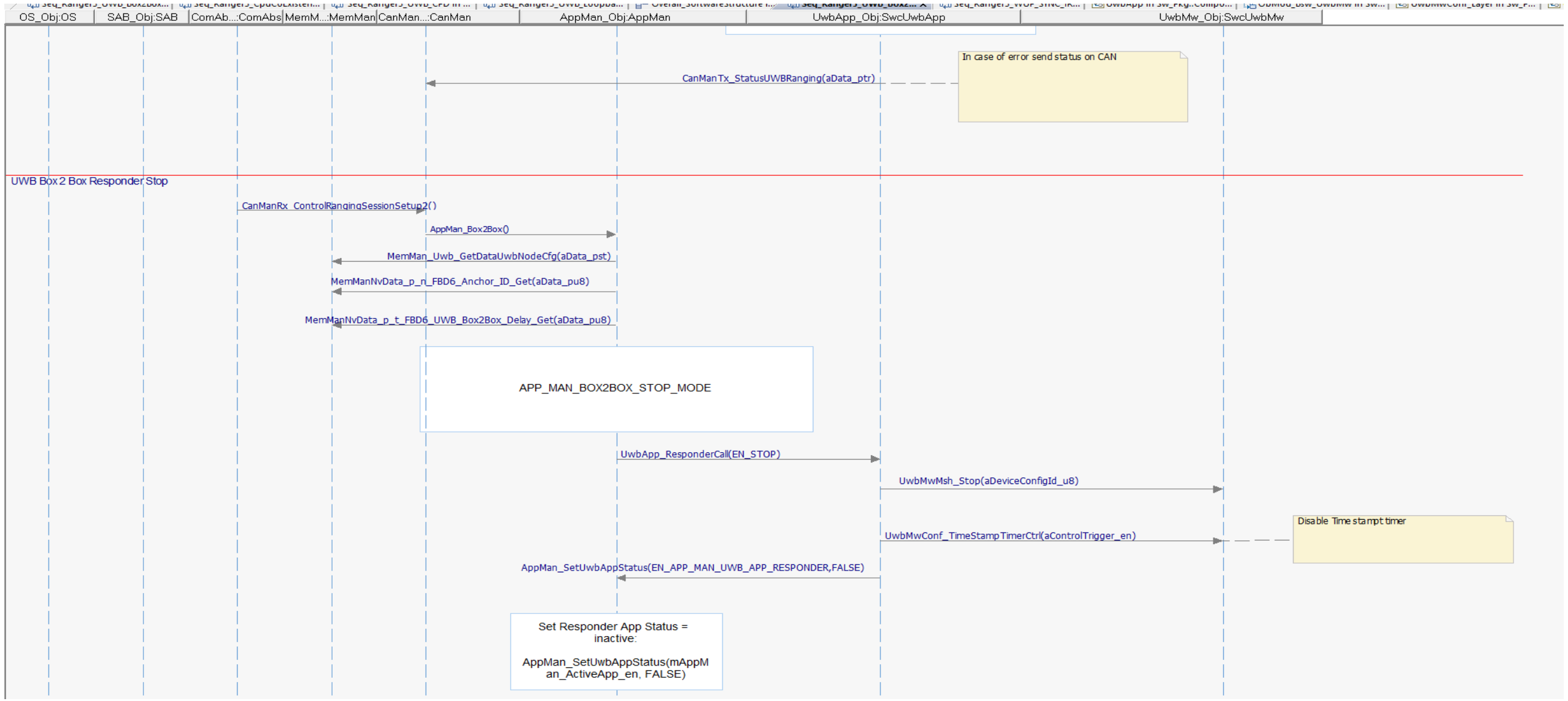
AppMan – Responder Start



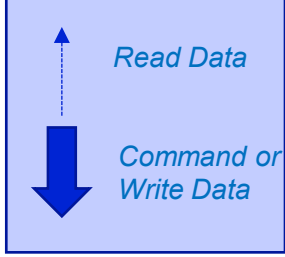
AppMan – Critical Event



App Man - Stop


OS_Obj:OS
SAB_Obj:SAB
ComAb...:ComAbs
MemM...:MemMan
CanMan...:CanMan
AppMan_Obj:AppMan
UwbApp_Obj:SwcUwbApp
UwbMw_Obj:SwcUwbMw
In case of error send status on CAN
CanManTx_StatusUWBRanging(aData_ptr)
UWB Box 2 Box Responder Stop
CanManRx_ControlRangingSessionSetup2()
AppMan_Box2Box()
MemMan_Uwb_GetDataUwbNodeCfg(aData_pst)
MemManNvData_p_n_FBD6_Anchor_ID_Get(aData_pu8)
MemManNvData_p_t_FBD6_UWB_Box2Box_Delay_Get(aData_pu8)
APP_MAN_BOX2BOX_STOP_MODE
UwbApp_ResponderCall(EN_STOP)
UwbMwMsh_Stop(aDeviceConfigId_u8)
Disable Time stampt timer
UwbMwConf_TimeStampTimerCtrl(aControlTrigger_en)
AppMan_SetUwbAppStatus(EN_APP_MAN_UWB_APP_RESPONDER,FALSE)
Set Responder App Status = inactive:
AppMan_SetUwbAppStatus(mAppMan_ActiveApp_en, FALSE)

UWB Callbacks




```
1104  void UwbMwConf_ActiveCallback(void)
1105  {
1106      /*#[ operation UwbMwConf_ActiveCallback() */
1107          /*disabled due to causing issues to cpd coexistence*/
1108          /*mCallbackStatusFlags_st.BB_CB_ACTIVE_EVENT = TRUE;*/
1109          /*OS_Service_EnterCriticalEventReq(OS_EVENT_CRITICAL_UWB, true);*/
1110      /*#]*/
1111  }
1112
1113  /*#[ ignore */
1114  /**
1115   * Operation for handling the callback category Idle.
1116   *
1117   * \return void
1118   */
1119  /*#]*/
1120  /*## operation UwbMwConf_IdleCallback() */
1121  void UwbMwConf_IdleCallback(void)
1122  {
1123      /*#[ operation UwbMwConf_IdleCallback() */
1124      mCallbackStatusFlags_st.BB_CB_IDLE_EVENT = TRUE;
1125      OS_Service_EnterCriticalEventReq(OS_EVENT_CRITICAL_UWB, true);
1126      /*#]*/
1127  }
1128
1129  /*#[ ignore */
1130  /**
1131   * Operation for handling the callback category Ready.
1132   *
1133   * \return void
1134   */
1135  /*#]*/
1136  /*## operation UwbMwConf_ReadyCallback() */
1137  void UwbMwConf_ReadyCallback(void)
1138  {
1139      /*#[ operation UwbMwConf_ReadyCallback() */
1140      mCallbackStatusFlags_st.BB_CB_READY_EVENT = TRUE;
1141      OS_Service_EnterCriticalEventReq(OS_EVENT_CRITICAL_UWB, true);
1142      /*#]*/
1143  }
1144
```

RTOS

```
void AppMan_CriticalEvent(void)
{
    /*#[ operation AppMan_CriticalEvent() */
    Std_ReturnType lReturnValue_t = E_OK;
    CallbackEvents_t lCallbackStatus_pst;
    switch(mAppMan_ActiveApp_en)
    {
      case EN_APP_MAN_UWB_APP_TESTMODES:
          /* call baseband callback function of testmode app */
          UwbAppTstMd_BasebandCallback();
          break;

      case EN_APP_MAN_UWB_APP_INITIATOR:
          /* call baseband callback function of initiator app */
          (void)UwbApp_InitiatorEventCheck();
          break;

      case EN_APP_MAN_UWB_APP_RESPONDER:
      case EN_APP_MAN_UWB_APP_CPD:
      case EN_APP_MAN_UWB_CO_EXISTENCY:
      /* call baseband callback function of responder and CPD app (coexistency) */

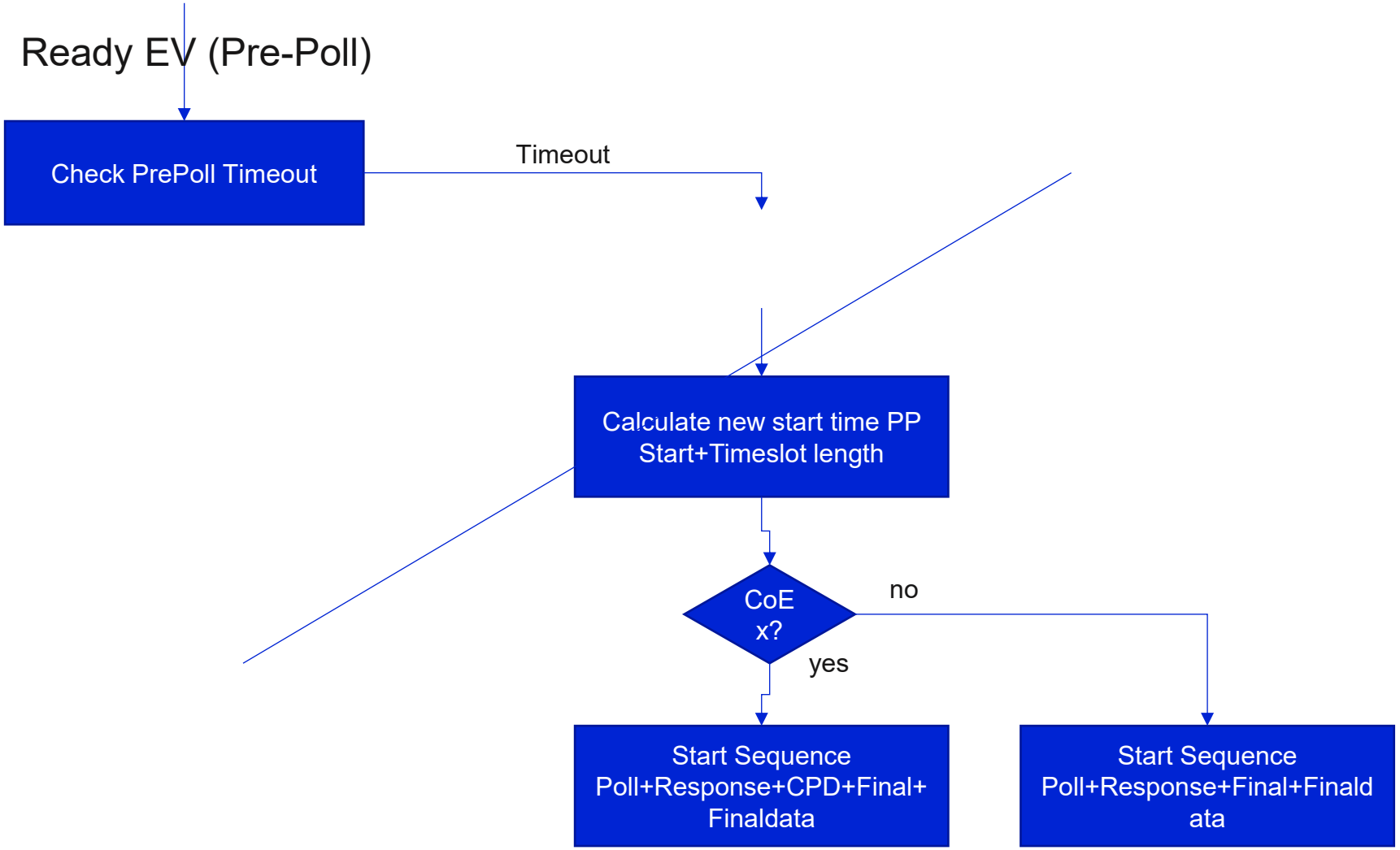
          lReturnValue_t = UwbApp_ResponderEventCheck();
          break;

      case EN_APP_MAN_UWB_APP_LOOPBACK:
                (void)UwbApp_LoopbackEventCheck();
                break;
      case EN_APP_MAN_UWB_APP_NONE:
      default:
           /* Delete obsolete Ready/Idle-Flags */
           UwbMwMsh_ClearCallbackStatus();

          break;
    }

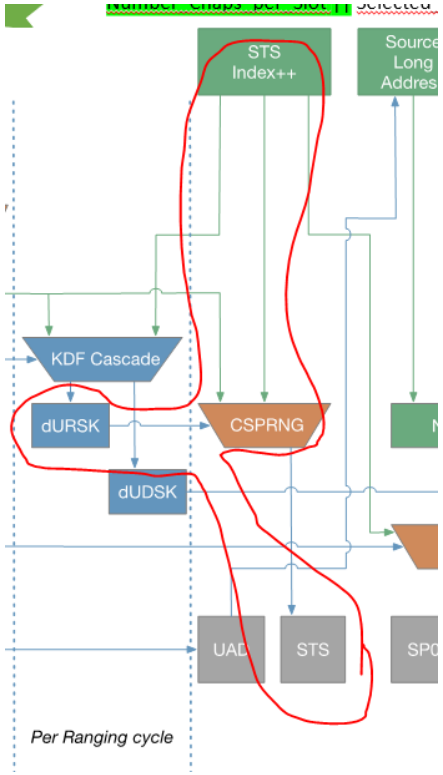
    /* Check return values - only start restart timer if ResponderState is NOT in STOP
```

Ranging without PrePoll

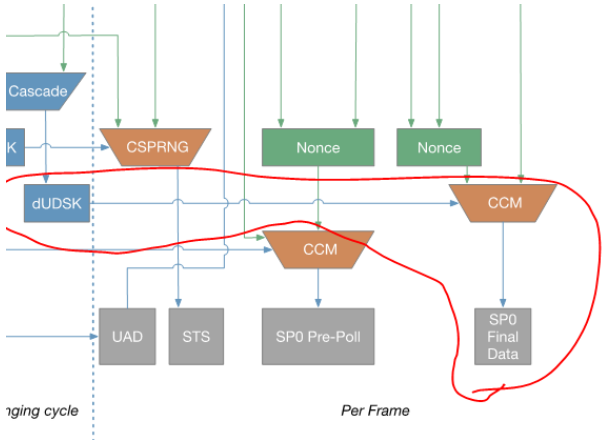


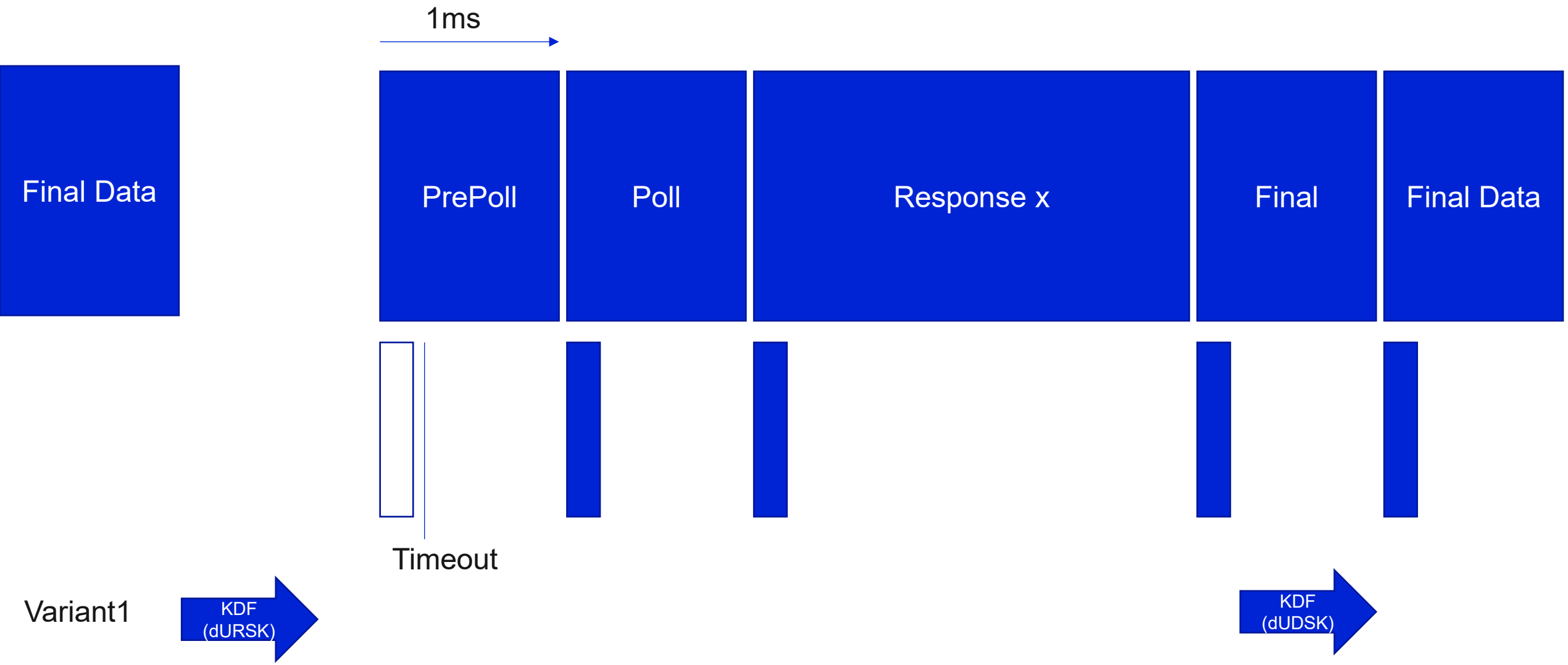
Calculation dURSK and dUDSK

Generate dURSK after Final Data



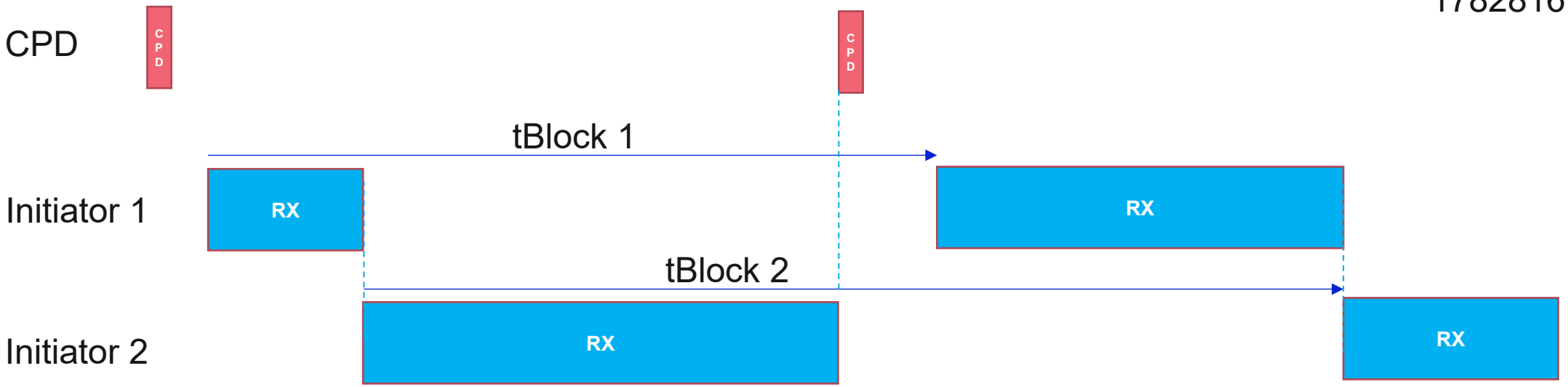
Generate dUDSK after Final message




1ms
Final Data
PrePoll
Poll
Response x
Final
Final Data
Timeout
Variant1
KDF
(dURSK)
KDF
(dUDSK)
Variant2

C0 → Time 0 is invalid (Ranging without IPB)

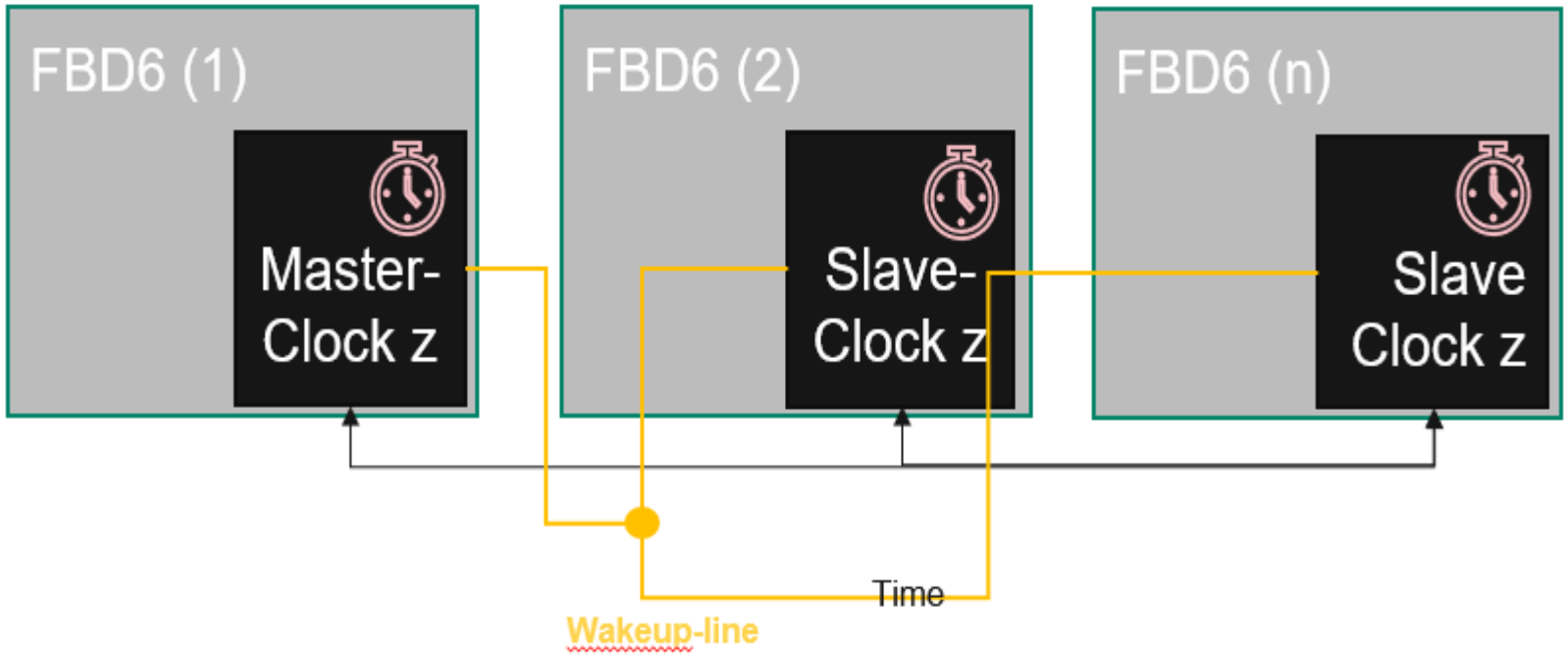
CR 1782816



Wenn Time0 gültig ist, dann ist der Ablauf identisch. Lediglich die RX Einschaltzeit wird berechnet (ist kleiner).
Bei ungültigen Time0 wird bis zur berechneten nächsten Einschaltzeit das RX Fenster geöffnet

Mit Daniel G. klären, ob es max. erlaubte RX On Zeit gibt

Time Sync & Window Calculation

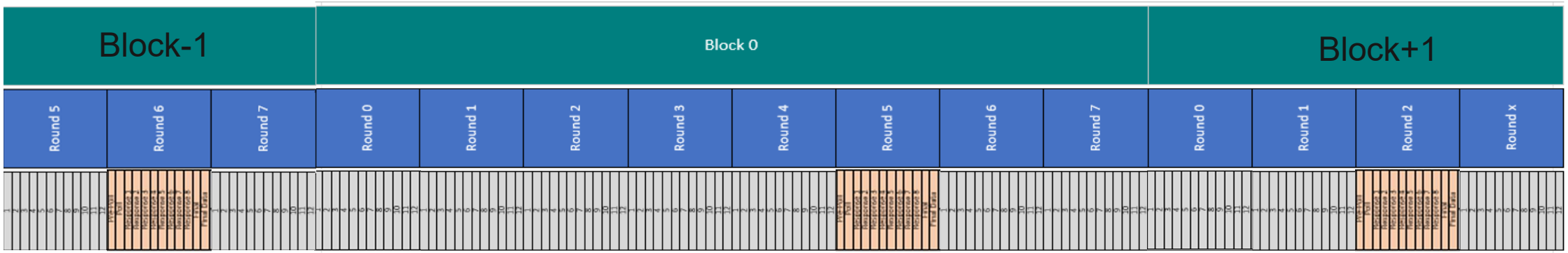


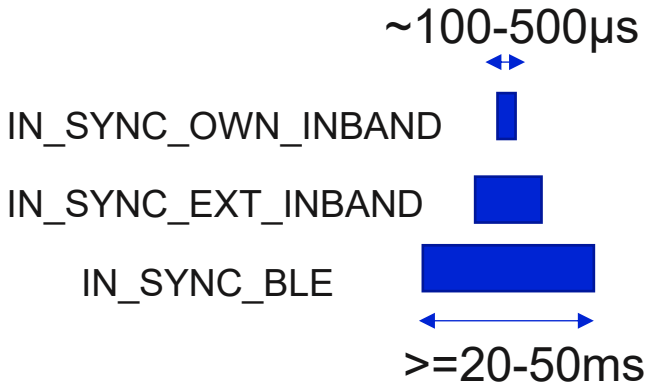
Time Sync Master:
CAN message with system time is sent
As soon as the transmitted system time is reached, WUP is activated

Time Sync Slave Slave:
Receive CAN message with system time
If WUP is activated, the transmitted system time is adopted

RX Window Calculation

Block+1

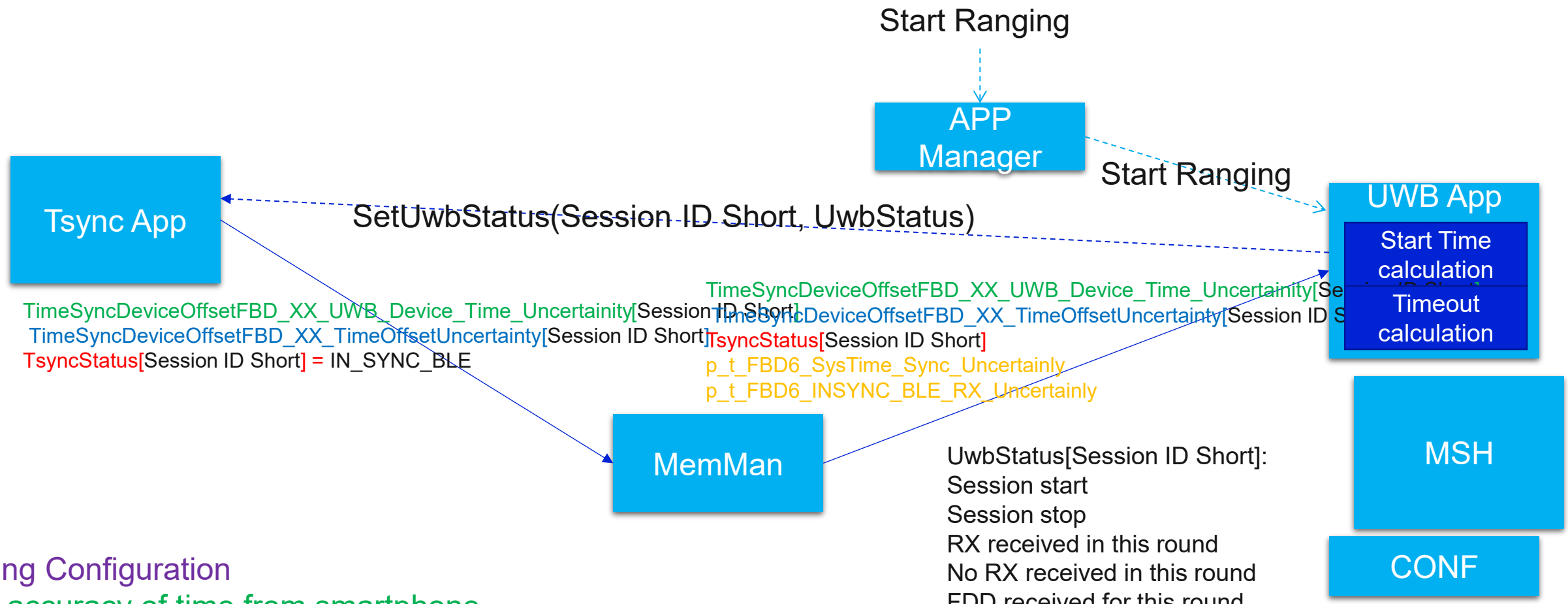




Requirement 6859
When the function "UWB-RX window management" is in state "IN_SYNC_BLE' then it shall «open the UWB- for the specific Device at:

Valid for IN_SYNC_BLE

UWB-Time0 - "TimeSyncDeviceOffsetFBD_XX_UWB_Device_Time_Uncertainity" - ("TimeSyncDeviceOffsetFBD_XX_TimeOffsetUncertainty" - "p_t_FBD6_SysTime_Sync_Uncertainly") - "p_t_FBD6_INSYNC_BLE_RX_Uncertainly"
» .



Ranging Configuration
Static accuracy of time from smartphone
Deviation from own system
NVM Parameter

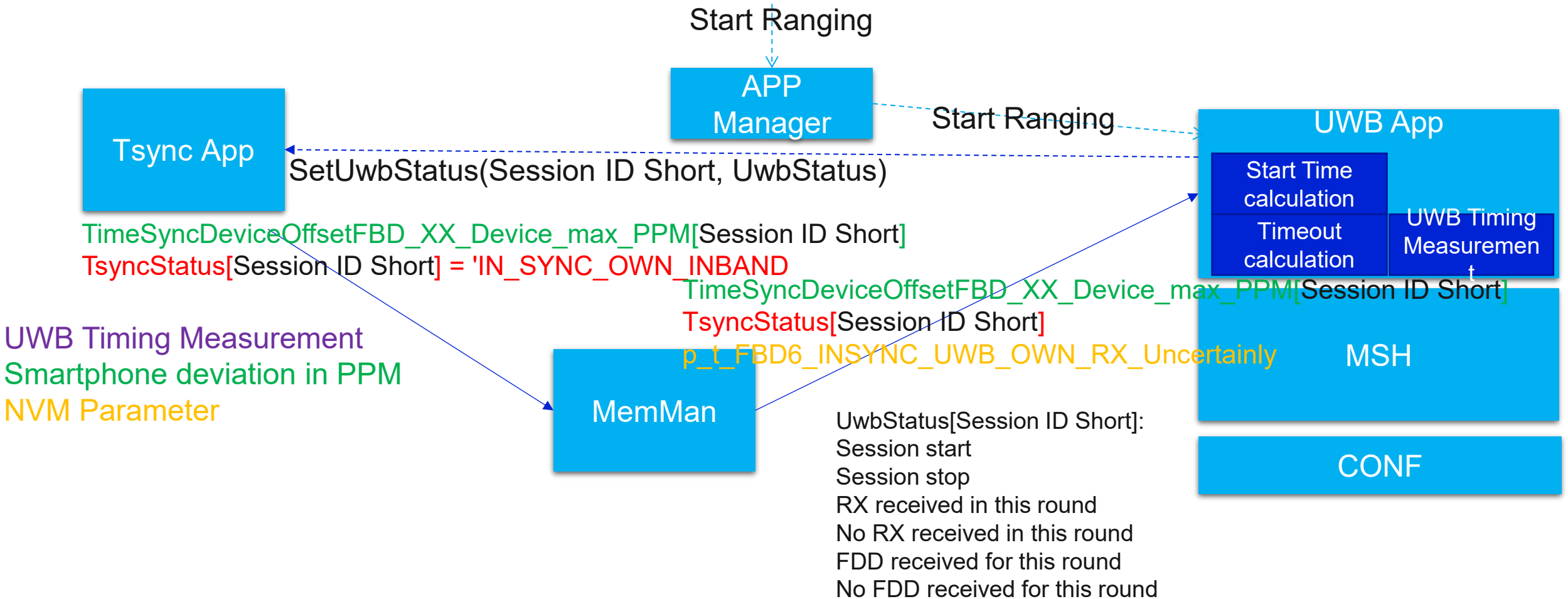
Correct UWB-Time0 →
Calculated RX = UWB-Time0+ + x*t_Block +/-t_Hopping

Requirement 6977

Valid for IN_SYNC_OWN_INBAND

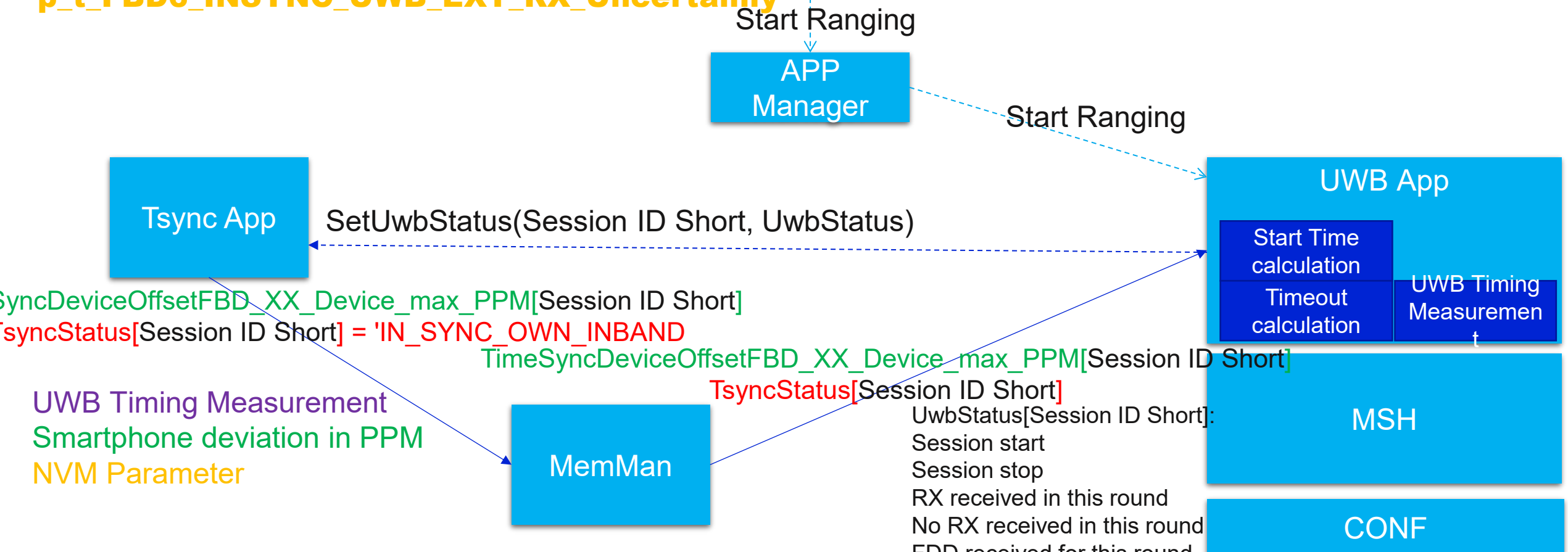
When the function "UWB-RX window management" is in state 'IN_SYNC_OWN_INBAND' then it shall «open the UWB-RX for the specific Device at:

Calculated_RX-Frame - ((Calculated RX-Frame-Timestamp - timestamp of last UWB-RX-Receive)*30ppm) - ((Calculated RX-Frame-Timestamp - timestamp of last Received UWB-RX)*"TimeSyncDeviceOffsetFBD_XX_Device_max_PPM")- "p_t_FBD6_INSYNC_UWB_OWN_RX_Uncertainly" » .



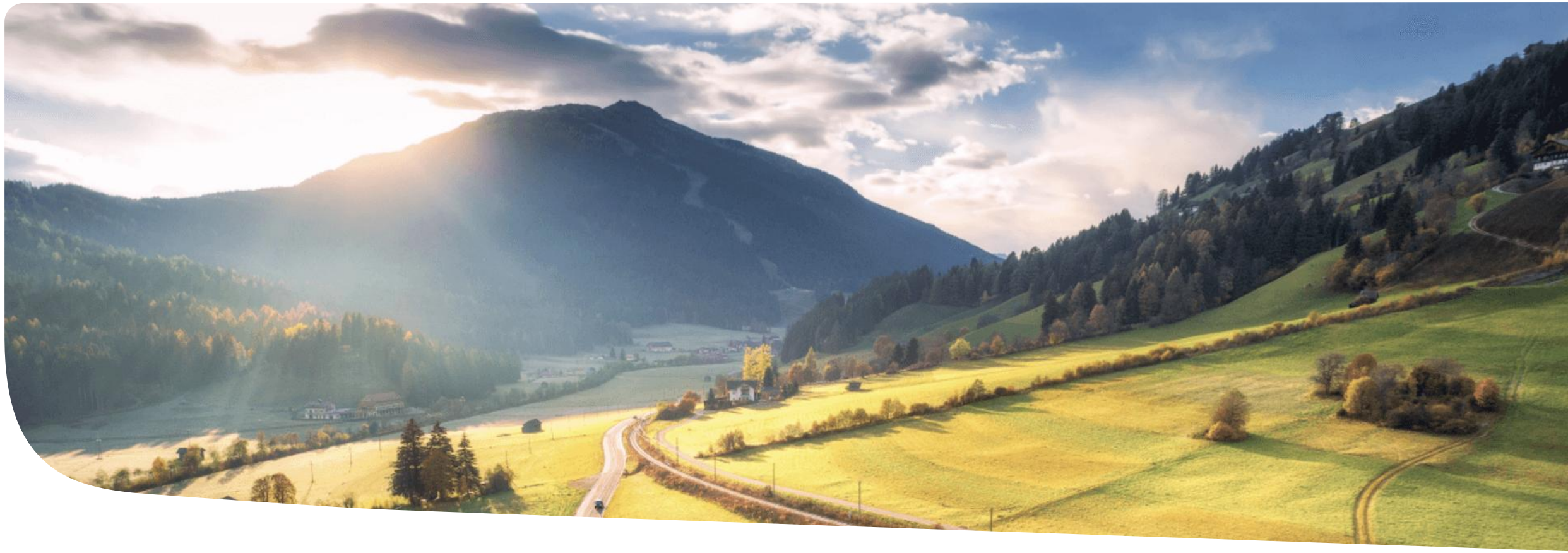
Valid for IN_SYNC_EXT_INBAND

Requirement 6978
When the function "UWB-RX window management" is in state 'IN_SYNC_EXT_INBAND' then it shall «open the UWB-RX for the specific Device at:
Calculated_RX-Frame - ((Calculated RX-Frame-Timestamp - timestamp of last Systemtimesync)*30ppm - "p_t_FBD6_SysTime_Sync_Uncertainly") - ((Calculated RX-Frame-Timestamp - timestamp of last Received UWB-RX)*"TimeSyncDeviceOffsetFBD_XX_Device_max_PPM")-"p_t_FBD6_INSYNC_UWB_EXT_RX_Uncertainly"



NXP Examples

NXP Delivery

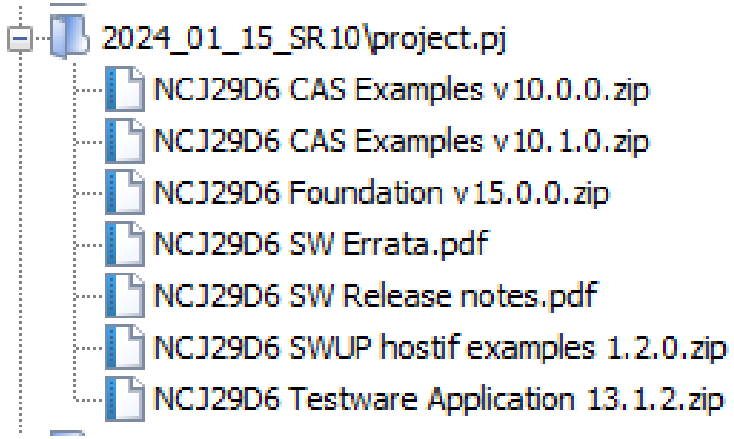


NXP Delivery

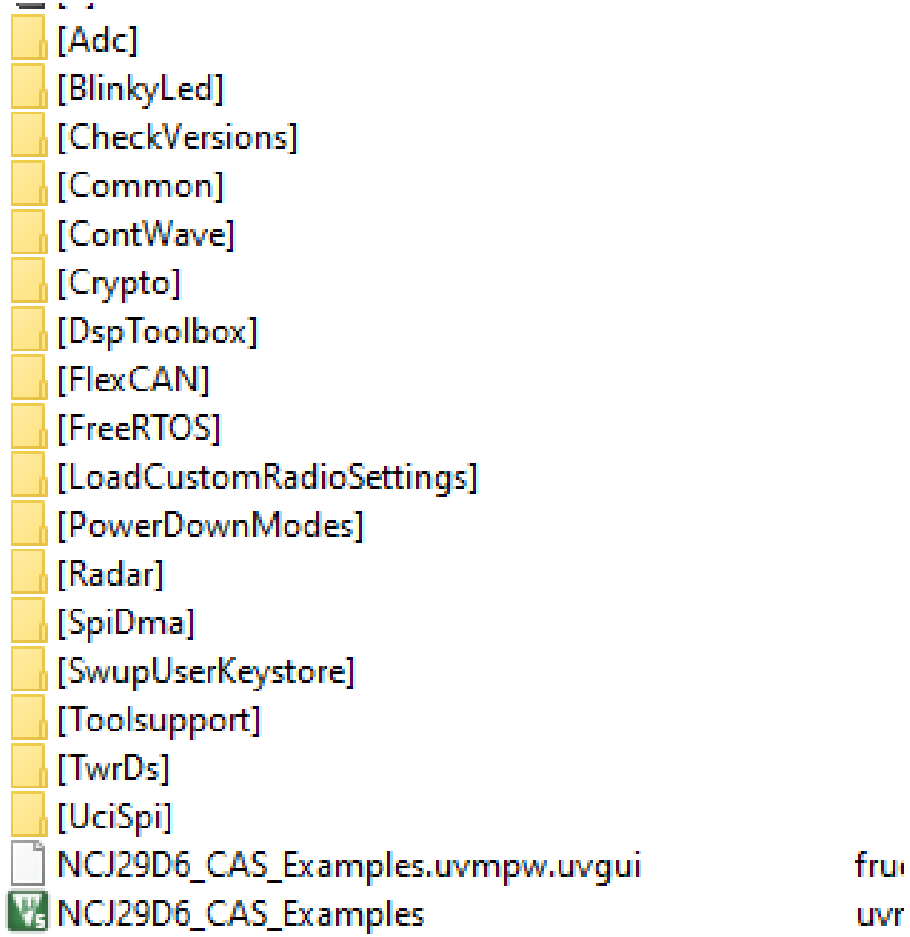
NXP Releases:

e:/Projects/CAPE/BMW/FBD6/50_FBD6/41_SW_UWB/60_Delivery/NXP/

e.g. NXP Release SR10.0:



NCJ29D6 CAS Examples



NCJ29D6 Foundation Documentation

API:

NCJ29D6 Watchdog	chm
NCJ29D6 UCI	chm
NCJ29D6 SWUP API	chm
NCJ29D6 SPI	chm
NCJ29D6 SBE API	chw
NCJ29D6 SBE API	chm
NCJ29D6 GpTimers	chm
NCJ29D6 GPIO	chm
NCJ29D6 GpADC	chm
NCJ29D6 FlexCAN	chm
NCJ29D6 DSP TOOLBOX	chm
NCJ29D6 DMA	chm
NCJ29D6 CRC	chm
NCJ29D6 ComManager	chm
NCJ29D6 COM	chm
NCJ29D6 BB API	chw
NCJ29D6 BB API	chm

User manual:

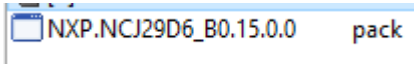
UM12008 NCJ29D6 Radio Settings Generator User Manual	pdf
UM11984 NCJ29D6 Foundation User Manual	pdf
UM11897 NCJ29D6 DSP Toolbox MATLAB Interface User Manual	pdf
UM11895 NCJ29D6 SW Update Package Generator User Manual	pdf
UM11850 NCJ29D6 User Flash Driver Lauterbach User Manual	pdf
UM11849 NCJ29D6 User Flash Driver SEGGER JLink User Manual	pdf

NCJ29D6 Foundation Packages

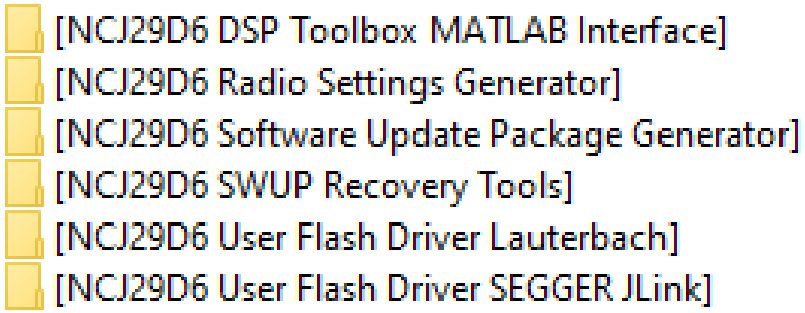
NDP SW-Packages:



Keil µVision IDE software development pack:



NCJ29D6 Foundation - Tools



Lauterbach Trace32
NXP Scripts

› Flash NXP Packages:

B::CD.PEDIT "C:\Projekte\BMW_FBD6\50_FBD6\41_SW_UWB\60_Delivery\NXP\2024_01_15_SR10\NCJ29D6 Foundation v15.0.0\Tools\NCJ29D6 User Flash Driver Lauterbach\Scripts\NCJ29D6_b0_swup_apply_pkg.cmm"

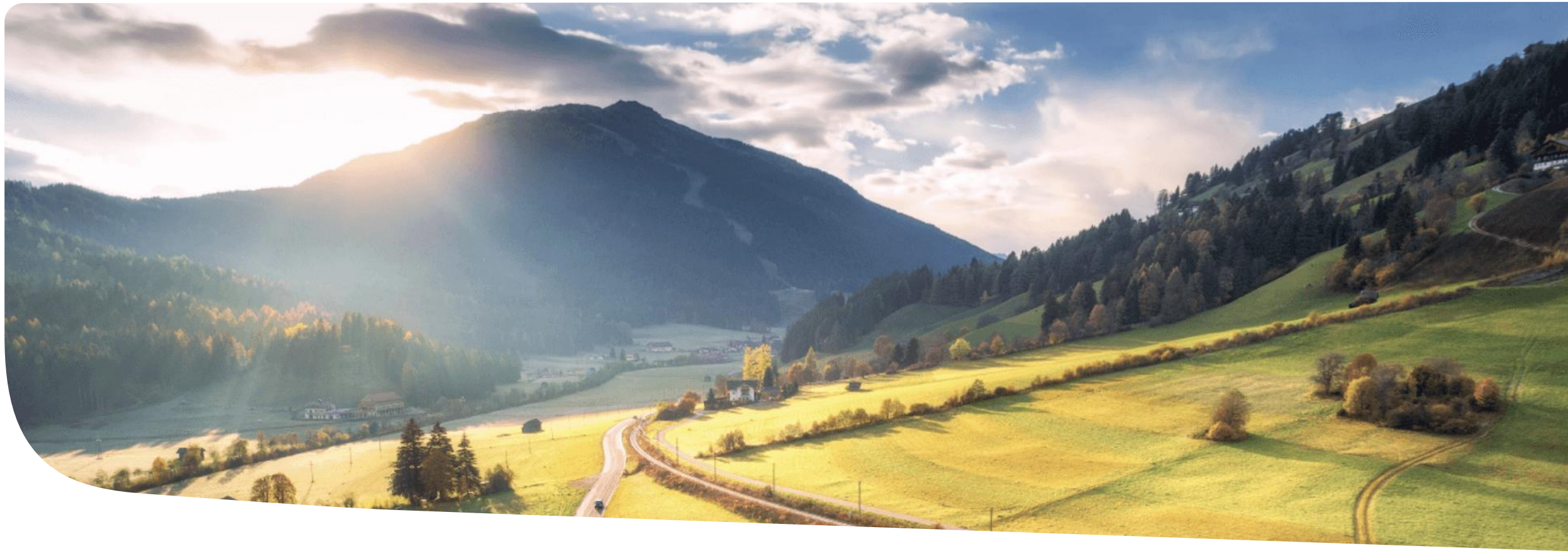
Setup... | Save | Save As.. | Quit | Find... | Do | Debug

```
1  ;
2  ;  Copyright 2023 NXP
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5  ;  and may only be used strictly in accordance with the applicable license terms.
6  ;  By expressly accepting such terms or by downloading, installing, activating
7  ;  and/or otherwise using the software, you are agreeing that you have read, and
8  ;  that you agree to comply with and are bound by, such license terms. If you do
9  ;  not agree to be bound by the applicable license terms, then you may not
10 ;  retain, install, activate or otherwise use the software.
11 ;
12 ; ----------------------------------------------------------------------
13 ;
14 ; NCJ29D6 applies a software update package (via SWUP) on B0 sample;
15 ;
16 ; ------------------------------------------------------------------------------------------
17 ; Step: Check the size of component chunk
18 ;
19 ; the size must be aligned with page size (4KB) .
20 ;
21 GLOBAL &chunk_size
```

NCJ29D6 DSP	pkg
NCJ29D6 RadioSettings	pkg
NCJ29D6 SBE	pkg
NCJ29D6 SWUP	pkg

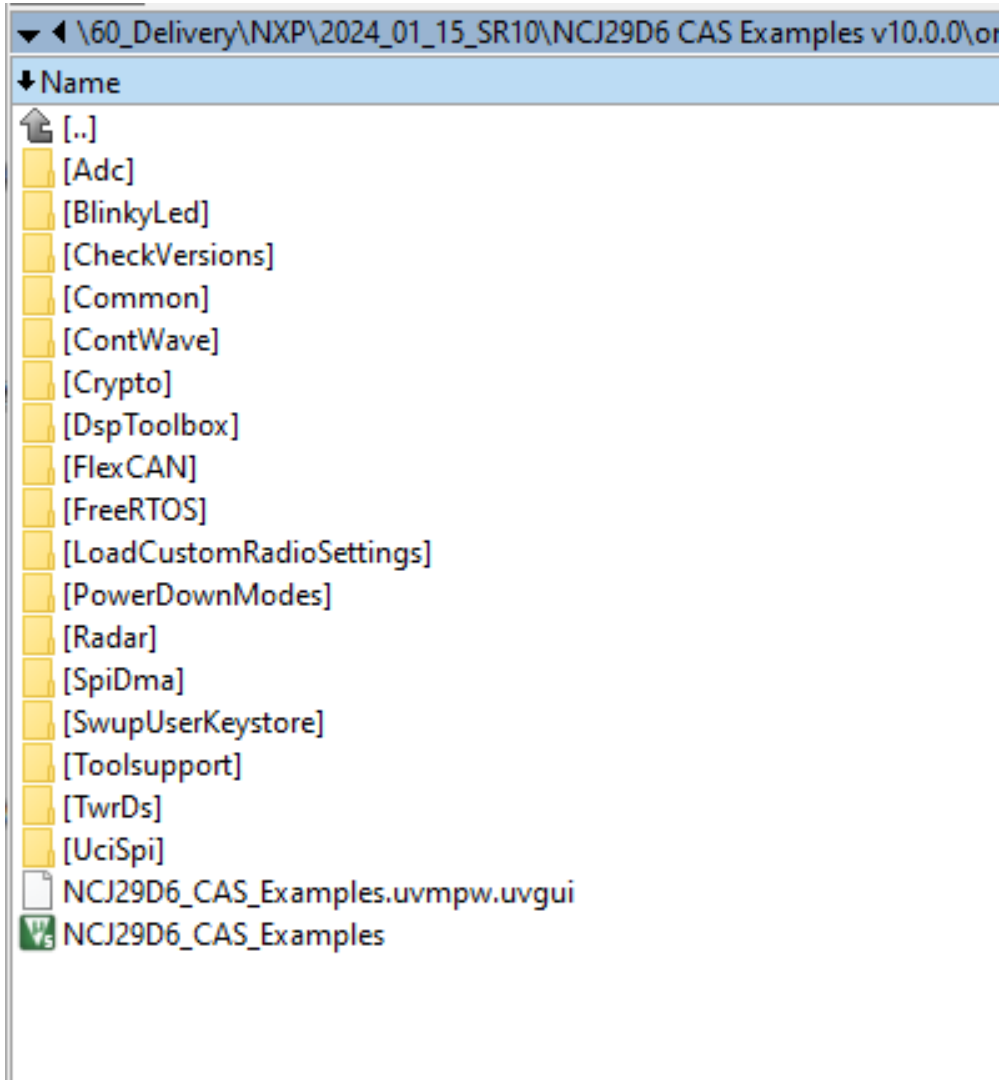
NXP Examples

NXP Examples

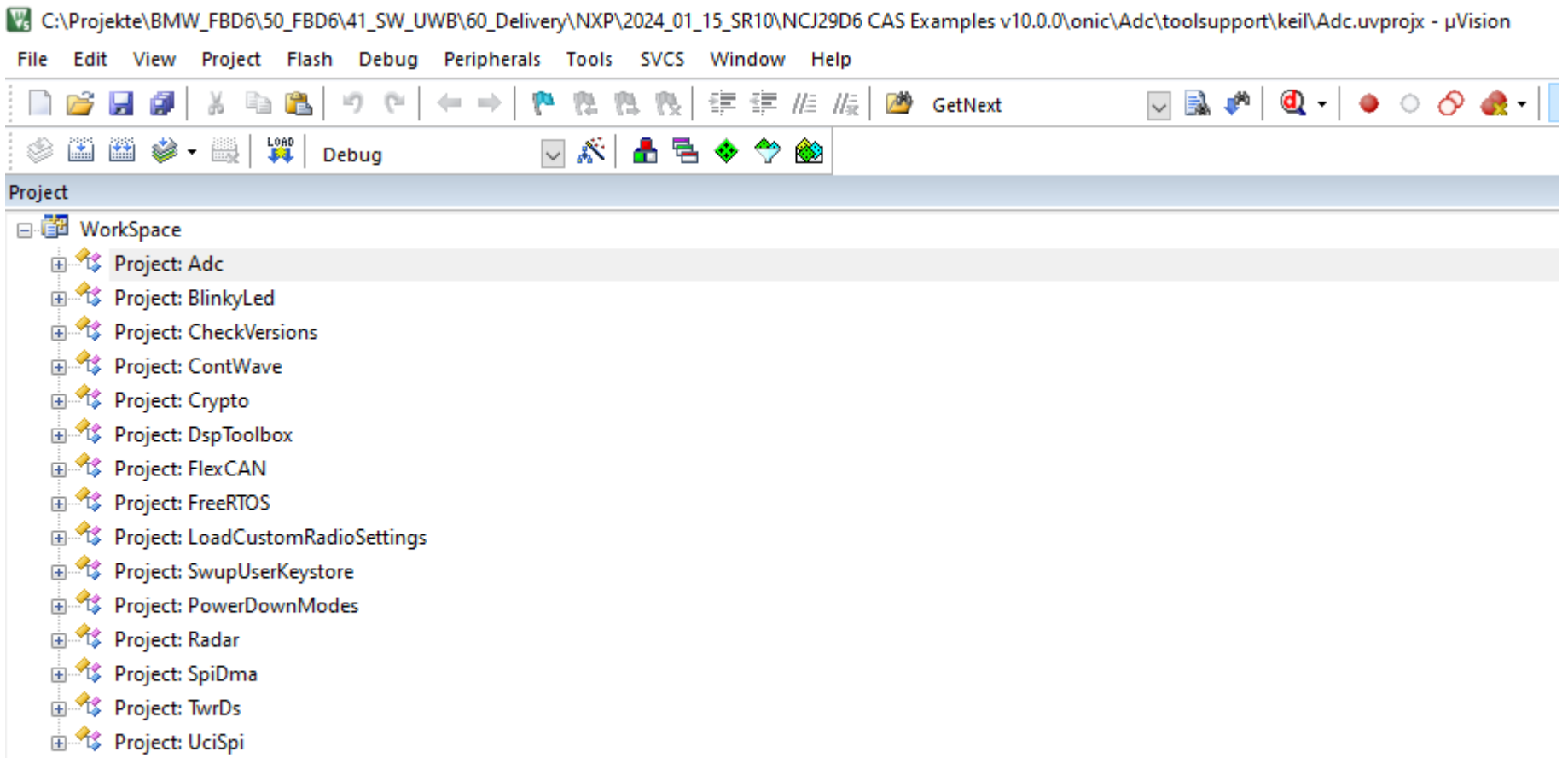


NCJ29D6 CAS Examples

Release Notes:
\50_FBD6\41_SW_UWB\60_Delivery\NXP\2024_01_15_SR10\NCJ29D6 CAS Examples v10.0.0\NCJ29D6_CAS_Examples_Release_Notes.pdf



CAS Examples Keil



Radio settings in „NCJ29D6 RadioSettings.pkg“

Table 73. Order of radio configuration slots

Slot	Scenario	Description
RX		
0	Rx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Rx_Sc406	High Performance (min phase 1 GHz)
2	Rx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Rx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Rx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Rx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Rx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Rx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Rx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Rx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)
TX		
0	Tx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Tx_Sc406	High Performance (min phase 1 GHz)
2	Tx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Tx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Tx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Tx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Tx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Tx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Tx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Tx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)

Radio Setting Generator

User Manual (SR10):

..\41_SW_UWB\60_Delivery\NXP\2024_01_15_SR10\NCJ29D6 Foundation v15.0.0\Documentation\User manual\UM12008 NCJ29D6 Radio Settings Generator User Manual.pdf

1.1 Requirements

MATLAB® installation of version R2019b or later is required. If a MATLAB® installation is not available, alternatively, the MATLAB® Runtime version 9.7 (R2019b) can be installed. Download the Windows 64-bit version of the MATLAB® Runtime for R2019b from the MathWorks Web site by navigating to http://www.mathworks.com/products/compiler/mcr/index.html.

Get License Information:
NCJ29D6_RadioSettingsGenerator.exe –about

Generating an Example Configuration Script
NCJ29D6_RadioSettingsGenerator.exe

Generating the TX and RX Configuration .bin Files:
NCJ29D6_RadioSettingsGenerator.exe <config_script_name.m>

Radiosetting Generator

NCJ29D6 B0 Radio Settings Generator

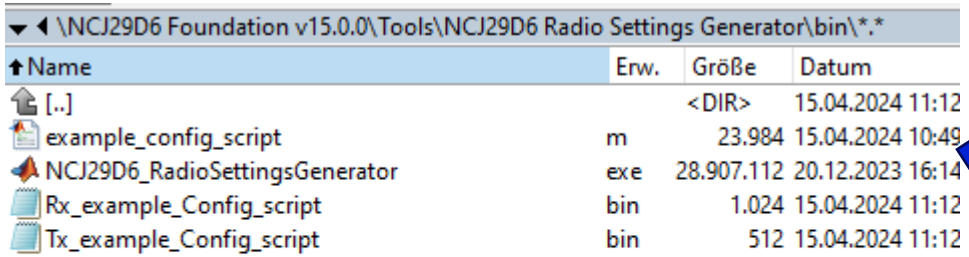
User Manual

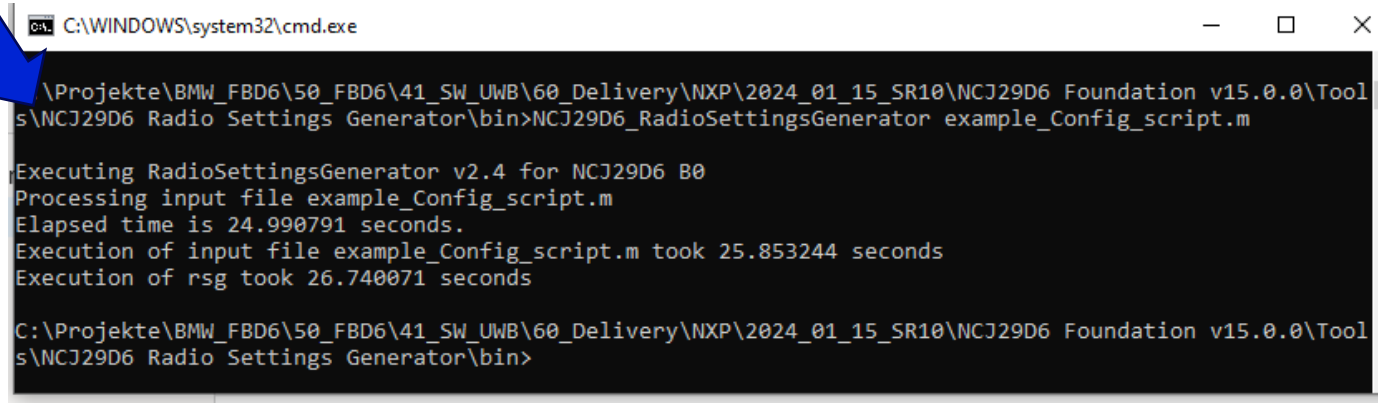
Rev. 1.0 — 22 November 2023 User manual

Document Information

Information	Content
Keywords	NCJ29D6 B0 Radio Settings Generator
Abstract	This document describes the usage of the NCJ29D6 B0 Radio Settings Generator command line tool.

Radiosetting Generator





DSP Toolbox

User Manual (SR10):

..\41_SW_UWB\60_Delivery\NXP\2024_01_15_SR10\NCJ29D6 Foundation v15.0.0\Documentation\User manual\UM11897 NCJ29D6 DSP Toolbox MATLAB Interface User Manual.pdf

Radio Setting Generator

User Manual (SR10):

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NXP Demoboard

/50_FBD6/30_HW/00_Datasheets/NXP/UWB/NCJ29D6_DemoBoard_UserManual.pdf

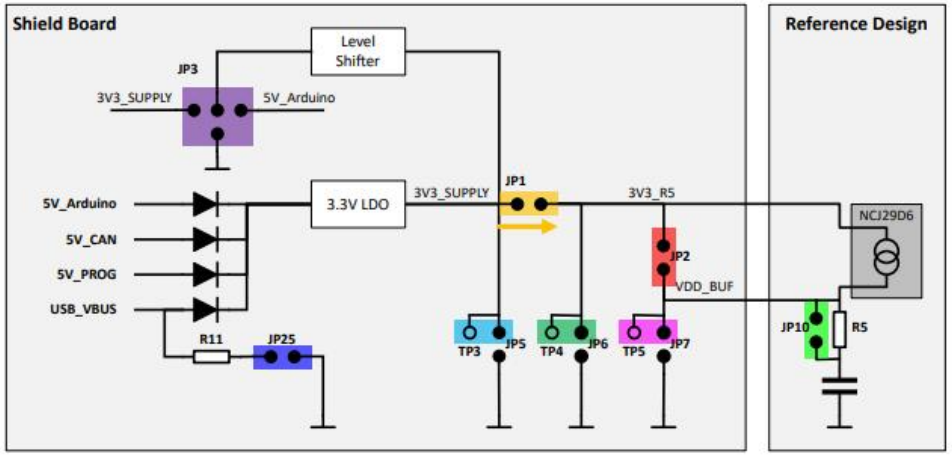


Figure 5. Overview of the power options on the NCJ29D6 demo board.

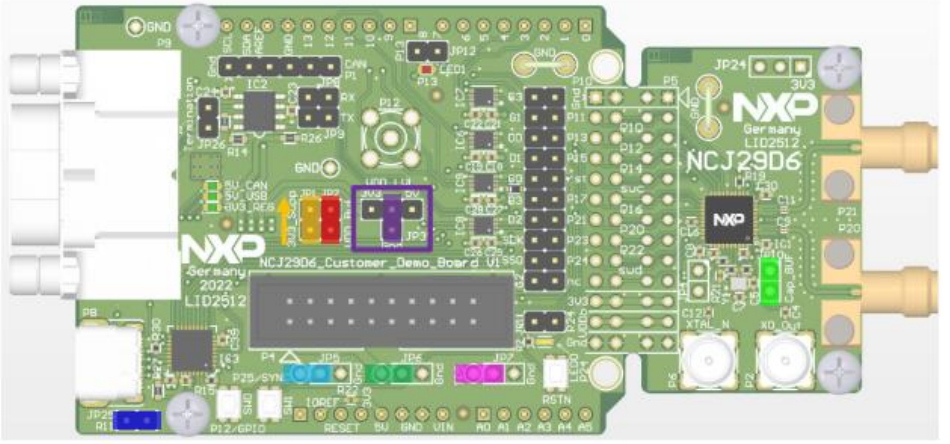
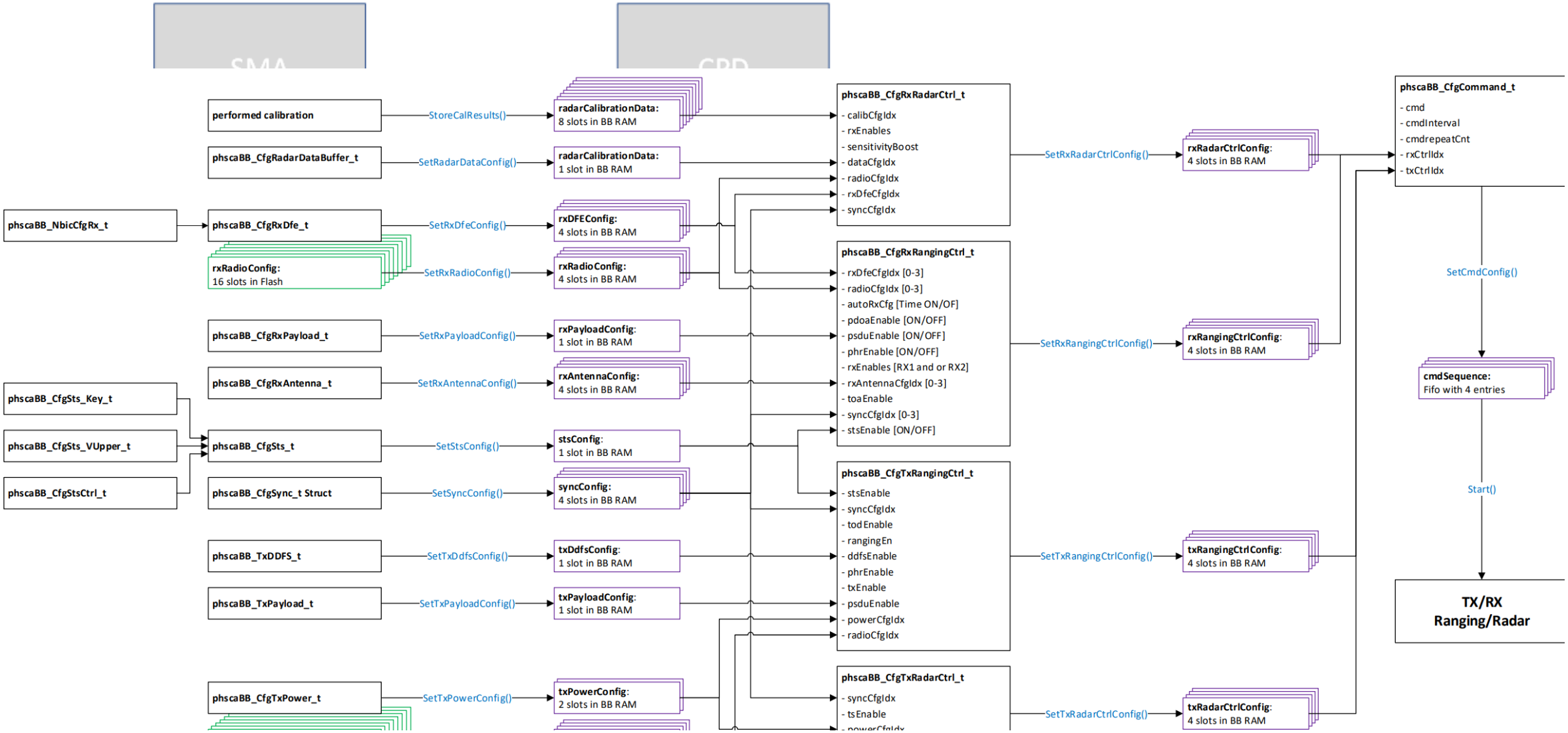
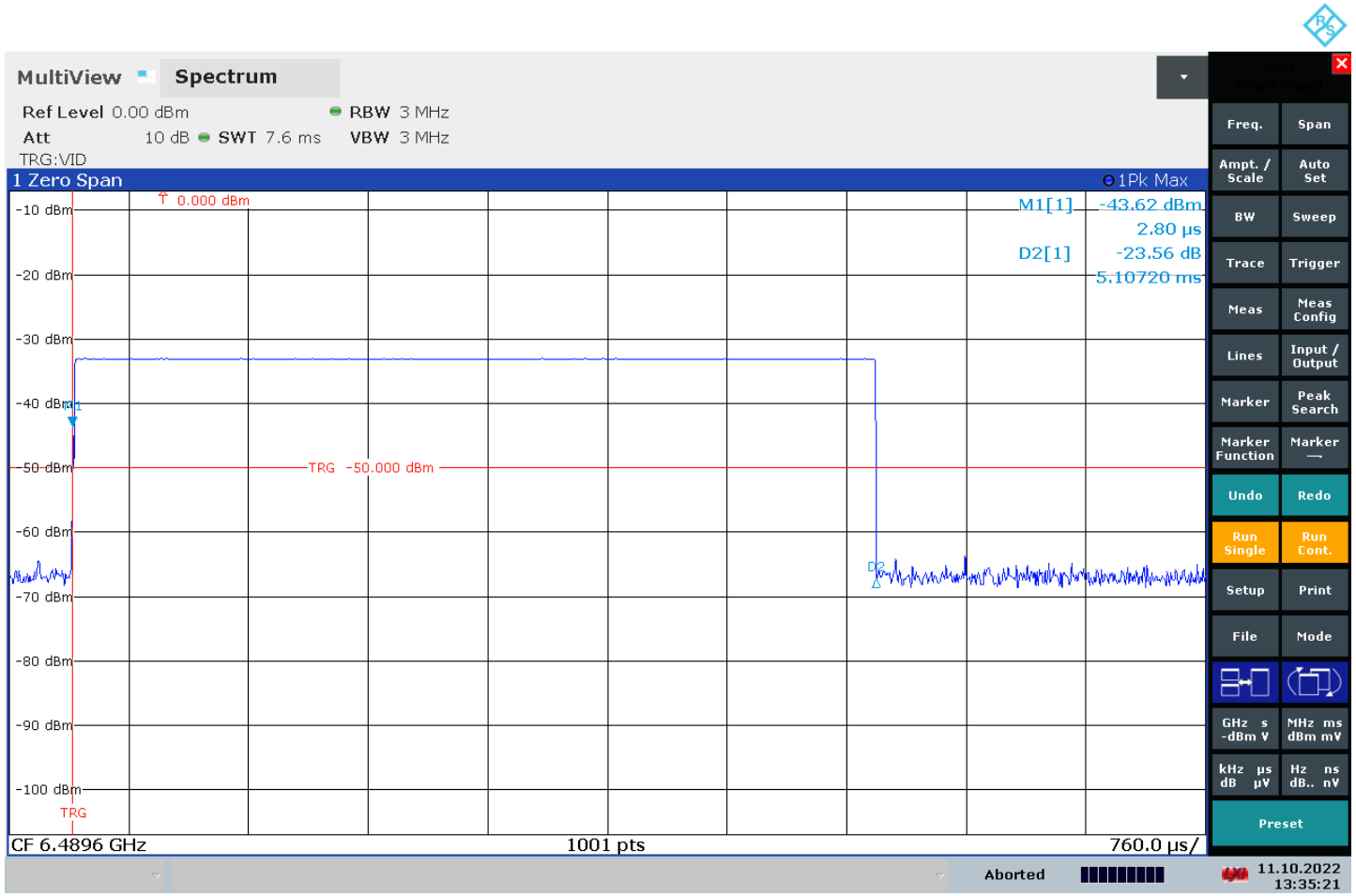


Figure 6. Location of the jumpers and test points on the NCJ29D6 demo board.

Message Handler & Sequence Handling (Timing CPD&Ranging)

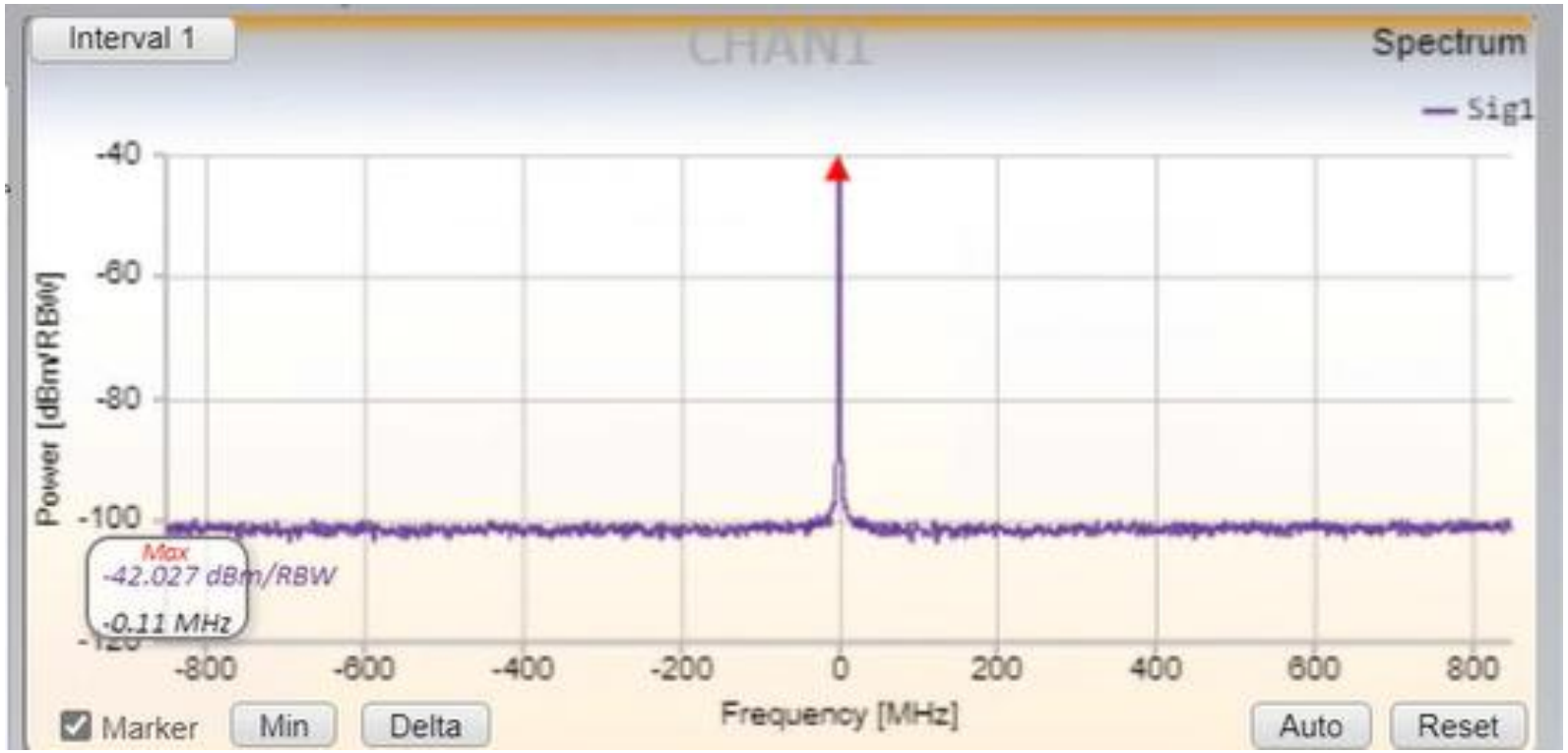


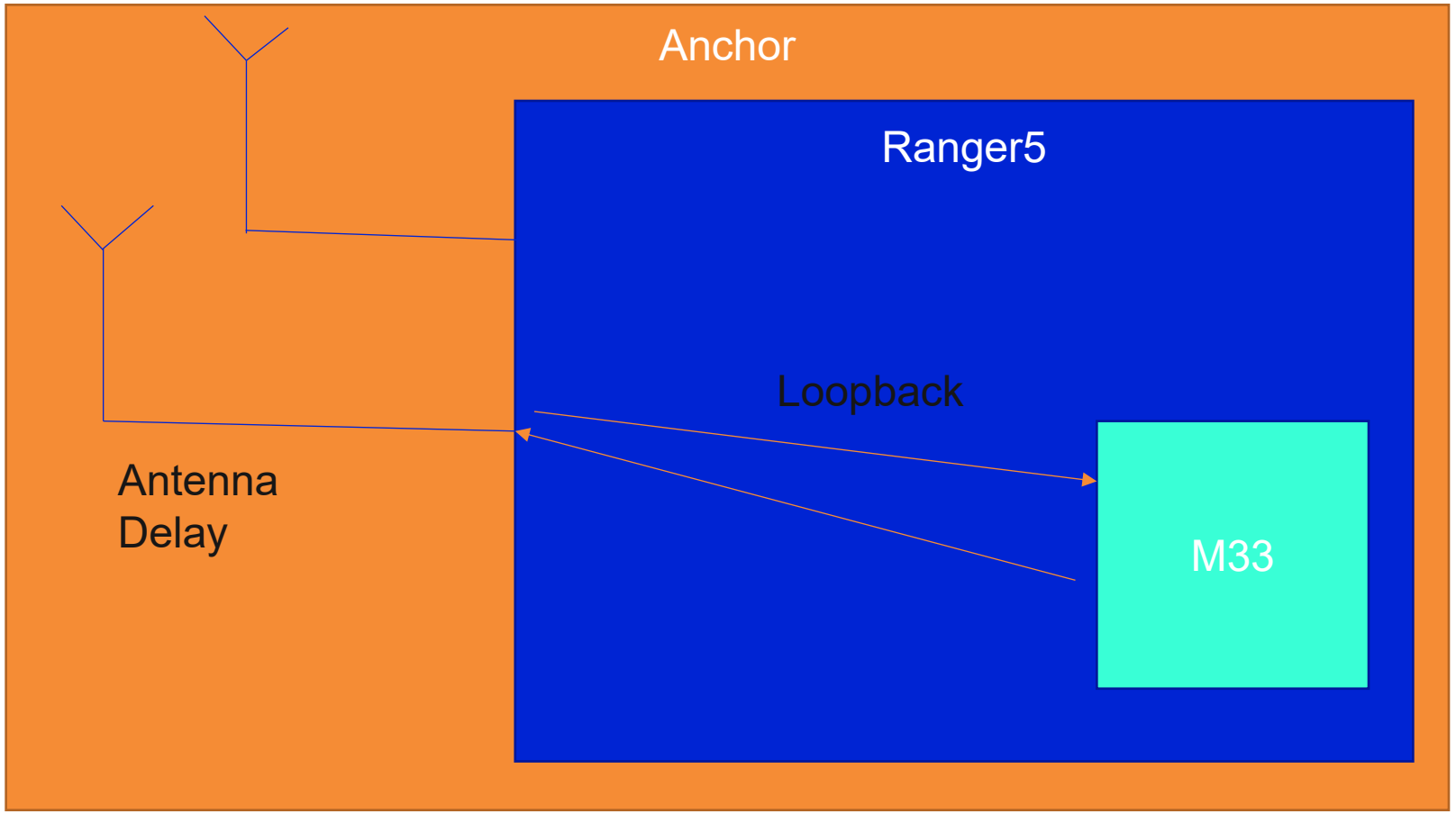
CW Pulse (time)

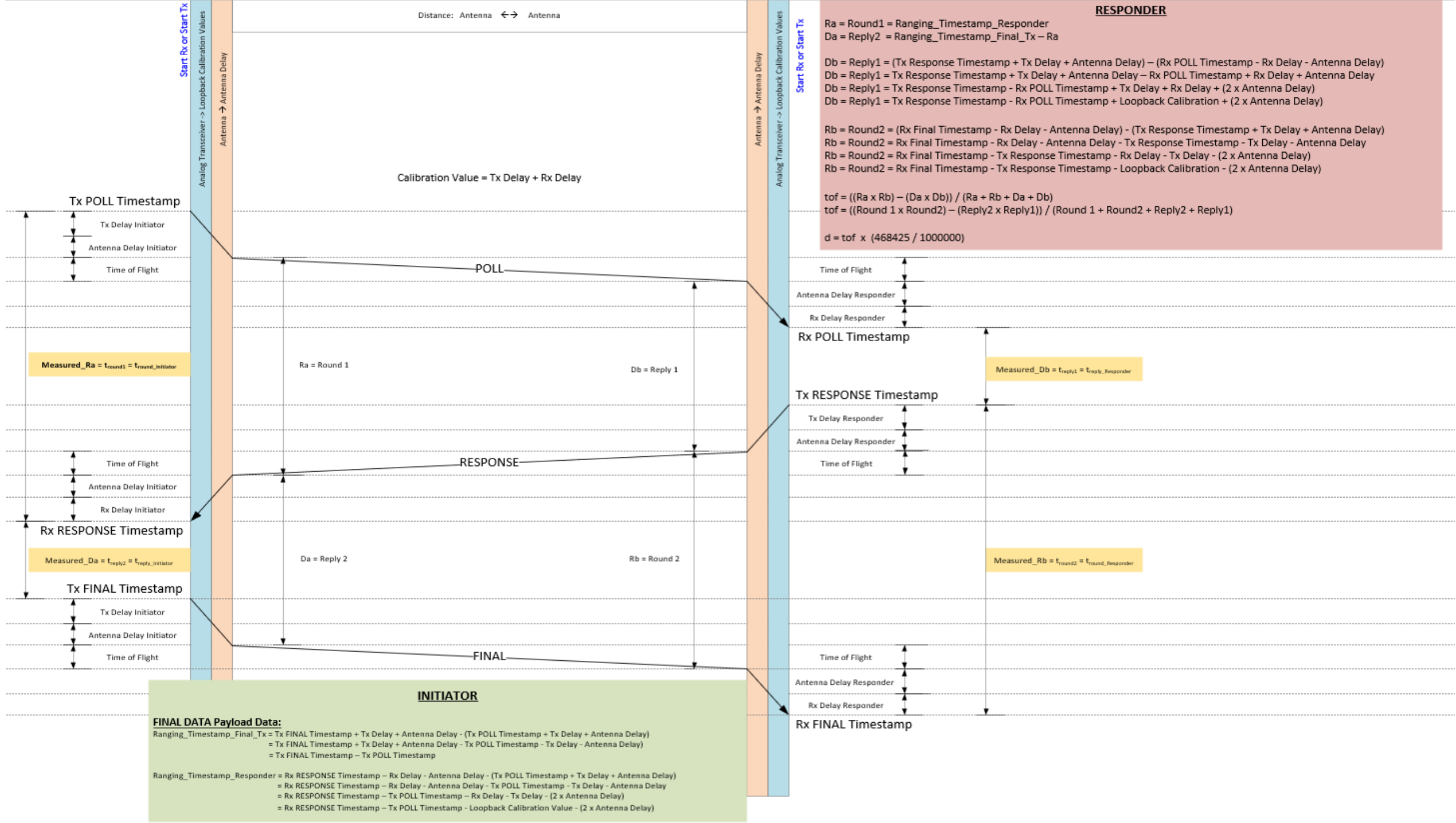


13:35:22 11.10.2022

CW Pulse (frequency)




Anchor
Ranger5
Loopback
Antenna
Delay
M33



RESPONDER

Ra = Round1 = Ranging_Timestamp_Responder
Da = Reply2 = Ranging_Timestamp_Final_Tx – Ra

Db = Reply1 = (Tx Response Timestamp + Tx Delay + Antenna Delay) – (Rx POLL Timestamp - Rx Delay - Antenna Delay)
Db = Reply1 = Tx Response Timestamp + Tx Delay + Antenna Delay – Rx POLL Timestamp + Rx Delay + Antenna Delay
Db = Reply1 = Tx Response Timestamp - Rx POLL Timestamp + Tx Delay + Rx Delay + (2 x Antenna Delay)
Db = Reply1 = Tx Response Timestamp - Rx POLL Timestamp + Loopback Calibration + (2 x Antenna Delay)

Rb = Round2 = (Rx Final Timestamp - Rx Delay - Antenna Delay) - (Tx Response Timestamp + Tx Delay + Antenna Delay)
Rb = Round2 = Rx Final Timestamp - Rx Delay - Antenna Delay - Tx Response Timestamp - Tx Delay - Antenna Delay
Rb = Round2 = Rx Final Timestamp - Tx Response Timestamp - Rx Delay - Tx Delay - (2 x Antenna Delay)
Rb = Round2 = Rx Final Timestamp - Tx Response Timestamp - Loopback Calibration - (2 x Antenna Delay)

tof = ((Ra x Rb) – (Da x Db)) / (Ra + Rb + Da + Db)
tof = ((Round 1 x Round2) – (Reply2 x Reply1)) / (Round 1 + Round2 + Reply2 + Reply1)

d = tof x (468425 / 1000000)

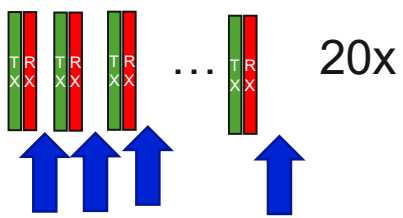
INITIATOR

FINAL DATA Payload Data:

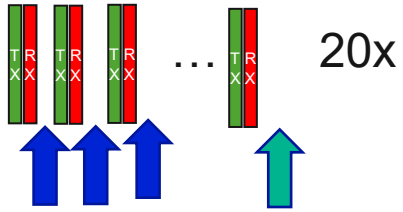
Ranging_Timestamp_Final_Tx = Tx FINAL Timestamp + Tx Delay + Antenna Delay - (Tx POLL Timestamp + Tx Delay + Antenna Delay)
= Tx FINAL Timestamp + Tx Delay + Antenna Delay - Tx POLL Timestamp - Tx Delay - Antenna Delay)
= Tx FINAL Timestamp – Tx POLL Timestamp

Ranging_Timestamp_Responder = Rx RESPONSE Timestamp – Rx Delay - Antenna Delay - (Tx POLL Timestamp + Tx Delay + Antenna Delay)
= Rx RESPONSE Timestamp – Rx Delay - Antenna Delay - Tx POLL Timestamp - Tx Delay - Antenna Delay
= Rx RESPONSE Timestamp – Tx POLL Timestamp – Rx Delay - Tx Delay - (2 x Antenna Delay)
= Rx RESPONSE Timestamp – Tx POLL Timestamp - Loopback Calibration Value - (2 x Antenna Delay)

1. UwbApp_LoopbackCall
2. Start Measurement Pulse Shape 0 and 2
   → UwbMwMsh_Start()



Ready Event



Idle Event

UwbApp_LoopbackEventCheck()

Ready-Event:
→ e.g. Read Out Timestamps()

Idle-Event:
e.g. Read Out Timestamps()
    Calculation Loopback Value 0 and 2
    Storage Nv Data

```
case EN_SEQ_LOOPBACK:
UwbMwConf_SetFrequency(lUwbDeviceRangingCfg_pst->FrequencyChannelIndex_u8);

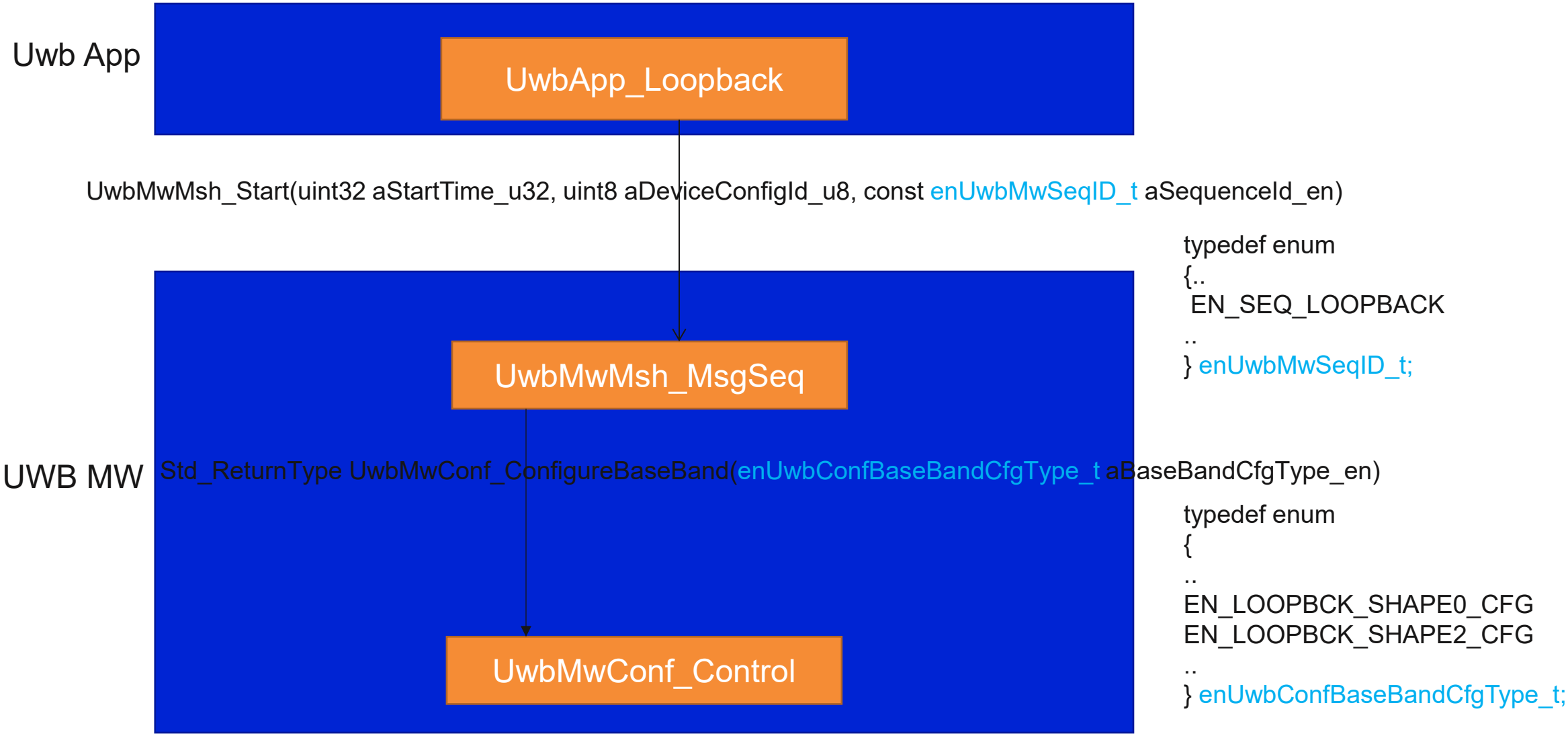
/* Start Measurement Pulsshape 0 */
lFifo_st[0].cmd = PHSCABB_CMD_RANGING_TRX;
UwbMwConf_ConfigureBaseBand(EN_LOOPBCK_SHAPE0_CFG);

lFifo_st[0].rxCtrlIdx    = 0x02u;                                    /* Index to select Rx control configuration */
lFifo_st[0].txCtrlIdx    = 0x02u;                                     /* Index to select Tx control configuration */
lFifo_st[0].cmdRepeatCnt = 20u;                                    /* Number of repetitions of the commands */
lFifo_st[0].cmdRepeatInterval = (uint32_t)(2000*1248u)/10u; /* Interval between commands  */
lFifo_st[0].cmdSequenceInterval = (uint32_t)(CONFIG_RANGING_GRID*1248u)/10u;                 /* Interval between sequences */

UwbMwConf_ConfigureBaseBand(EN_LOOPBCK_SHAPE2_CFG);

/* Start Measurement Pulsshape 2 */
lFifo_st[1].cmd = PHSCABB_CMD_RANGING_TRX;
lFifo_st[1].rxCtrlIdx    = 0x03u;                                    /* Index to select Rx control configuration */
lFifo_st[1].txCtrlIdx    = 0x03u;                                     /* Index to select Tx control configuration */
lFifo_st[1].cmdRepeatCnt = 20u;                                    /* Number of repetitions of the commands */
lFifo_st[1].cmdRepeatInterval = (uint32_t)(2000*1248u)/10u; /* Interval between commands  */
lFifo_st[1].cmdSequenceInterval = (uint32_t)(CONFIG_RANGING_GRID*1248u)/10u;                 /* Interval between sequences */
```

Loopback UWB Stack



Pulse Shape

21.5.3 *PulseShape Combinations*

Compliant receivers shall be able to receive and suitably process both symmetrical an precursor-free pulses. It is up to the implementer whether the same receiver configurat for PulseShape 0x2 and PulseShape 0x1, or whether reception of PulseShape 0x2 is e optimized via a separate receiver configuration.

Possible transmitter PulseShape combinations (PulseShape_Combo) for the initiator a responder are shown in Table 21-4. The responder's and initiator's receive configurati be set according to the PulseShape selections made by the initiator's and responder's transmitters, respectively.

Table 21-4: Overview of PulseShape_Combo values and the associated transmit pulse shap

PulseShape_Combo	Transmit PulseShape	
	RX	TX
0x00	0x0	0x0
0x01	0x0	0x1
0x02	0x0	0x2

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Confidential

PulseShape_Combo	Transmit PulseShape	
	RX	TX
0x10	0x1	0x0
0x11	0x1	0x1
0x12	0x1	0x2
0x20	0x2	0x0
0x21	0x2	0x1

PulseShape_Combo	Transmit PulseShape	
	Initiator	Responder
0x00	0x0	0x0
0x01	0x0	0x1
0x02	0x0	0x2
0x10	0x1	0x0
0x11	0x1	0x1
0x12	0x1	0x2
0x20	0x2	0x0
0x21	0x2	0x1
0x22	0x2	0x2

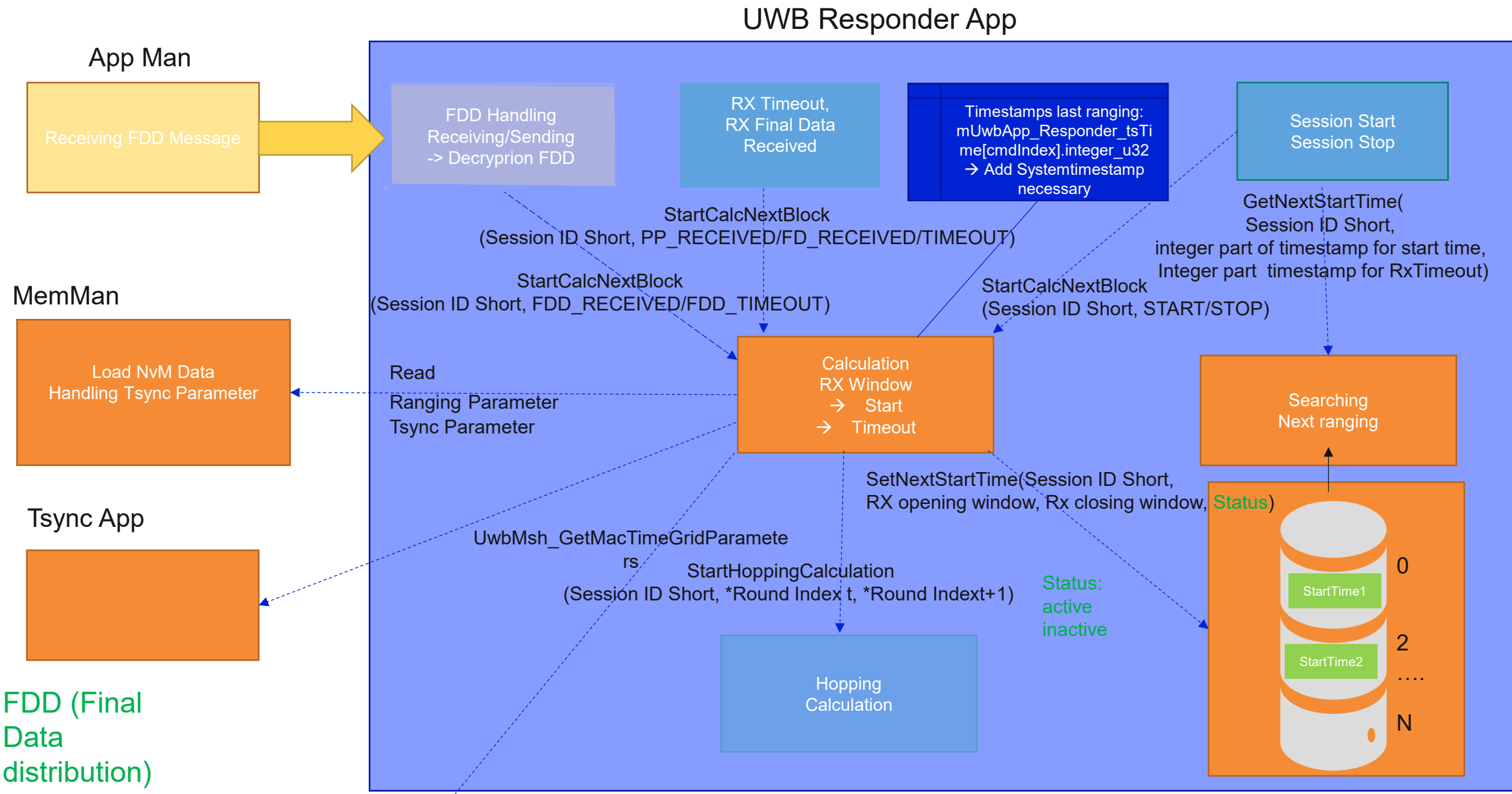


Initiator (Radioslots)	Responder (Radioslot)
TX: 0 / RX: 0	Tx: 0 / RX: 0
Tx: 0 / RX: 8	Tx: 8 / RX: 0
Tx: 0 / RX: 8	Tx: 8 / RX: 0
Tx: 8 / RX: 0	Tx: 0 / RX: 8
Tx: 8 / RX: 8	Tx: 8 / RX: 8
Tx: 8 / RX: 8	Tx: 8 / RX: 8
Tx: 8 / RX: 0	Tx: 0 / RX: 8
Tx: 8 / RX: 8	Tx: 8 / RX: 8
Tx: 8 / RX: 8	Tx: 8 / RX: 8

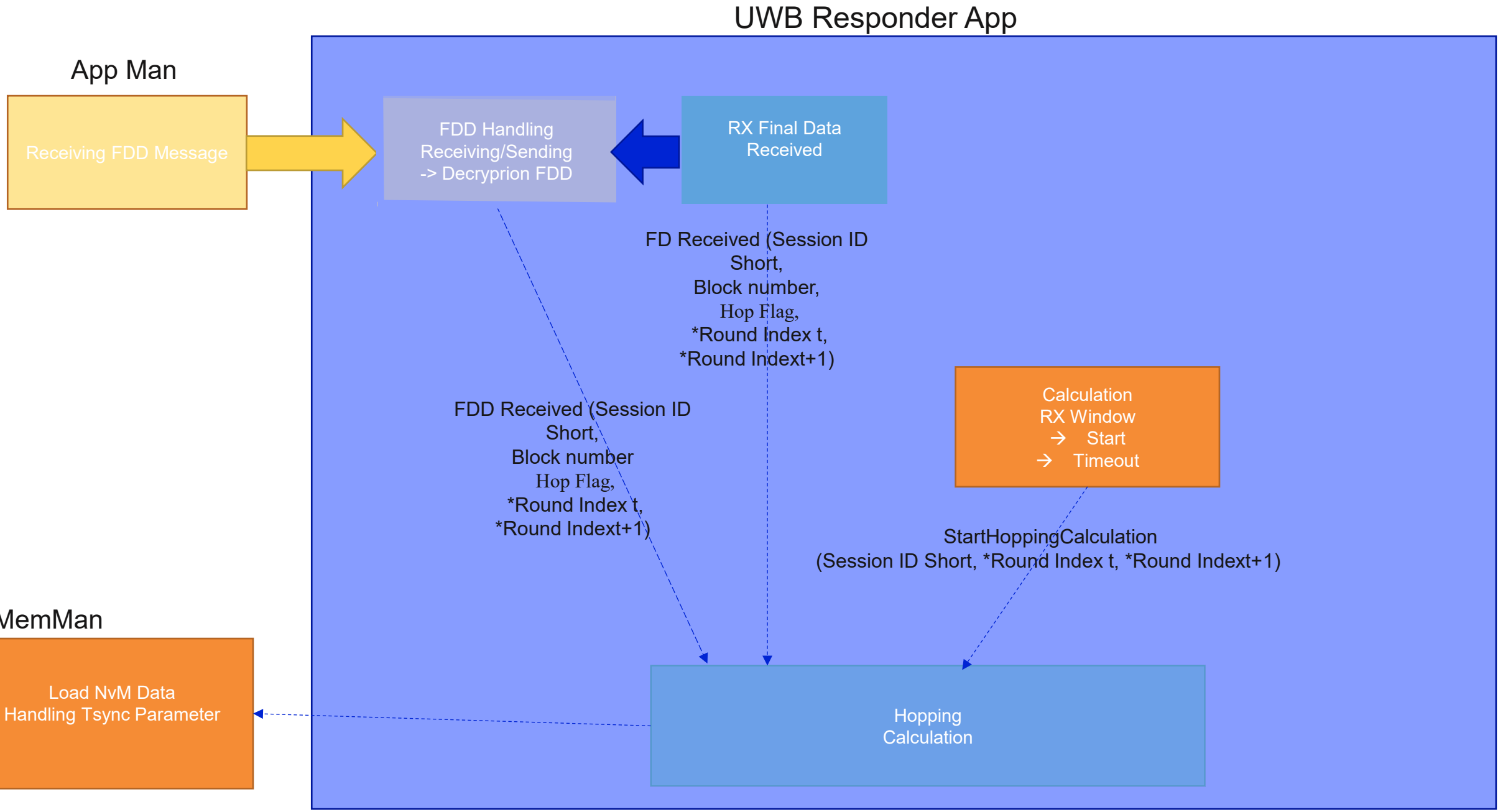
UwbMwConf_ConfigureBaseBand

Slot	Scenario	Description
RX		
0	Rx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Rx_Sc406	High Performance (min phase 1 GHz)
2	Rx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Rx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Rx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Rx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Rx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Rx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Rx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Rx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)
TX		
0	Tx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Tx_Sc406	High Performance (min phase 1 GHz)
2	Tx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Tx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Tx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Tx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Tx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Tx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Tx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Tx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)

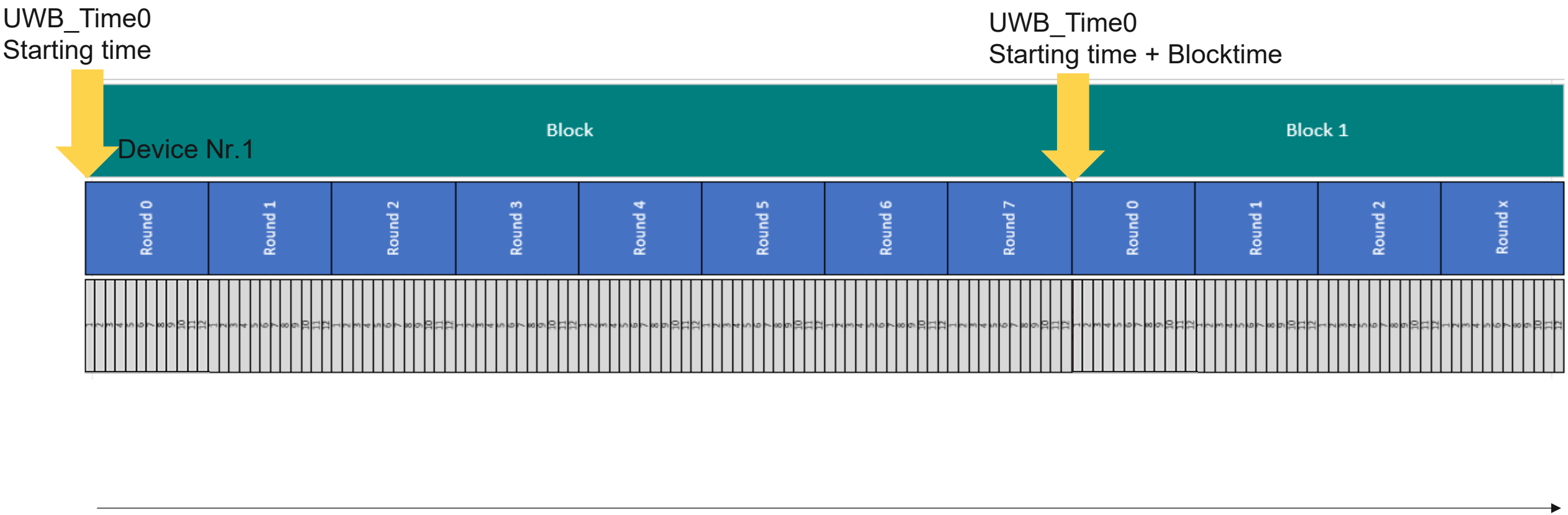
Workpackages Uwb WindowCalculation



Workpackages Uwb WindowCalculation

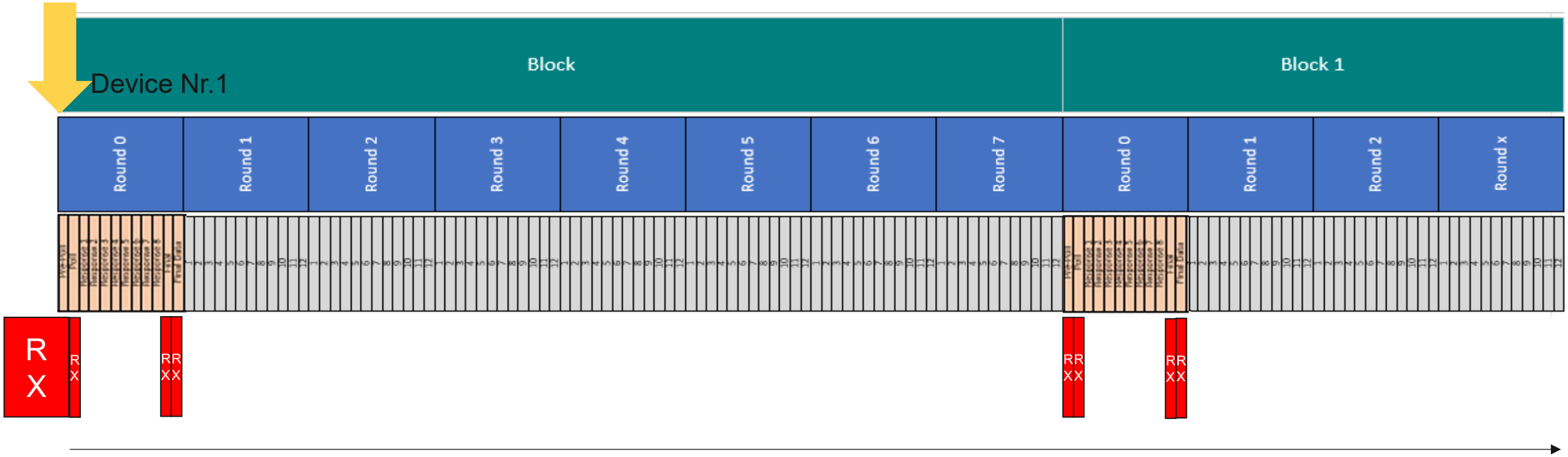


UWB Time0
Ranging not possible

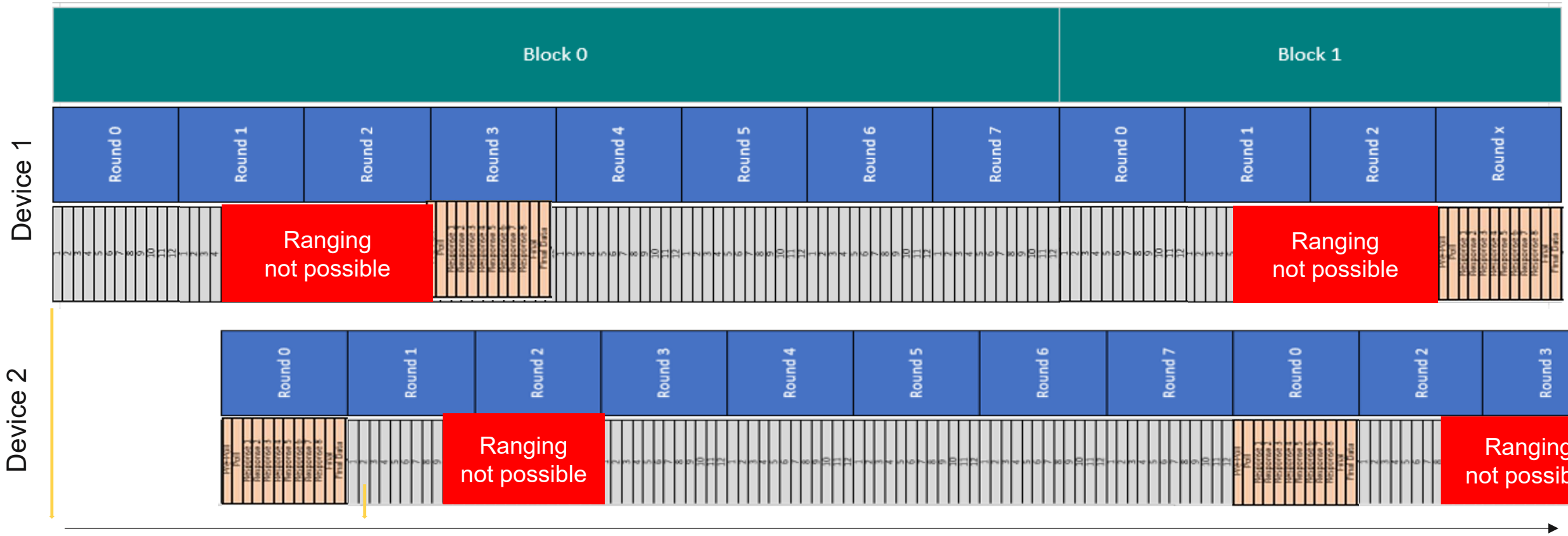


UWB Time0
Ranging possible

UWB_Time0
Starting time



UWB Collision



BLE

19.2.5.4 Ranging Session Response (RS-RS)

This message is sent by the device according to Section 20.5.2.

Message	Message ID	Parameters
Ranging_Session_RS	4	RAN_Multiplier, Slot_BitMask SYNC_Code_Index_BitMask Channel_BitMask Hopping_Config_Bitmask

Table 19-24: Ranging_Session_RS message and its parameters

Hopping_Config_Bitmask	1	0-255	[b7 b6 b5] bitmask of hopping modes, the device offers to use in this ranging session 100 No Hopping 010 Continuous Hopping 001 Adaptive Hopping [b4 b3 b2 b1 b0] bit mask of hopping sequences the device offers to use in this ranging session b4=1 is always set since the default hopping sequence and support thereof is mandatory and defined in Appendix A. b3=1 is set when the optional AES-based hopping sequence, which is defined in Appendix A, is supported Rest of the bits are reserved for future hopping sequences.

b7	b6	b5	b4	b3	b2	b1	b0
100 No Hopping 010 Continuous Hopping 001 Adaptive Hopping			bit mask of hopping sequences the device offers to use in this ranging session				
			is always set since the **default hopping** sequence and support thereof is mandatory	b3=1 is set when the optional **AES-based hopping** sequence			

Examples of a valid Selected_Hopping_Config_BitMask

[10000000] no hopping mode

[01010000] continuous hopping with default hopping sequence

[00110000] adaptive hopping with default hopping sequence

[01001000] continuous hopping with AES-based optional hopping sequence

[00101000] adaptive hopping with AES-based optional hopping sequence

Default/AES based Hopping

This section describes the method to generate the hopping sequence

- **Default Hopping Sequcence:**

  This method is mandatory and shall be supported by any DCK device.

  For the *k*-th ranging session with N_{Round}^{k} ranging rounds per block and hoping key $HOP_Key_RW^{k}$, the following function is used to generate the hopping round S^{k} index for ranging block i:

$$S^{k}\left(i, HOP_Key_RW^{k}, N_{Round}^{k}\right) = \left(\left(\left(\left(i + HOP_Key_RW^{k}\right)\ \&\ 0\text{xFFFF}\right)^{2} \bmod \left(2^{16} - 15\right)\right) N_{Round}^{k}\right) >> 16$$

  The initiator shall generate and send the hopping key $HOP_Key_RW^{k}$ to the responder device during the ranging session initiation process (see Section 20.5.2).

- **AES-Based Hopping Sequence**:

  This method is optional may be supported by any DCK device. It uses AES to generate the hopping sequence.

  For the *k*-th ranging session with N_{Round}^{k} ranging rounds per block and hopping key $HOP_Key_RW^{k}$, the following function is used to generate the AES-based hopping round S^{k} index for ranging block i:

$$S^{k}\left(i, HOP_Key_RW^{k}, N_{Round}^{k}\right) = \left(\left(\left(\text{AES}\left(i, HOP_Key_RW^{k}\right)\ \&\text{0xFFFF}\right) N_{Round}^{k}\right) >> 16\right)$$

  AES-128 shall be used for AES function. Block index i and hopping key $HOP_Key_RW^{k}$ shall be used for plaintext and key, respectively.

20.4 Hopping Flag and Round Index Determination

As stated above, the initiator in the *k*-th ranging session in any given RAN shall start the UWB ranging procedure in the first ranging round (ranging round number 0) in the first ranging block (ranging block number 0) by default. This assumes that the responder-device has either achieved block synchronization by OOB method or, if not, is permanently listening for Pre-Poll messages.

At the initiator, and assuming no resource conflict occurs, the $Hop_Flag^k(i+1)$ and $Round_Idx^k(i+1)$ for the next ranging block (ranging block *i*+1) will depend on the hopping mode selected during ranging session setup as follows:

- If the hopping mode is set to "no hopping", then the initiator shall continue to use the same ranging round in ranging block *i* and $Round_Idx^k(i+1)$ is set as:

$$Round_Idx^k(i+1) = Round_Idx^k(i) = 0$$

  and $Hop_Flag^k(i+1)$ is set to 0. Note that in this case, $Hop_Flag^k(i+1)$ is irrelevant.

- If the hopping mode is set to "continuous hopping", the initiator uses the S_{i+1}^k -th ranging round

$$Round_Idx^k(i+1) = S_{i+1}^k$$

  and $Hop_Flag^k(i+1)$ is set to 1. Again $Hop_Flag^k(i+1)$ is irrelevant in this case.

- If the hopping mode is set to "adaptive hopping", then at the initiator, the $Hop_Flag^k(i+1)$ and $Round_Idx^k(i+1)$ for the next ranging block (ranging block *i*+1) shall be set as follows:
  - If the initiator determines that the round is clean, i.e. no interference, and shall stay in the current round

$$Hop_Flag^k(i+1) = 0 \quad \text{and} \quad Round_Idx^k(i+1) = Round_Idx^k(i)$$

  - If the initiator determines that there is some interference on the current round and shall hop to a different round:

$$Hop_Flag^k(i+1) = 1 \quad \text{and} \quad Round_Idx^k(i+1) = S_{i+1}^k$$

    If the initiator determines that there is little interference and ranging in the current round is successful, the initiator shall continue ranging in the current round, i.e.

$$Round_Idx^k(i+1) = Round_Idx^k(i) \quad \text{and} \quad Hop_Flag^k(i+1) = 0$$

    The initiator shall send the $Hop_Flag^k(i+1)$ and $Round_Idx^k(i+1)$ for next ranging block to the responder-device as part of the payload packet carrying time stamps *Final_Data* at the end of the ranging sequence.

Hopping Flag and Round Index

if *Final_Data* packet is received

{

$Hop_Mode^{k}(i+1) = Final_Data.Hop_Flag$

if ($Hop_Flag^{k}(i+1) = 0$)

{

set $Round_Idx^{k}(i+1) = Round_Idx^{k}(i)$

}

elseif ($Hop_Flag^{k}(i+1) = 1$)

{

set $Round_Idx^{k}(i+1) = S^{k}_{i+1}$

}

}

else

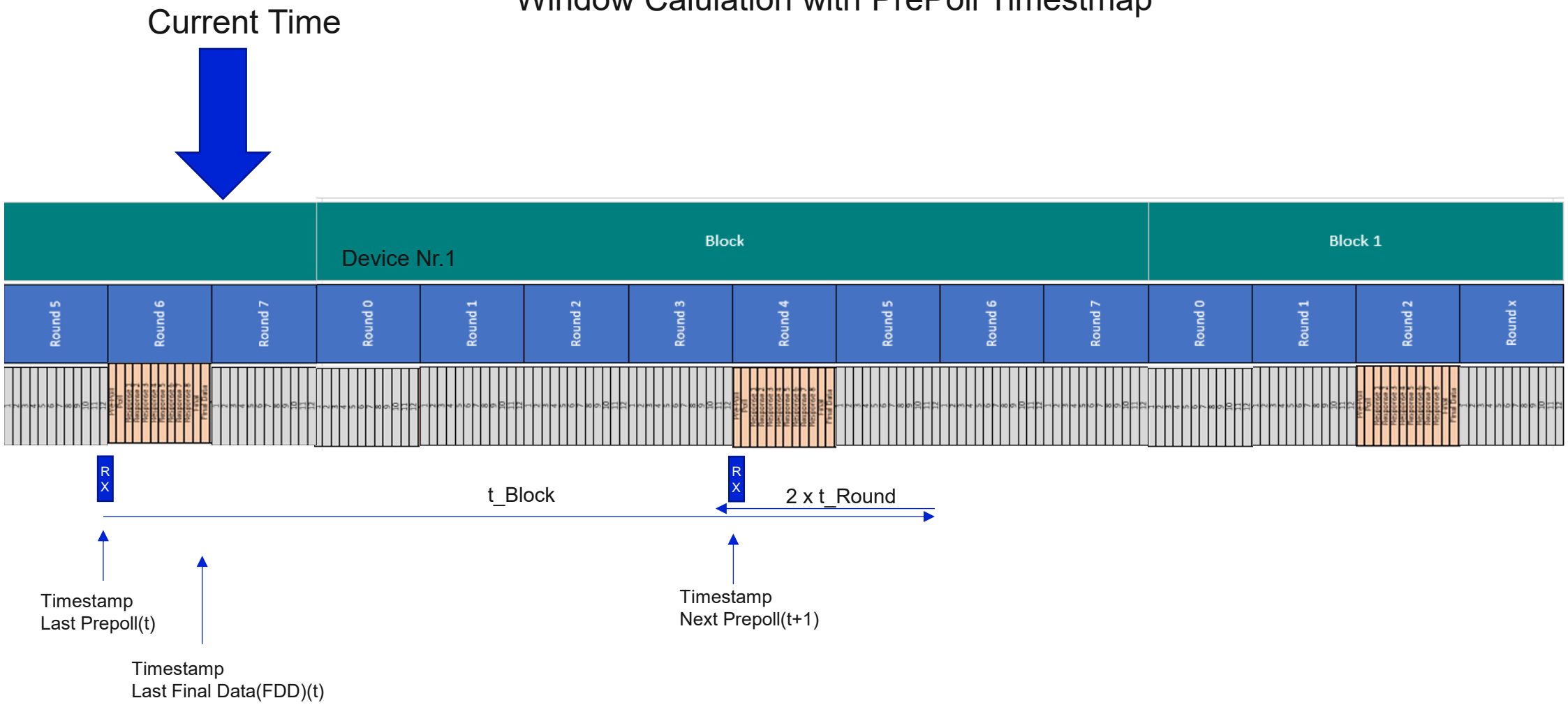
{

set $Round_Idx^{k}(i+1) = S^{k}_{i+1}$.

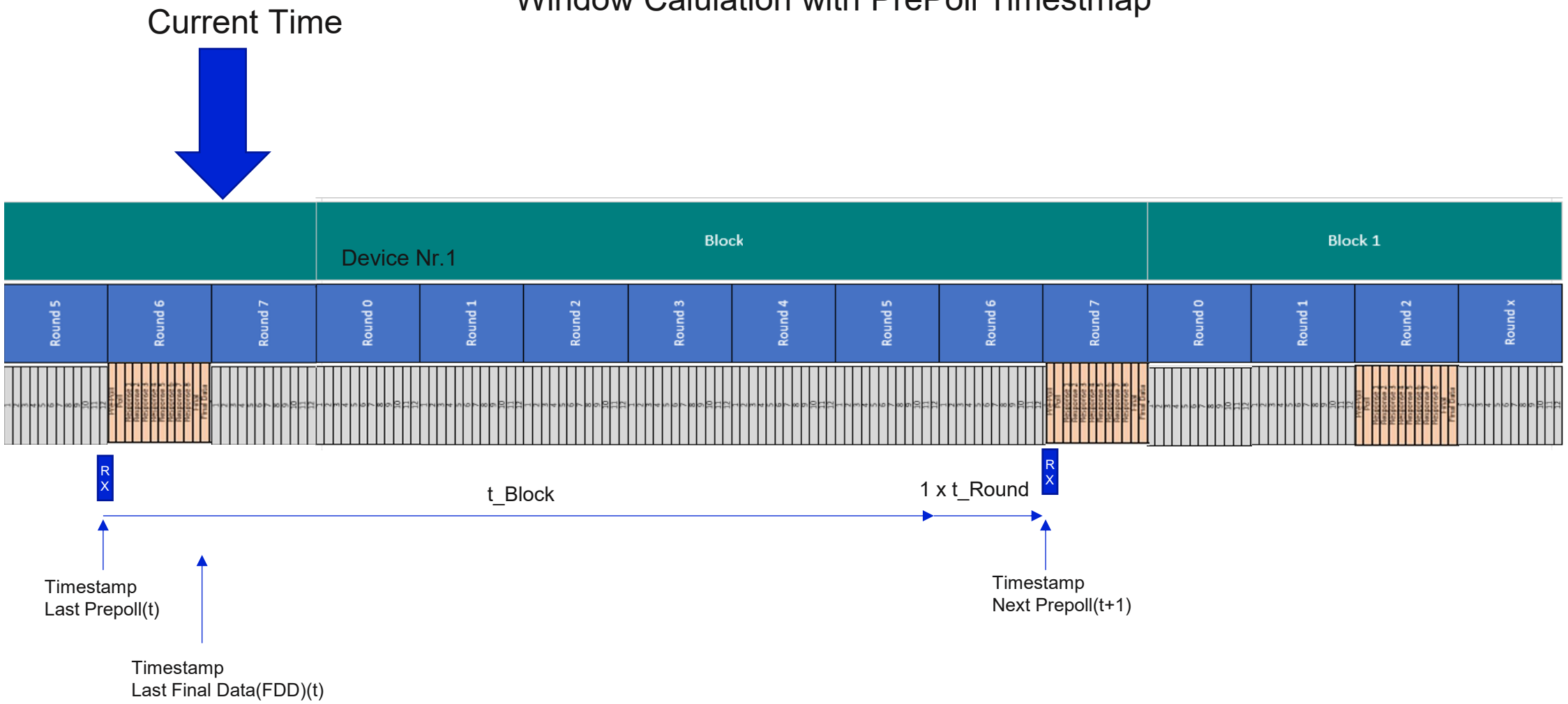
}

The initiator may decide to trigger the hopping function based on the interference level or packet collision rate that it measures over the current exchange and/or the number of responses that it receives back from the responder-device. In general, the exact criteria for triggering hopping at the initiator is out of scope of this specification and is left to each device manufacturer. However, the only requirement in any such criteria is that *if the device does not receive any responses from the responder-device, the device shall turn on hopping.*

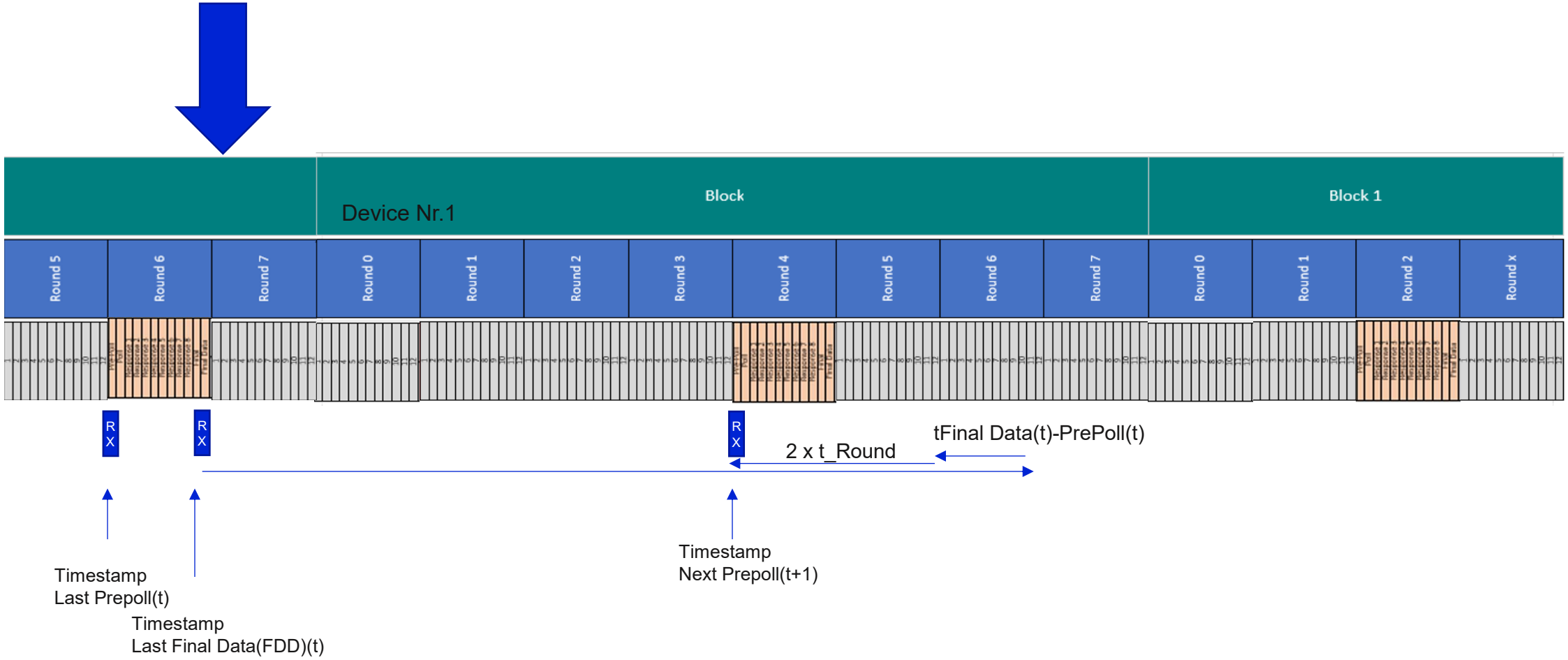
Window Calulation with PrePoll Timestmap



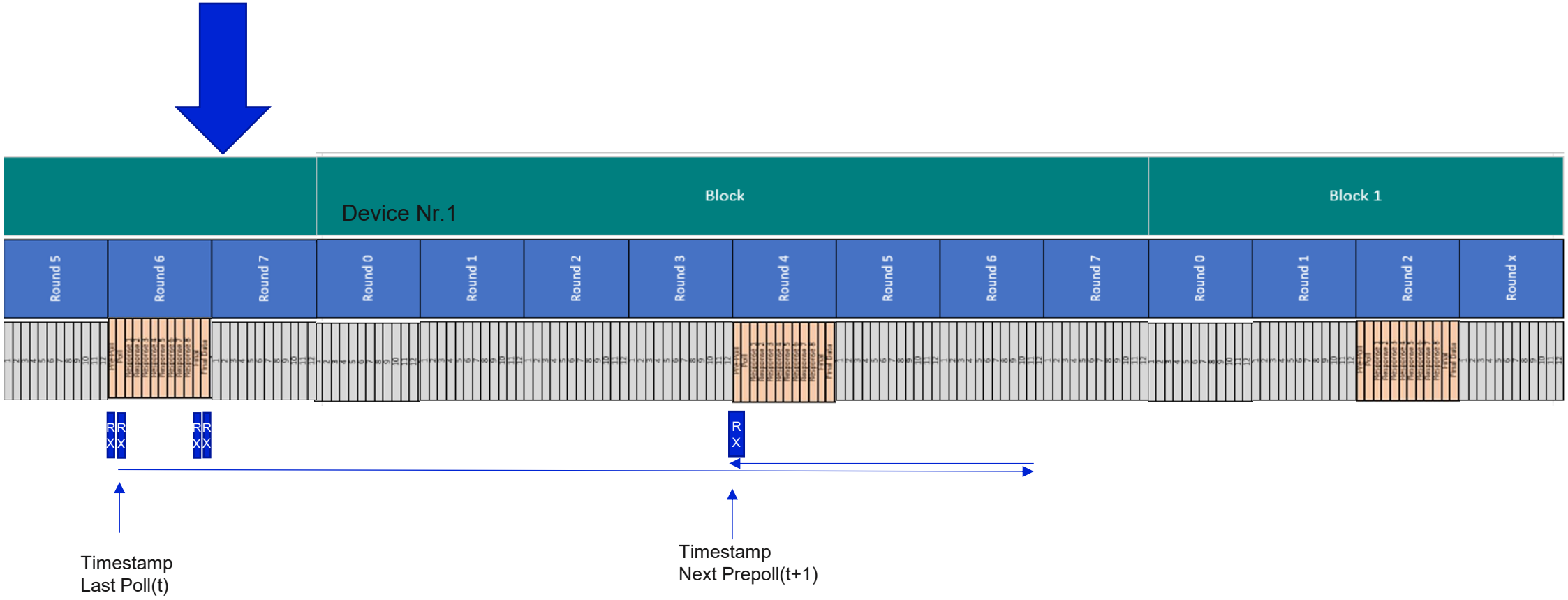
Window Calulation with PrePoll Timestmap



Window Calulation with Final Data Timestmap

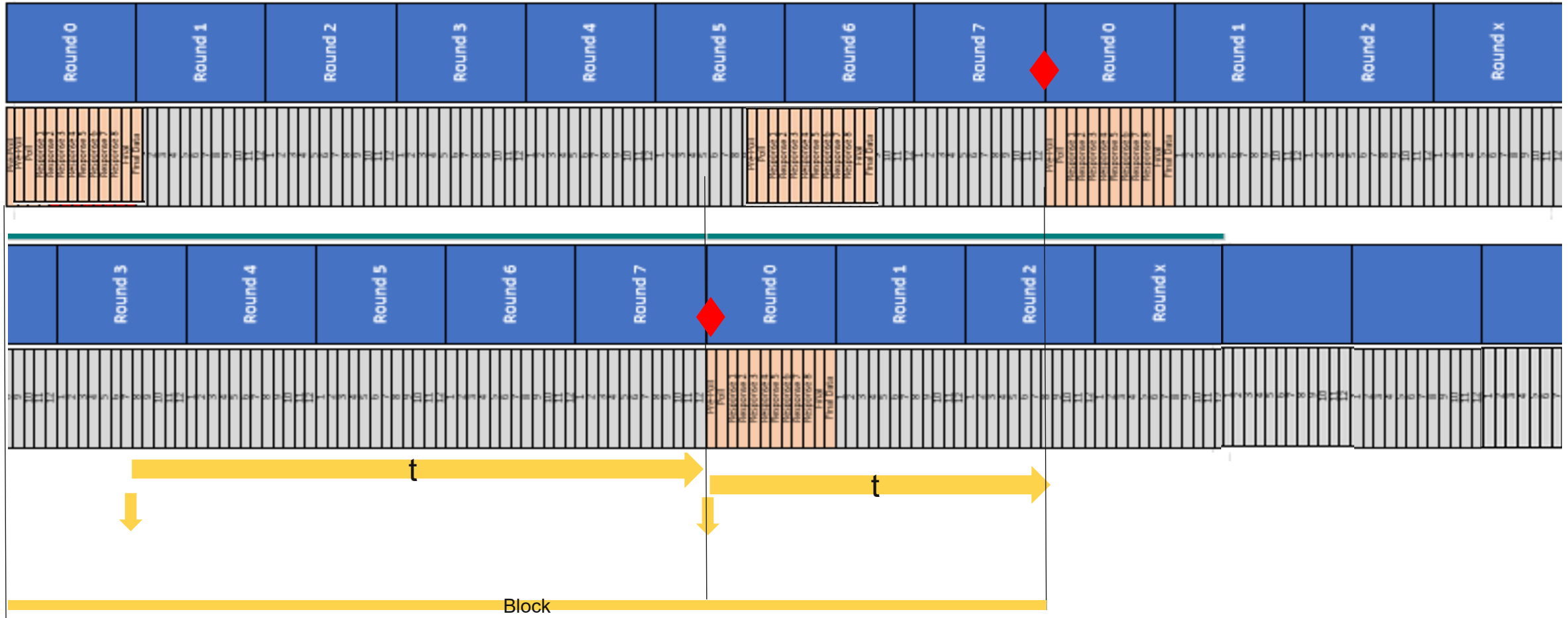


Window Calulation with Final Data Timestmap



2 Smartphones

2 Smartphone (Start Ranging ein mal aufgerufen)



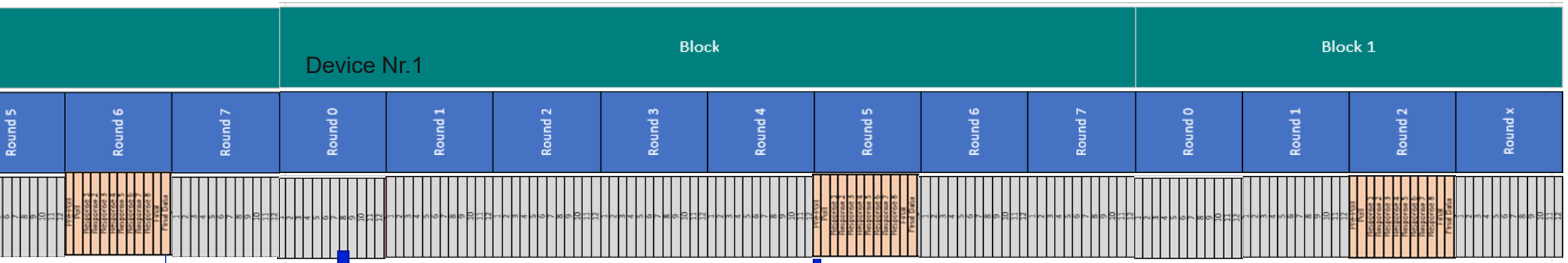
1: TX PrePoll

2: TX PrePoll

1: TX PrePoll

Valid for IN_SYNC_EXT_INBAND

Read System Time: GetSystemTime()



Step1
Calculated RX-Frame-Timestamp

Read NXP Timestamp = timestamp of last Systemtimesync

DeltaTimeFDD

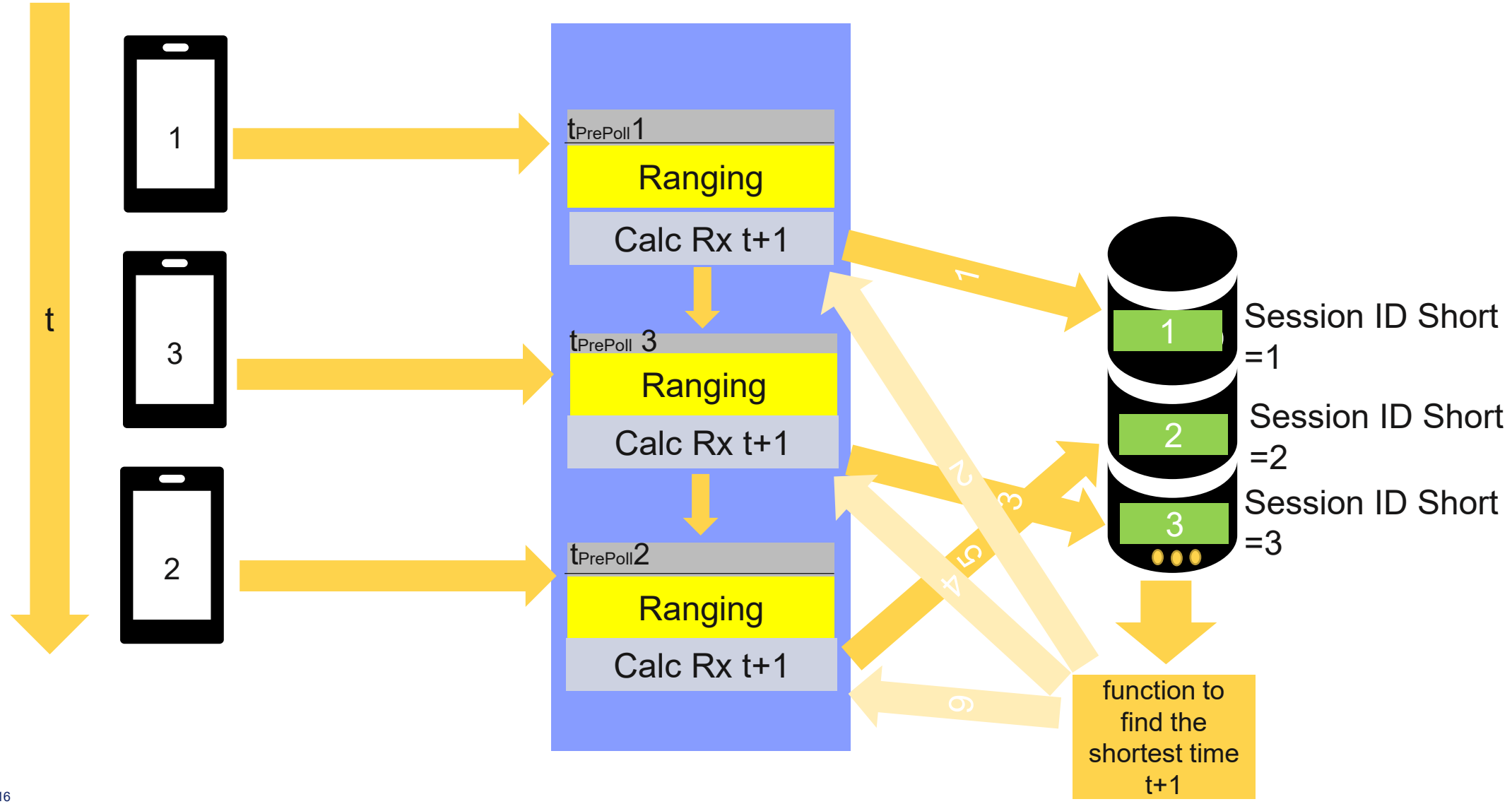
UWB

FDD

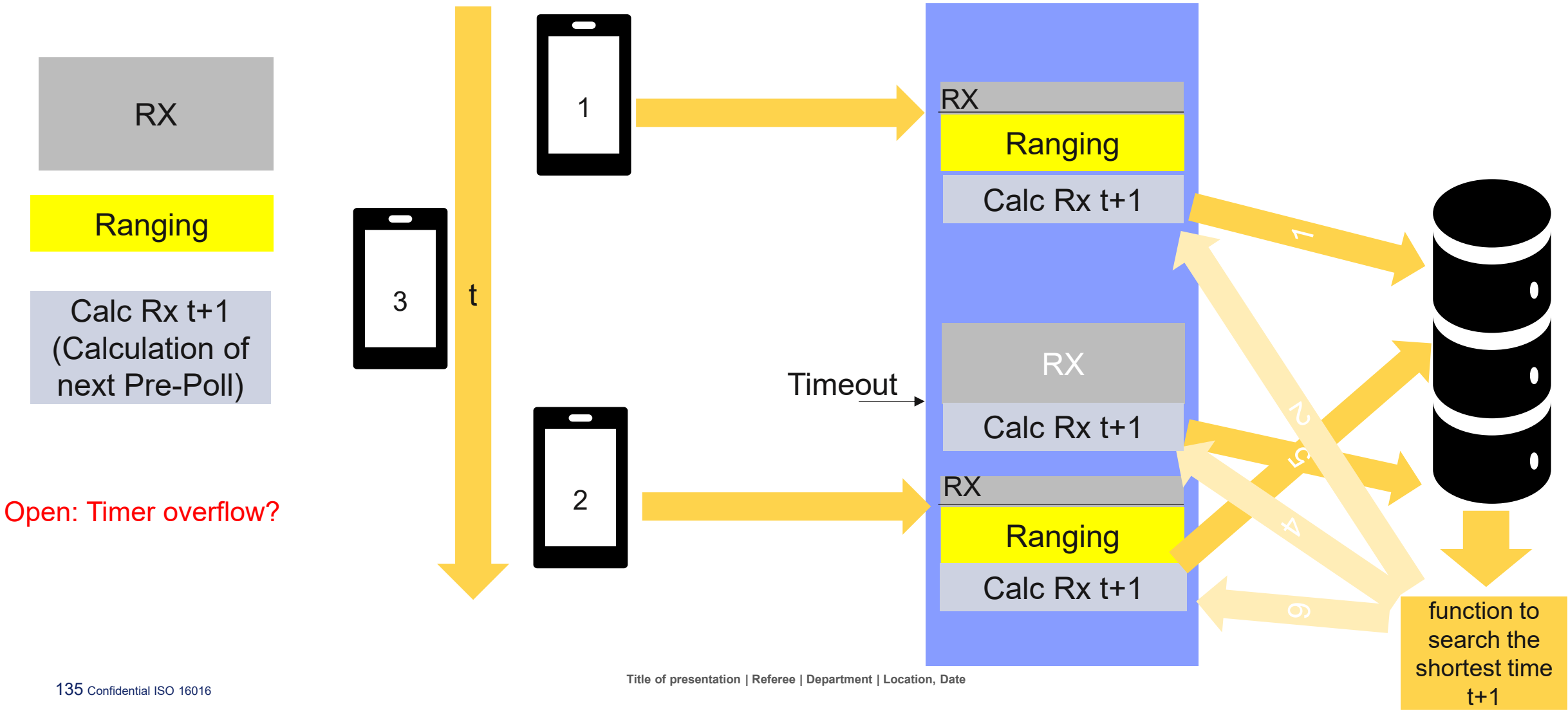
SYNC

TSync=24

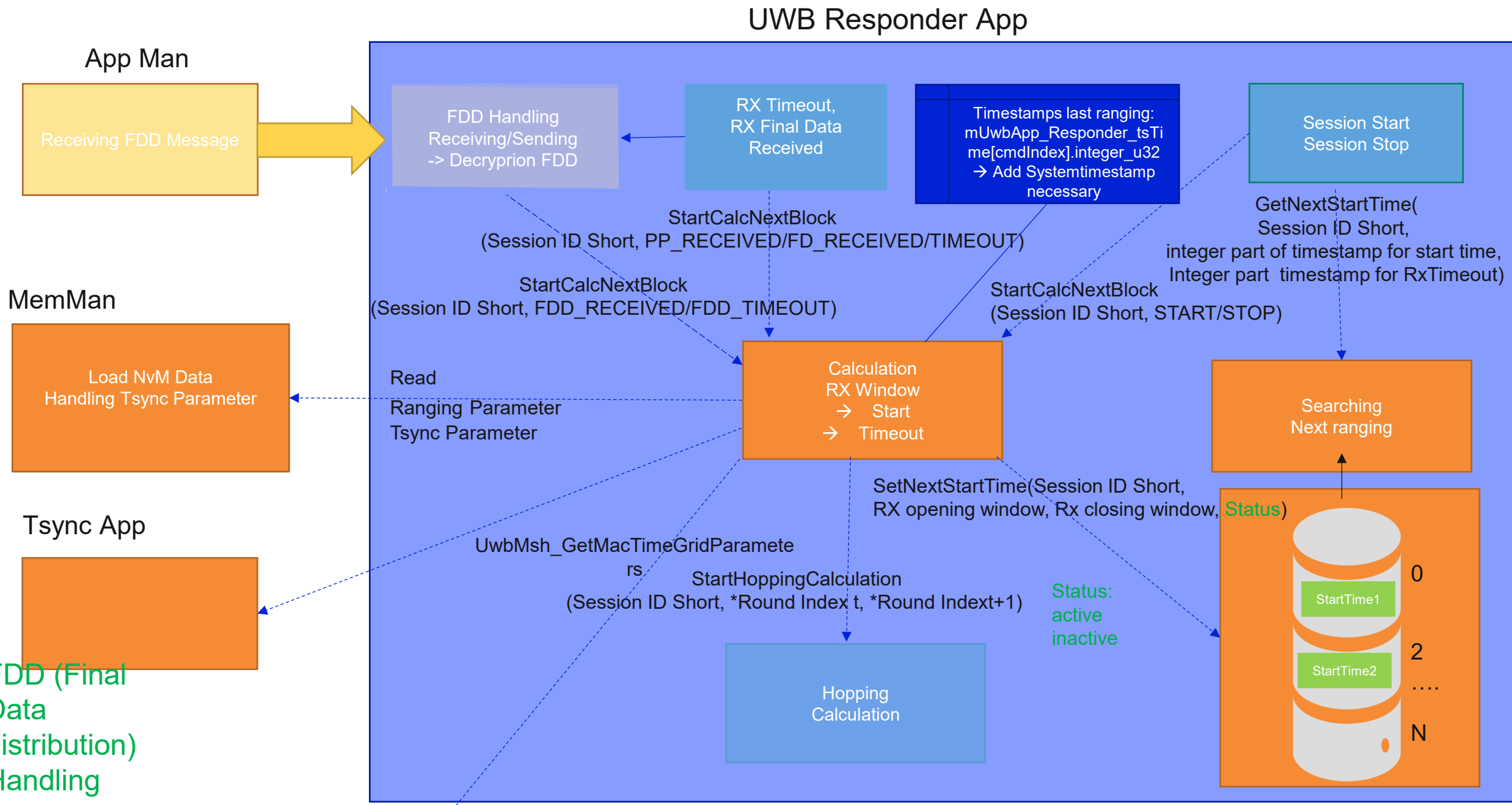
Ranging with 3 Smartphones



Ranging with 3 Smartphones and Ranging only with 2 Smartphones possible



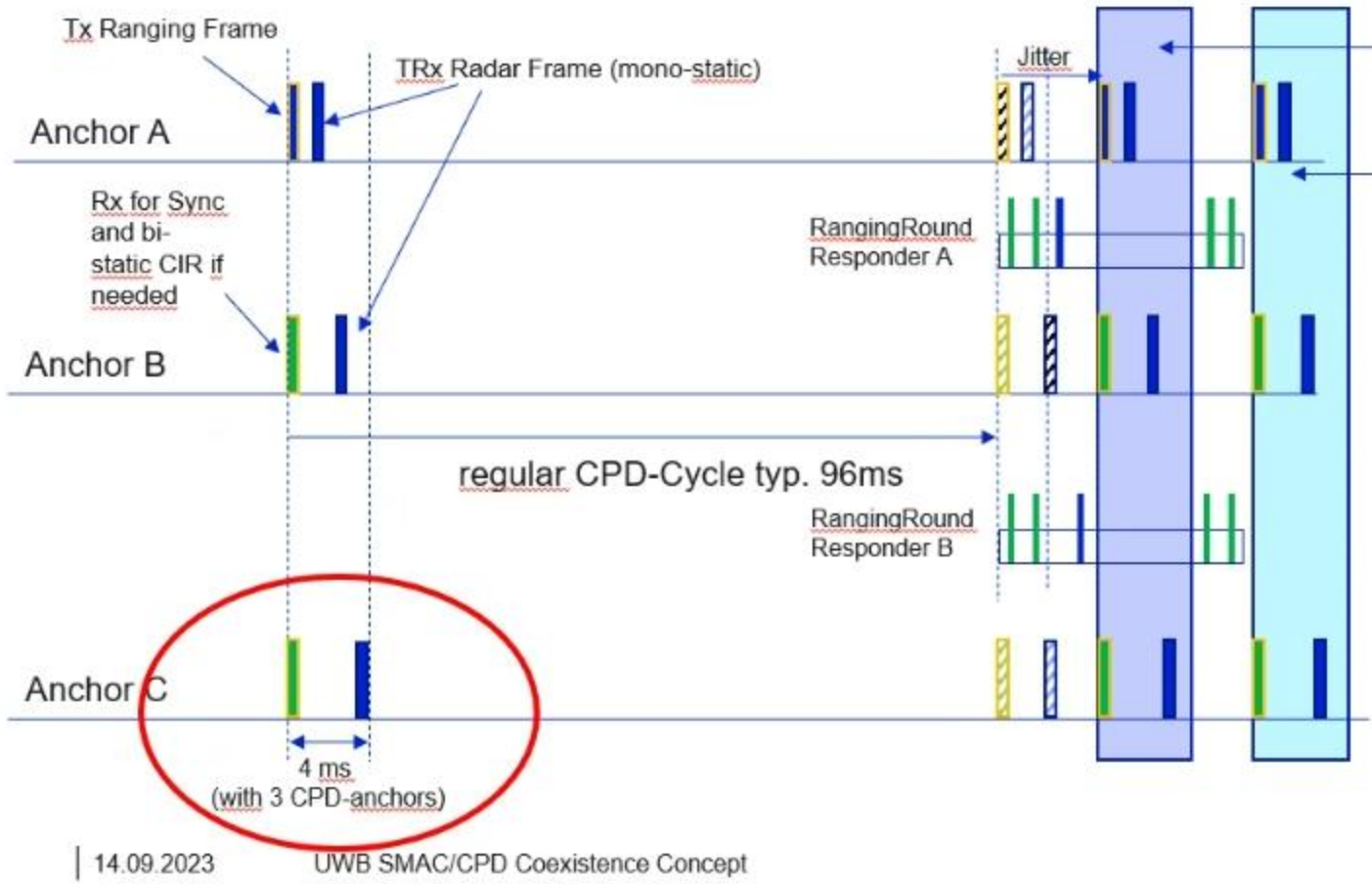
Workpackages Uwb WindowCalculation



BMW FBD6 Coexistence Concept

Three FBD6 scenario

CPD own cycle, delayed If needed to next free gap in ranging rounds



Case 1: CPD delay into Gap for 6 responses in ranging round >= 6ms

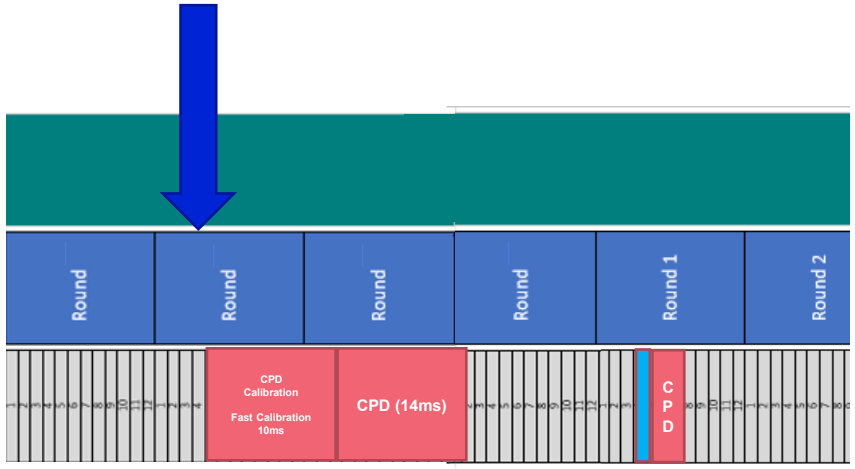
Case 2: delay after FinalData message

Assumptions:

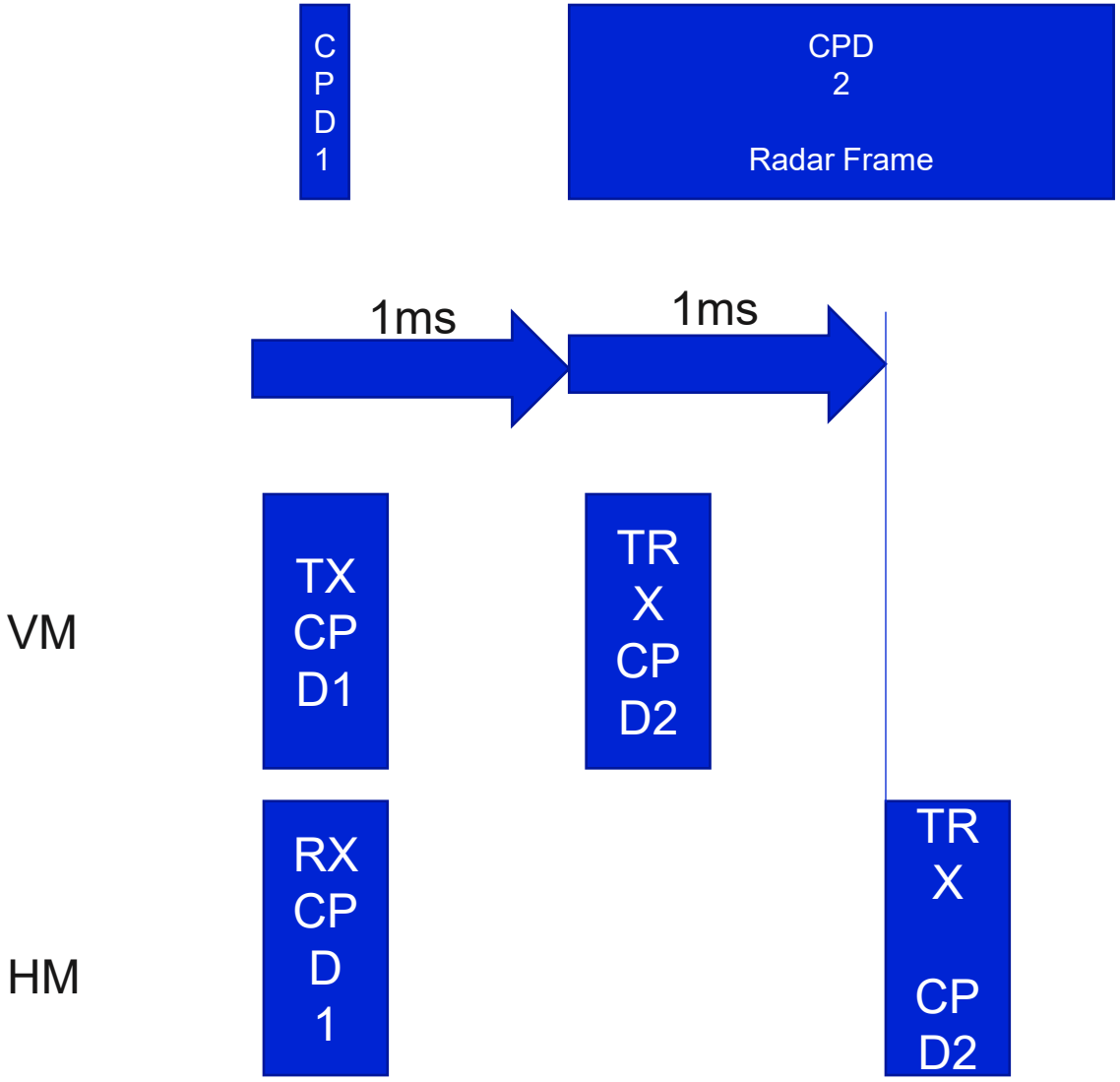
- CPD-cycle mutiple of 96 ms (CCC)
- Fastest CPD-Slot time 1ms
- DSP-Blocking for Radar ~300µs
- RadarFrame <= 500 µs
- 8 Responder in RangingRound
- CPD-Nodes are Responder #1 and #2
- Continous Hopping for RangingSession is active by default as requested in CCC

CPD Start via CAN

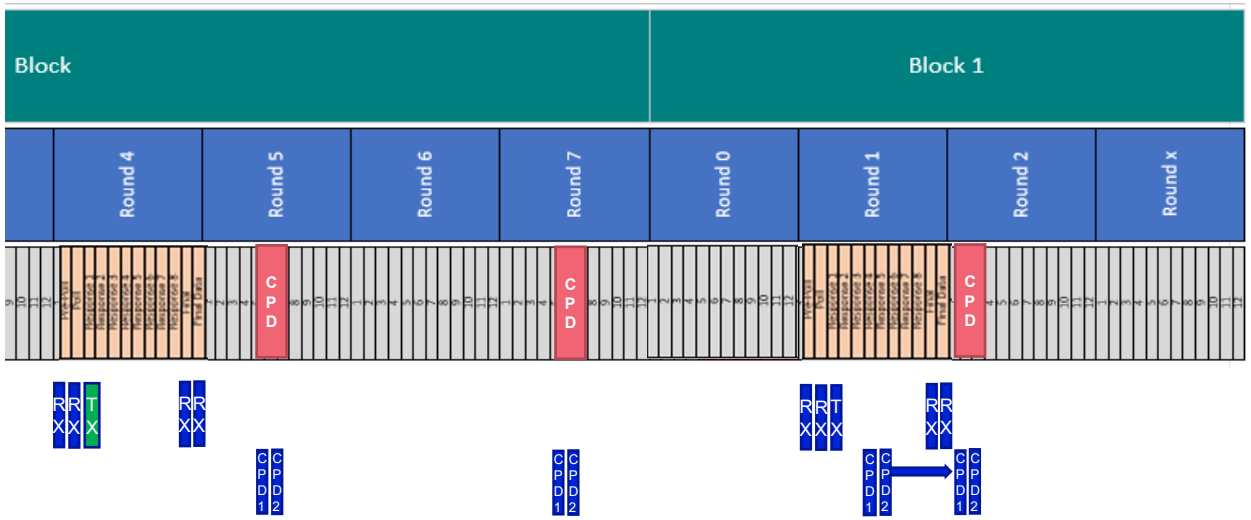
CAN Message „CPD Start“



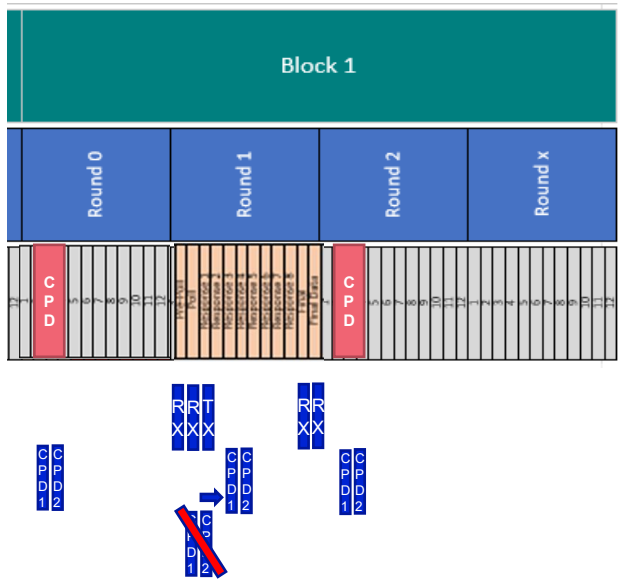
Cycle time:24(48/96)ms


CPD 1
CPD 2
Radar Frame
1ms
1ms
VM
HM
TX CPD1
TRX CPD2
RX CPD 1
TRX CPD2

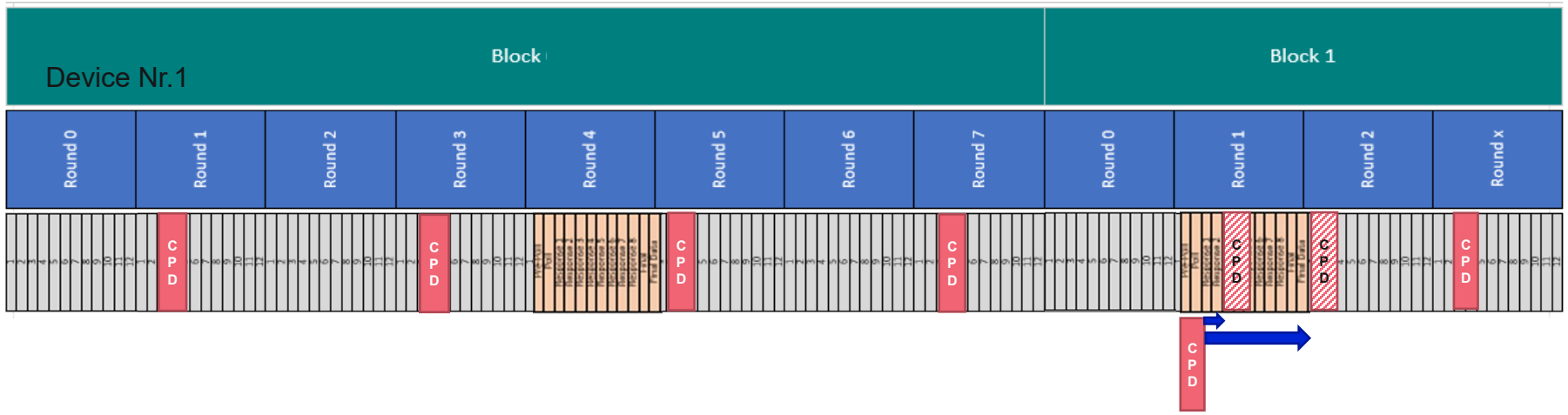
anging+CPD


Block
Block 1
Round 4
Round 5
Round 6
Round 7
Round 0
Round 1
Round 2
Round x
CPD
RRTX
RRXX
CPD1
CPD2

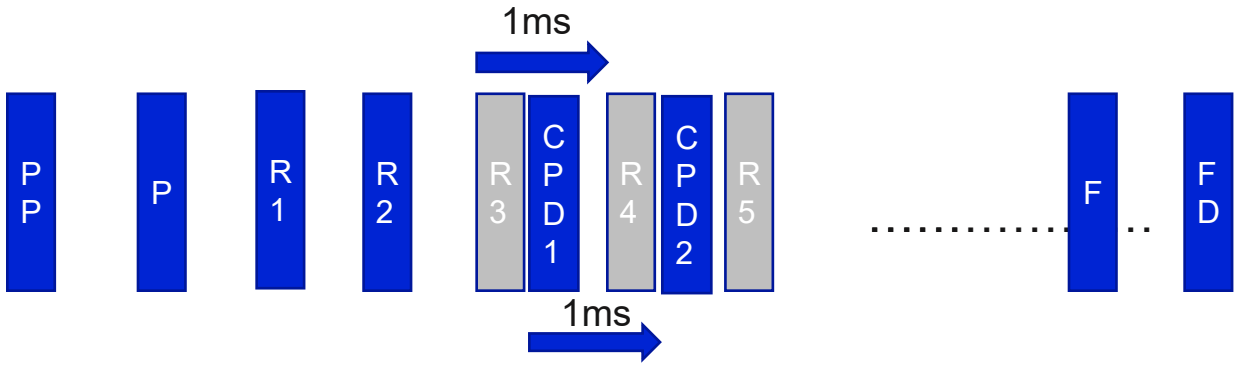
Ranging+CPD


Block 1
Round 0
Round 1
Round 2
Round x
C P D
C P D
R R X X
T X
R R X X
C P D 1
C P D 2
C P D 1
C P D 2
C P D 1
C P D 2

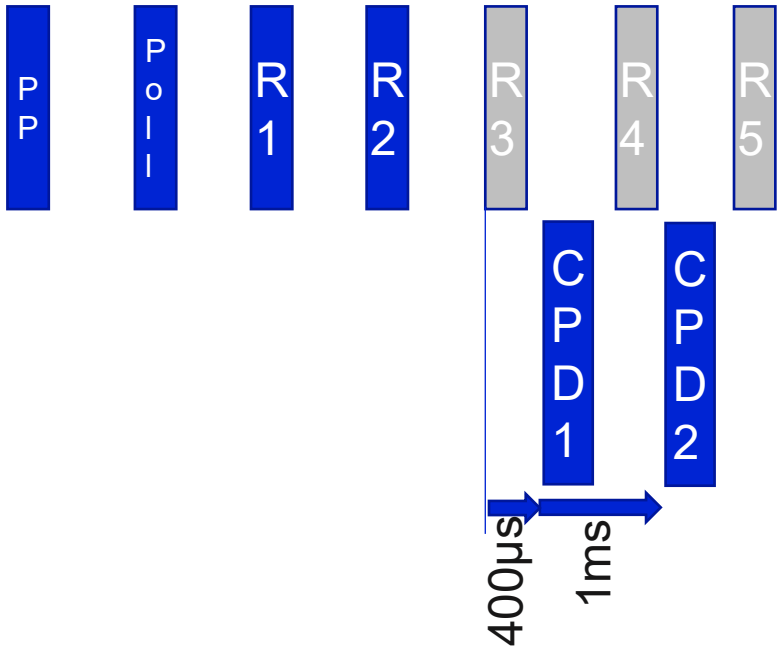
Window Calulation with CPD



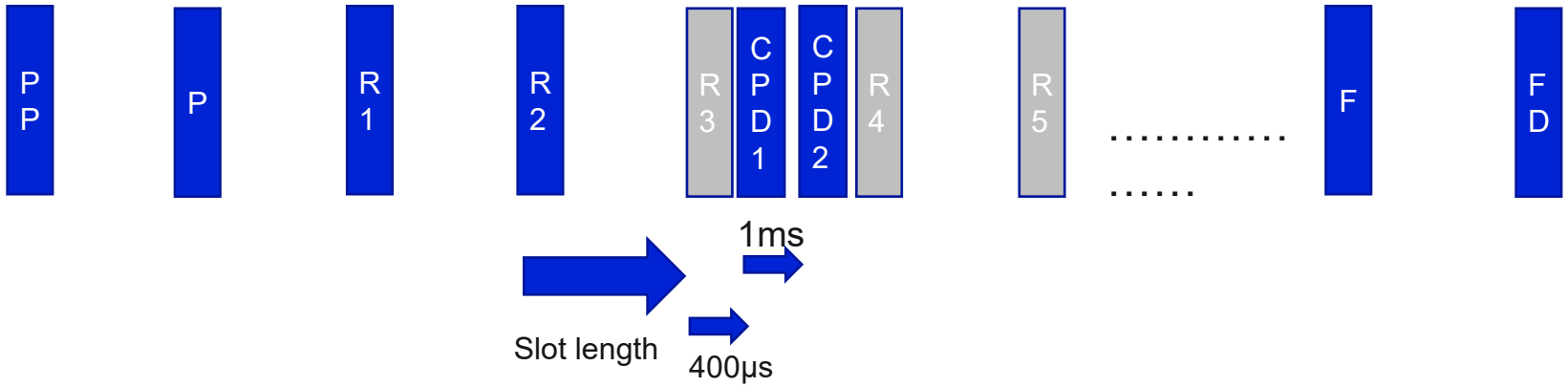
e.g. Slot length = 1ms



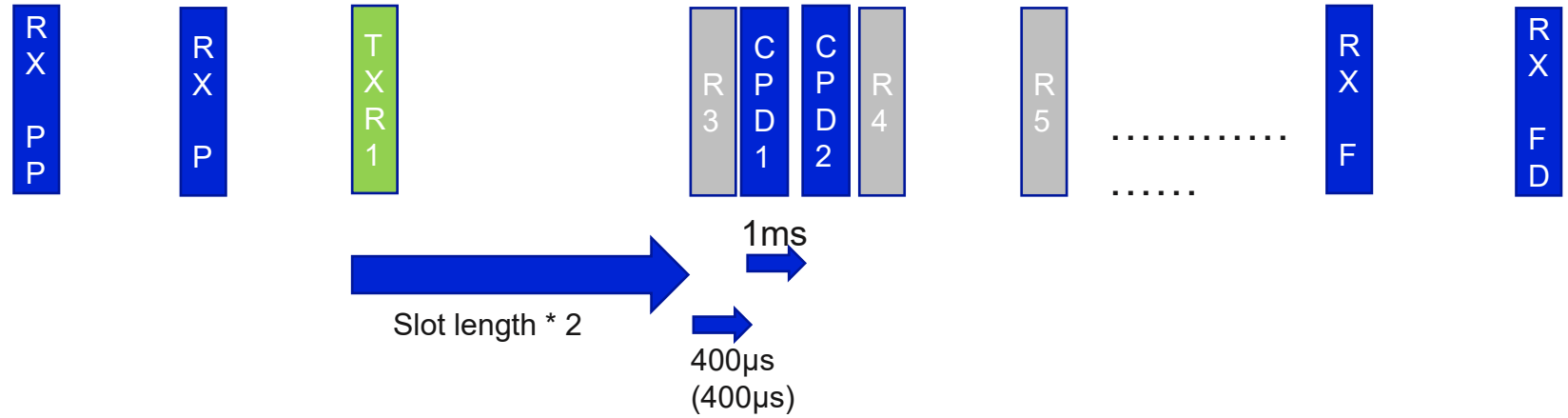
e.g. Slot length = 1ms



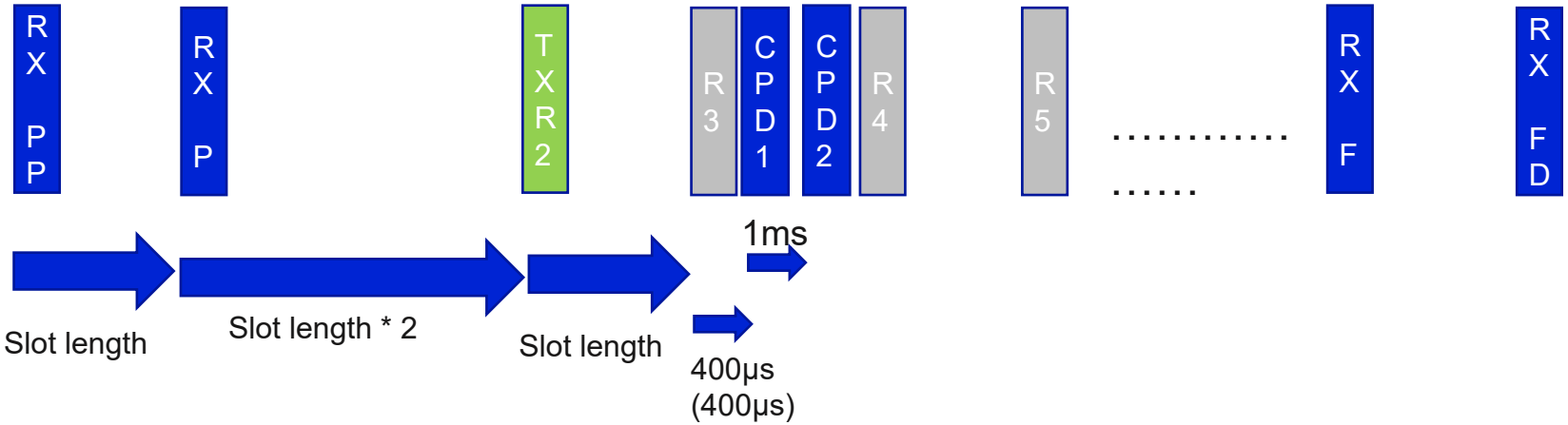
Slot Length >=2ms



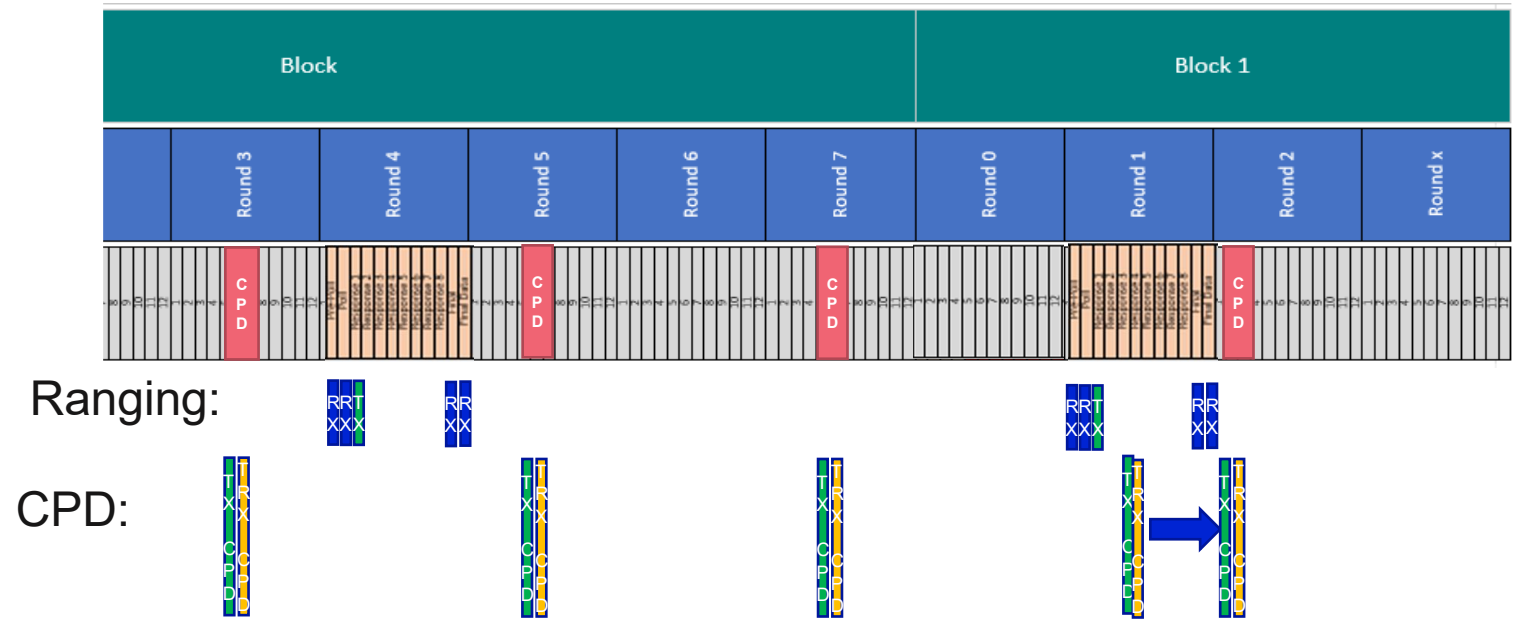
Responder 1:



Responder 2:



CPD Start via CAN



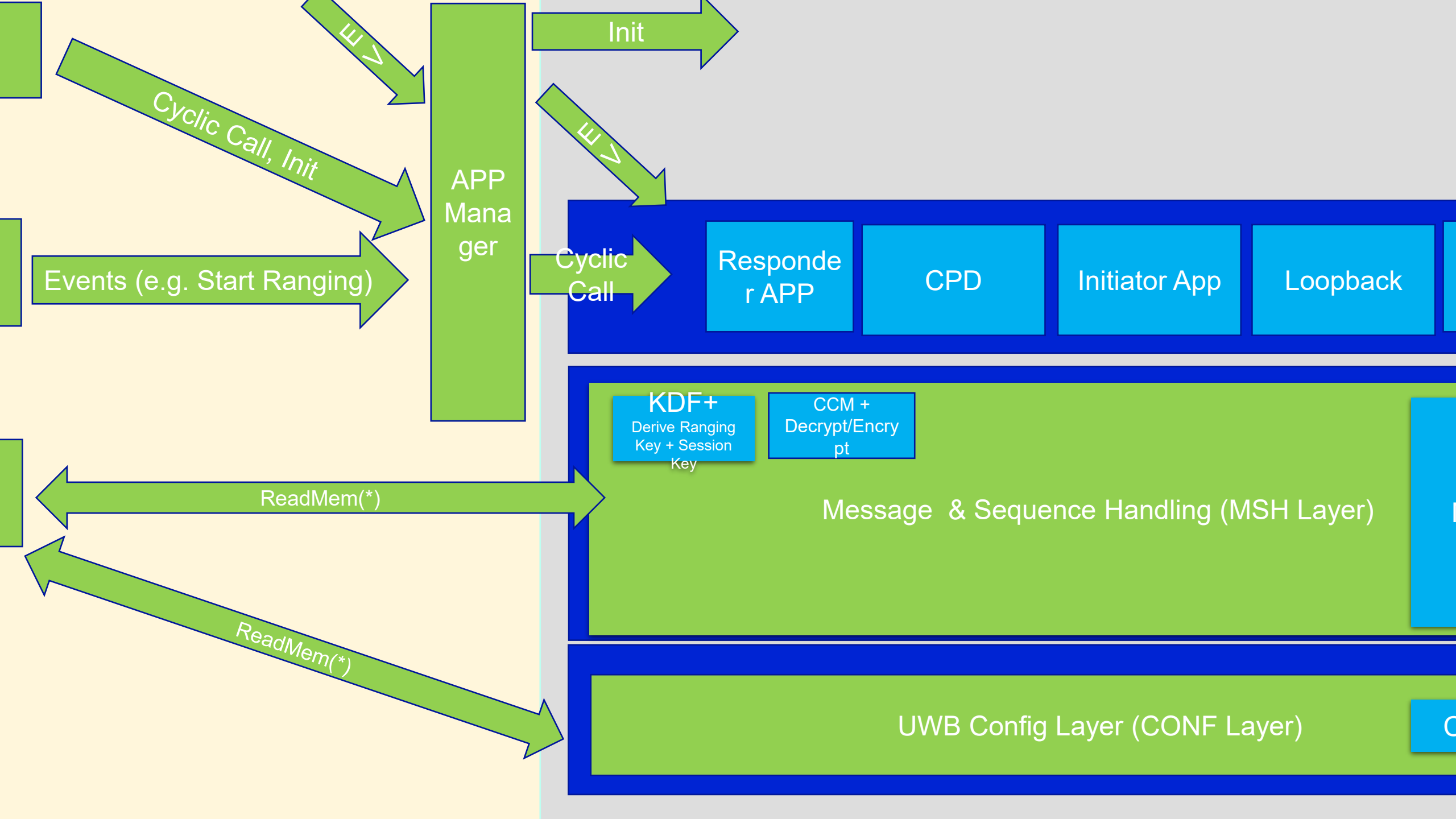
Msh Start

VM

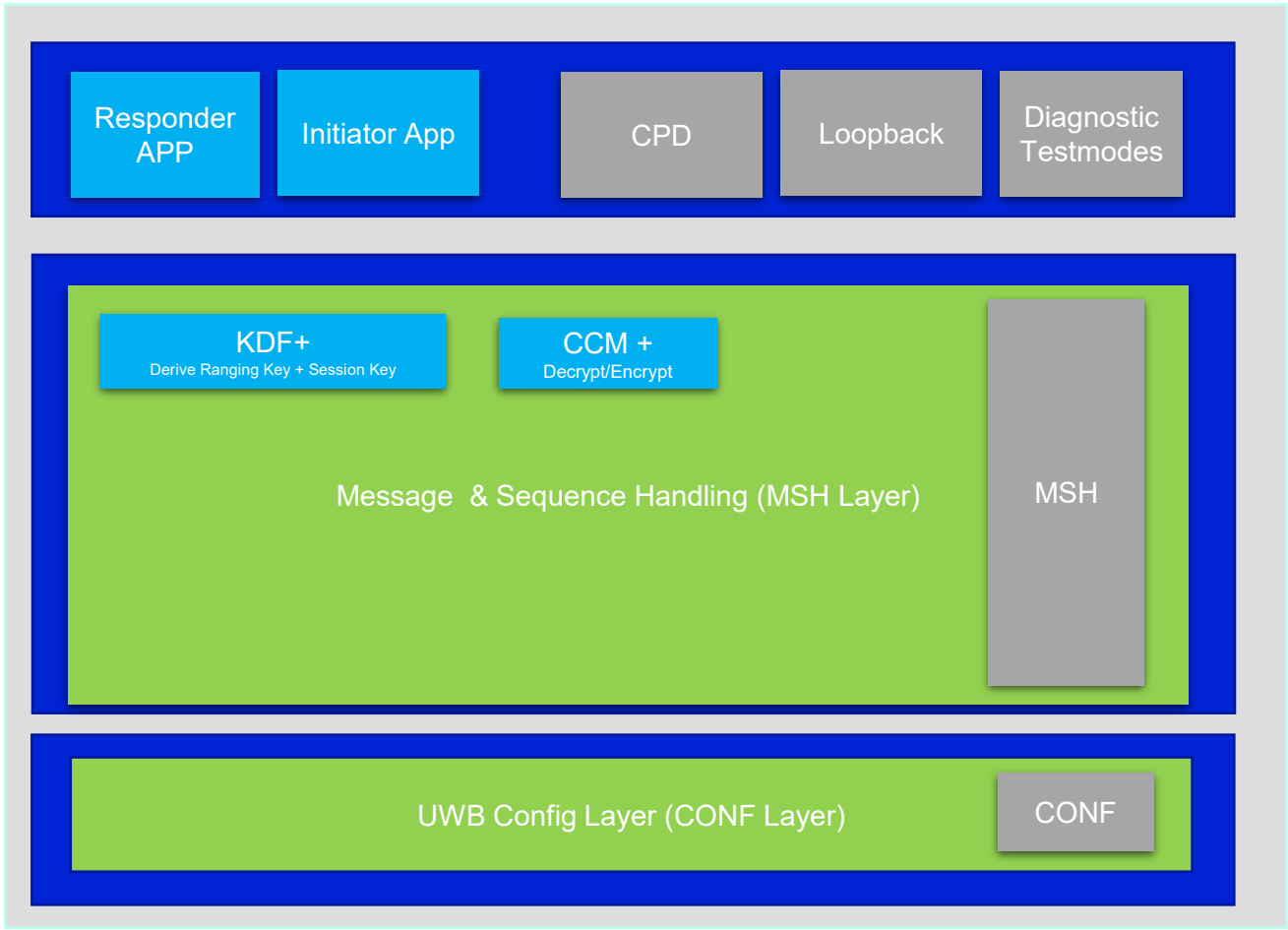
	PrePoll	Poll	Response	CPD1	CPD2	Final	Final Data
Cmd	RX	RX	TX	TX	TRX	RX	RX
rxCtrlIdx	2	1	Not used (0)	Not used (0)	0	1	3
txCtrlIdx	Not used (0)	Not used (0)	1	1	2	Not used (0)	Not used (0)
Frame Type	Data Frame	Ranging Frame	Ranging Frame	Ranging Frame	Radar Frame	Ranging Frame	Data Frame

HM

	PrePoll	Poll	Response	CPD1	CPD2	Final	Final Data
Cmd	RX	RX	TX	RX	TRX	RX	RX
rxCtrlIdx	2	1	Not used (0)	1	0	1	3
txCtrlIdx	Not used (0)	Not used (0)	1	Not used (0)	2	Not used (0)	Not used (0)
Frame Type	Data Frame	Ranging Frame	Ranging Frame	Ranging Frame	Radar Frame	Ranging Frame	Data Frame

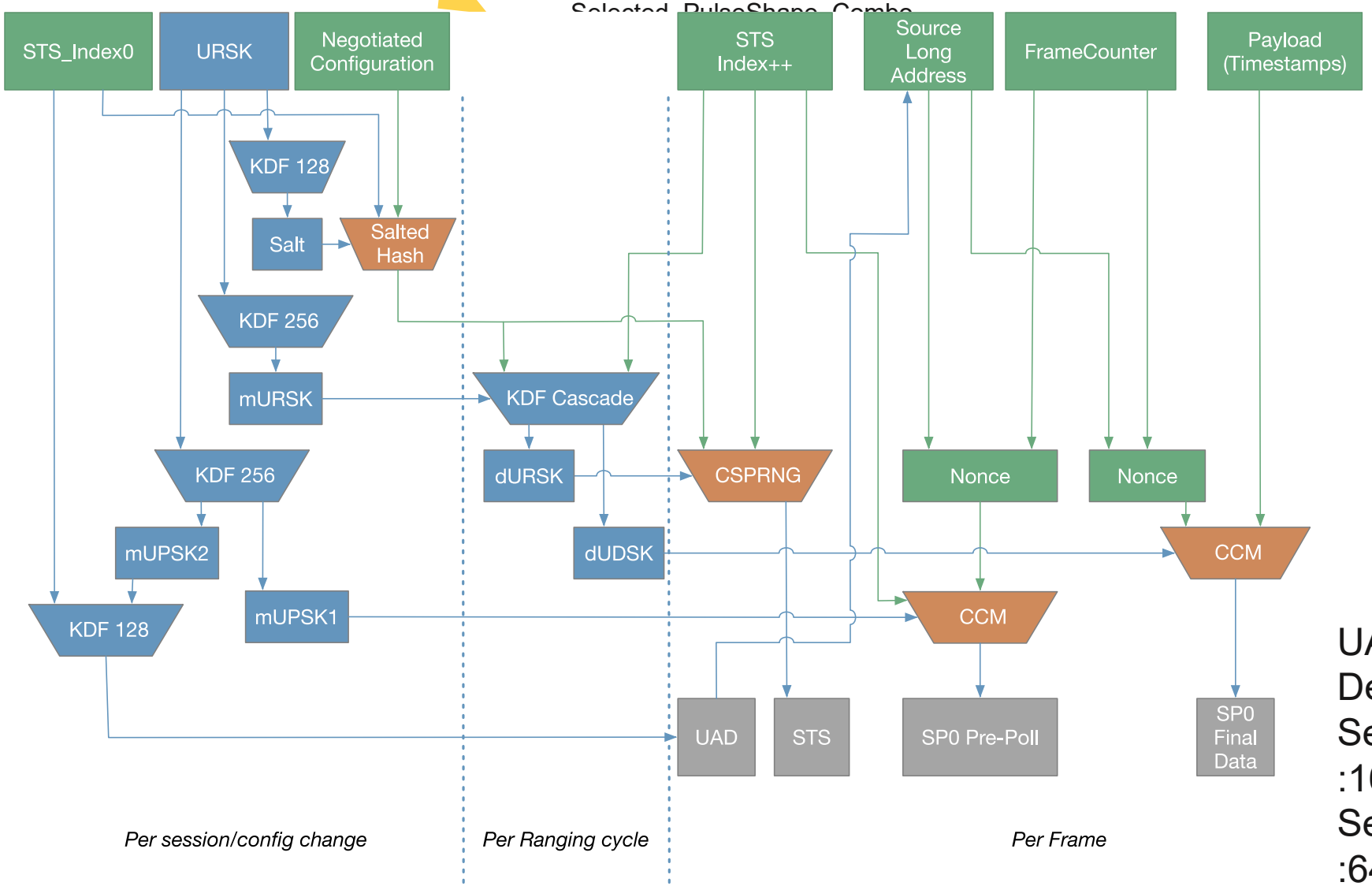

Init
Cyclic Call, Init
APP Manager
Events (e.g. Start Ranging)
Cyclic Call
Responder APP
CPD
Initiator App
Loopback
KDF+
Derive Ranging Key + Session Key
CCM + Decrypt/Encrypt
Message & Sequence Handling (MSH Layer)
ReadMem(*)
ReadMem(*)
UWB Config Layer (CONF Layer)

UWB Security



KDF

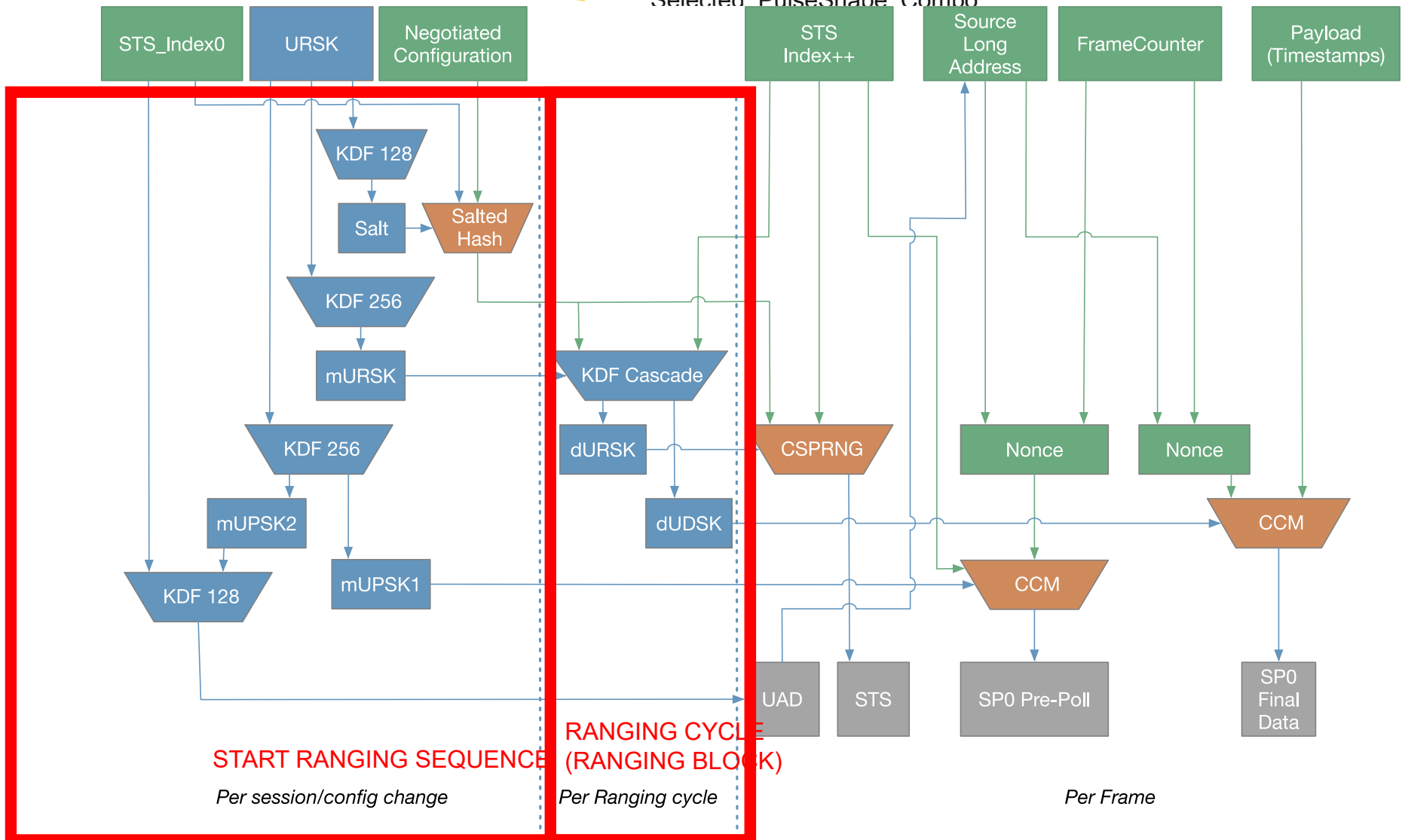
Selected_Protocol_Version ||
Selected_UWB_Config_Id ||
UWB_Session_Id || STS_Index0 ||
Number_Responder_Nodes ||
Session_RAN_Multiplier ||
Number_Slot_per_Round ||
Number_Chaps_per_Slot ||
Selected_PulseShape_Combo



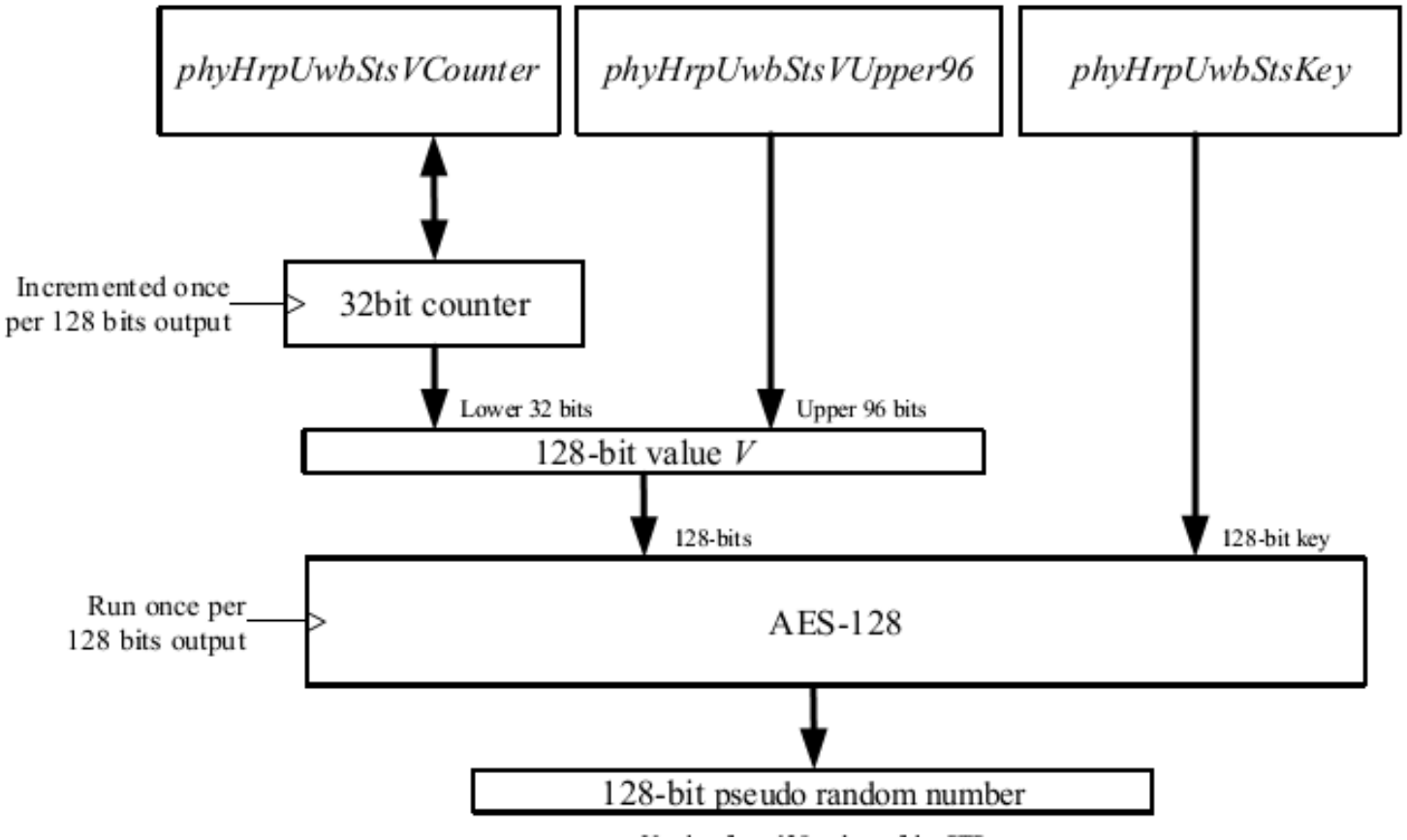
UAD: DestinationShortAddress[0:16] = SetUpperBitToZeroIfReserved(UAD[80:16]) SourceLongAddress[0:64] = SetUpperBitToZeroIfReserved(UAD[16:64])

KDF

Selected_Protocol_Version ||
Selected_UWB_Config_Id ||
UWB_Session_Id || STS_Index0 ||
Number_Responder_Nodes ||
Session_RAN_Multiplier ||
Number_Slot_per_Round ||
Number_Chaps_per_Slot ||
Selected_PulseShape_Combo



STS



phyHrpUwbStsVCounter = SaltedHash[0:32]

phyHrpUwbStsVUpper96 = SaltedHash([64:64])|| ((SaltedHash[32:32]+STS_Index) & 0xffffffff)

phyHrpUwbStsKey = dURSK[0:128]

Rules for STS incrementation

The following rules shall be used for incrementing the STS index:

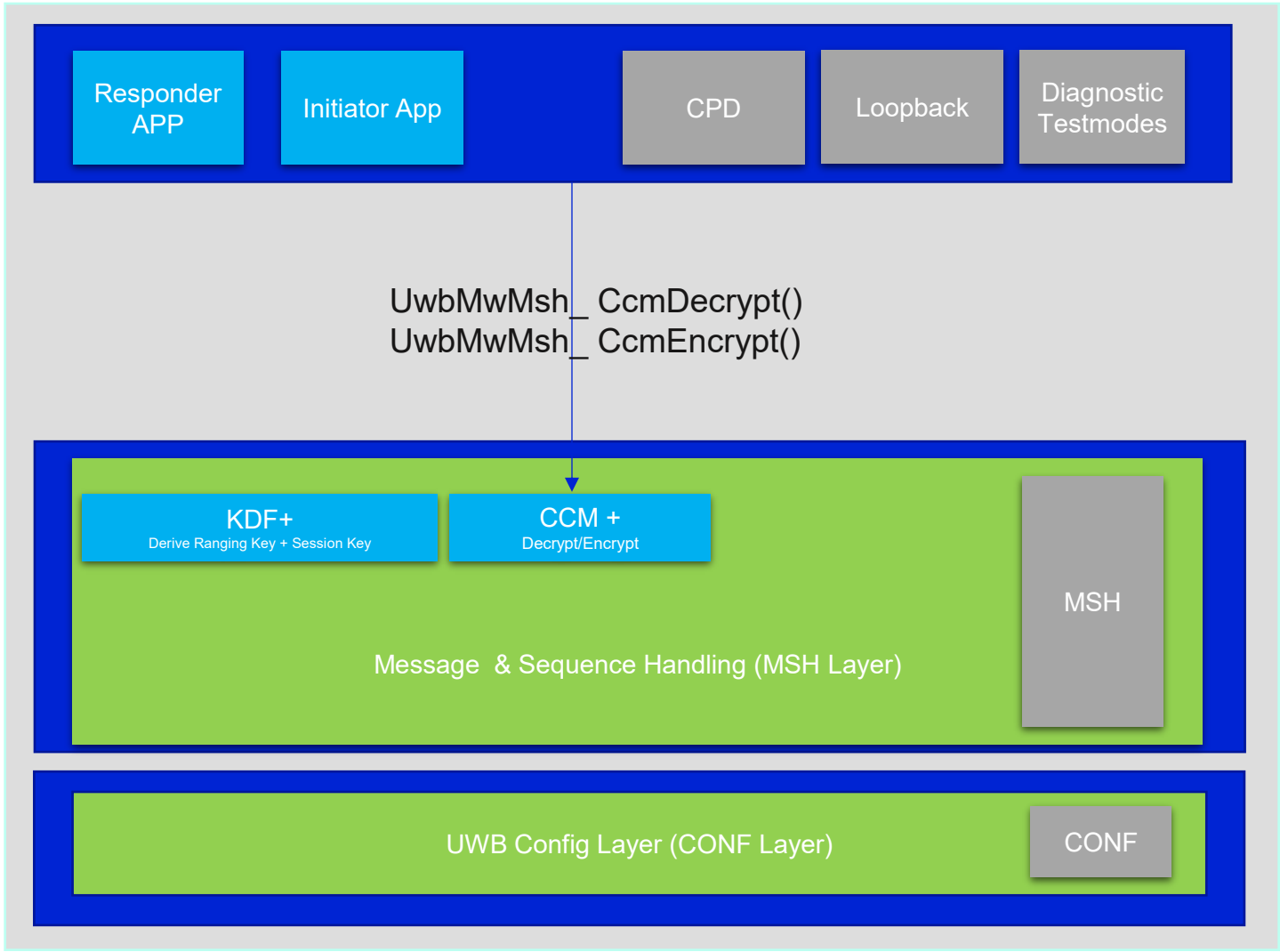
1. The STS index is managed separately for each ranging session of a given RAN.
2. A *k*-th ranging session starts at time UWB_{time0}^{k}.
3. The STS index of the k-th ranging session is referenced to the start of this ranging session:

   STS_Index0^{k} is mapped to ranging round 0 of ranging block 0. That is, for $i = 0$ and $Round_Idx^{k}(i = 0) = 0$

   $$STS_Index^{k}\left(i = 0, Round_Idx^{k}(i = 0) = 0\right) = STS_Index0^{k}$$

4. Within a ranging round, the STS index increments in every slot even if that slot is unmapped (i.e. not used).
5. The incrementation scheme is applied for every ranging round (even if it is not used). This results in:
   a. The STS index is incremented by $N_{Slot_per_Round}^{k}$ in every ranging round.
   b. The STS index is incremented by $N_{Slot_per_Round}^{k} \times N_{Round}^{k}$ in every ranging block.
6. The STS index is incremented for all slots in the round. However, data packets (Pre-POLL and Final_Data) reference the STS index in their payload as follows:
   a. Pre-POLL message shall carry the reference to the STS index that is used in the following POLL message.
   b. Final_Data message shall refer to the STS index of the preceding FINAL message.

CCM Architecture



UWB Config

DCK protocol			M/O	STS Packet Type 3 (SP3) IEEE 802.15.4z parameters				STS Packet Type 0 (SP0) IEEE 802.15.4z parameters						
UWB configuration	UWB Channel (IEEE 802.15.4)	SYNC preamble code (IEEE 802.15.4)	Mandatory/ Optional	correspondin IEEE BPRF Mandoary Set	phyHrpUwbPsr	SFD (phyHrpUwbSfdSelector)	STS (segment x symbols)	corresponding IEEE802.15.4z BPRF Set	phyHrpUwbPsr	SFD (phyHrpUwbSfdSelector)	STS	phyHrpUwbPhrDataRate	PHR bit rate	PSDU bit rate
0	5, 9	9,10,11,12 Optional: 13-16, 21-24	M	N/A	64	ternary (0)	1x 64	BPRF #1	64	ternary (0)	no	DRBM_LP	0.85 Mbps	6.81 Mbps
1	5, 9	9,10,11,12 Optional: 13-16, 21-24	Device: M Vehicle: O	BPRF #4	64	binary (2)	1x 64	BPRF #2	64	binary (2)	no	DRBM_LP	0.85 Mbps	6.81 Mbps

Table 21-1: Supported UWB Configurations

BPRF set #	SYNC PSR	SFD # per Table 15-7c	SFD length	STS number of segments	STS segment length (units of 512 chips)	PHR + data	Data rate (Mb/s)	Description
1	64	0	8	0	n/a	Yes	6.8	STS packet configuration 0: SHR, PHR and Data.
2	64	2	8	0	n/a	Yes	6.8	
3	64	2	8	1	64	Yes	6.8	STS packet configuration 1: SHR, STS, PHR and Data.
4	64	2	8	1	64	No	n/a	STS packet configuration 3: SHR, STS only.

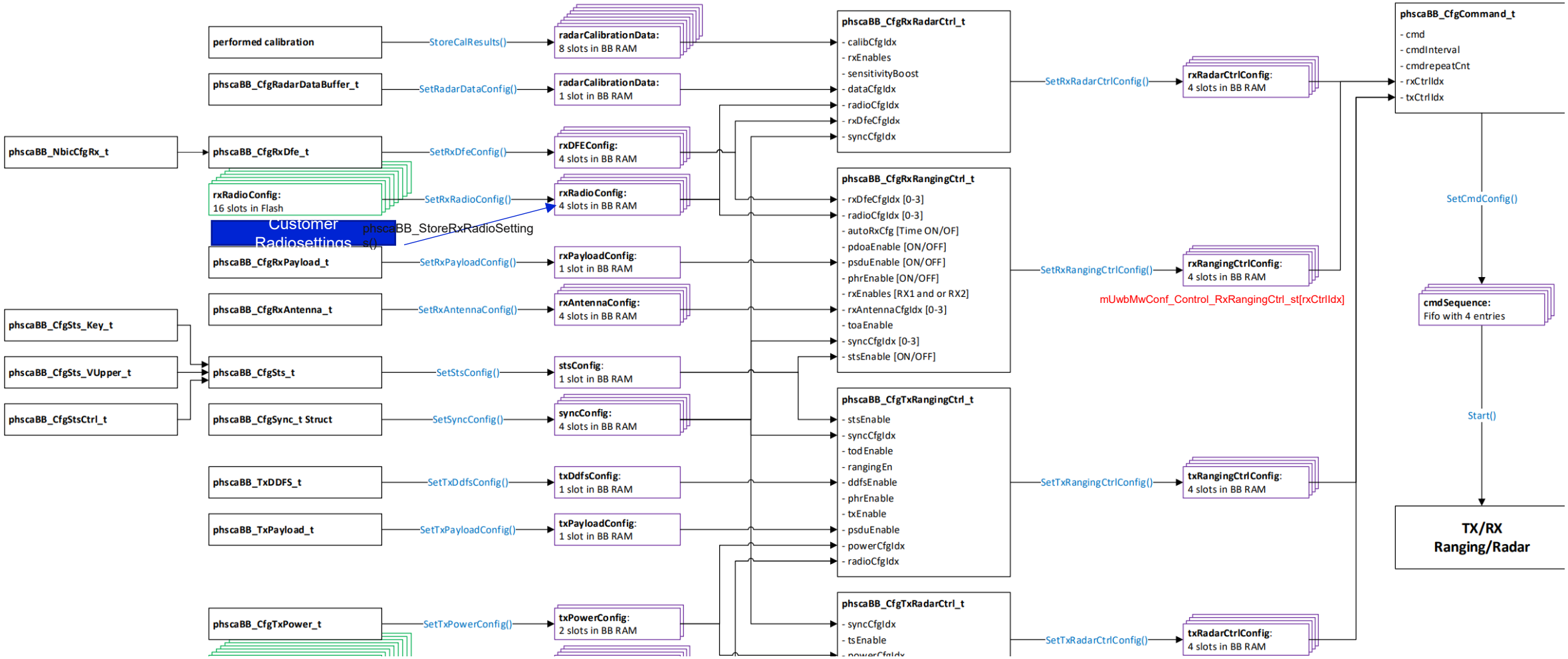
UWB Config 0/1
Unterscheidet sich durch die verwendete SFD
In In Config 0 sind hier 0-Werte (*4) enthalten.
Config 1 SFD enthält keine 0-Werte. Wodurch das Ranging eine bessere Performance hat.

Laut Renke: Beim iPhone kann man einstellen ob Config 0 oder Config 1 verwendet werden soll.
Samsung verwendet aktuell nur Config 0

UwbMwConf_ConfigureBaseBand

Slot	Scenario	Description
RX		
0	Rx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Rx_Sc406	High Performance (min phase 1 GHz)
2	Rx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Rx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Rx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Rx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Rx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Rx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Rx_Sc100	IEEE 6.8 (min phase 1 GHz)
9	Rx_Sc131_lin_phase_1G	ECO Std (lin phase 1 GHz)
TX		
0	Tx_Sc100_lin_phase_preamble_9	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 9
1	Tx_Sc406	High Performance (min phase 1 GHz)
2	Tx_Sc131_1G	ECO Std (min phase 1 GHz)
3	Tx_Sc136_1G	ECO Min (min phase 1 GHz)
4	Tx_Sc100_lin_phase_preamble_10	IEEE 6.8 (lin phase 1 GHz) with IEEE preamble code index 10
5	Tx_Sc131_500M	ECO Std (min phase 500 MHz)
6	Tx_Sc136_500M	ECO Min (min phase 500 MHz)
7	Tx_Sc131_kaiser_lin_phase_1G	ECO Std for radar (Gaussian, lin phase 1 GHz)
8	Tx_Sc100	IEEE 6.8 (min phase 1 GHz)

?


performed calibration
StoreCalResults()
radarCalibrationData:
8 slots in BB RAM
phscaBB_CfgRadarDataBuffer_t
SetRadarDataConfig()
radarCalibrationData:
1 slot in BB RAM
phscaBB_NbicCfgRx_t
phscaBB_CfgRxDfe_t
SetRxDfeConfig()
rxDFEConfig:
4 slots in BB RAM
rxRadioConfig:
16 slots in Flash
SetRxRadioConfig()
rxRadioConfig:
4 slots in BB RAM
Customer Radiosettings
phscaBB_StoreRxRadioSetting
s()
phscaBB_CfgRxPayload_t
SetRxPayloadConfig()
rxPayloadConfig:
1 slot in BB RAM
phscaBB_CfgRxAntenna_t
SetRxAntennaConfig()
rxAntennaConfig:
4 slots in BB RAM
phscaBB_CfgSts_Key_t
phscaBB_CfgSts_VUpper_t
phscaBB_CfgStsCtrl_t
phscaBB_CfgSts_t
SetStsConfig()
stsConfig:
1 slot in BB RAM
phscaBB_CfgSync_t Struct
SetSyncConfig()
syncConfig:
4 slots in BB RAM
phscaBB_TxDDFS_t
SetTxDdfsConfig()
txDdfsConfig:
1 slot in BB RAM
phscaBB_TxPayload_t
SetTxPayloadConfig()
txPayloadConfig:
1 slot in BB RAM
phscaBB_CfgTxPower_t
SetTxPowerConfig()
txPowerConfig:
2 slots in BB RAM
phscaBB_CfgRxRadarCtrl_t
- calibCfgIdx
- rxEnables
- sensitivityBoost
- dataCfgIdx
- radioCfgIdx
- rxDfeCfgIdx
- syncCfgIdx
SetRxRadarCtrlConfig()
rxRadarCtrlConfig:
4 slots in BB RAM
phscaBB_CfgRxRangingCtrl_t
- rxDfeCfgIdx [0-3]
- radioCfgIdx [0-3]
- autoRxCfg [Time ON/OF]
- pdoaEnable [ON/OFF]
- psduEnable [ON/OFF]
- phrEnable [ON/OFF]
- rxEnables [RX1 and or RX2]
- rxAntennaCfgIdx [0-3]
- toaEnable
- syncCfgIdx [0-3]
- stsEnable [ON/OFF]
SetRxRangingCtrlConfig()
rxRangingCtrlConfig:
4 slots in BB RAM
mUwbMwConf_Control_RxRangingCtrl_st[rxCtrlIdx]
phscaBB_CfgTxRangingCtrl_t
- stsEnable
- syncCfgIdx
- todEnable
- rangingEn
- ddfsEnable
- phrEnable
- txEnable
- psduEnable
- powerCfgIdx
- radioCfgIdx
SetTxRangingCtrlConfig()
txRangingCtrlConfig:
4 slots in BB RAM
phscaBB_CfgTxRadarCtrl_t
- syncCfgIdx
- tsEnable
SetTxRadarCtrlConfig()
txRadarCtrlConfig:
4 slots in BB RAM
phscaBB_CfgCommand_t
- cmd
- cmdInterval
- cmdrepeatCnt
- rxCtrlIdx
- txCtrlIdx
SetCmdConfig()
cmdSequence:
Fifo with 4 entries
Start()
TX/RX
Ranging/Radar

TX Power

Currently not in PTC:
\Desktop\BMW\TxPower_Calculation.xlsx

Data Fields

int8_t **txPowerBoost**

Tx power Boost: The user application with the knowledge on the shorter frames(<1 ms) may use this component to boost the power but still remain compliant with the regulatory limits specified as average power/ millisecs.
Range: 0 dBm to +16 dBm, Resolution : 0.17 dBm step.
This parameter is represented in two's-complement format.
Every 6 counts corresponds to +/-1 dBm of Tx power.
i.e. +0.17dBm(+1), +0.34dBm(+2), +0.51dBm(+3), +0.68dBm(+4), +0.85dBm(+5), +1.0dBm(+6)
. More...

int8_t **txPowerNominal**

Nominal Tx power. This is the nominal Tx power setting applied to the frame and doesn't account for any frame length criteria required for a boost
Range: -21 dBm to +14 dBm, Resolution : 0.17 dBm step.
This parameter is represented in two's-complement format.
Every 6 counts corresponds to +/-1 dBm of Tx power.
i.e. +0.17dBm(+1), +0.34dBm(+2), +0.51dBm(+3), +0.68dBm(+4), +0.85dBm(+5), +1.0dBm(+6)
-0.17dBm(-1), -0.34dBm(-2), -0.51dBm(-3), -0.68dBm(-4), -0.85dBm(-5), -1.0dBm(-6)
. More...

Few examples are represented here
+11.51dBm = +(11*6 + 0.51/0.17) = +(66 + 3)= +69
+8.85Bm = +(8*6 + 0.85/0.17) = +(48 + 5)= +53
+2.00dBm = +(2*6 + 0.0/0.17) = +(12 + 0)= +12
-8.85Bm = -(8*6 + 0.85/0.17) = -(48 + 5)= -53
-11.51dBm = -(11*6 + 0.51/0.17) = -(66 + 3)= -69

int8_t **txPowerOffset**

Tx power offset. This component is derived by executing the standalone Tx CW mode test with Nominal Tx power setting and observing the offset w.r.t Nominal Tx power with Spectral analyzer.
Range: -3 dBm to +3 dBm, Resolution : 0.17 dBm step.
This parameter is represented in two's-complement format.
Every 6 counts corresponds to +/-1 dBm of Tx power.
i.e. +0.17dBm(+1), +0.34dBm(+2), +0.51dBm(+3), +0.68dBm(+4), +0.85dBm(+5), +1.0dBm(+6)
-0.17dBm(-1), -0.34dBm(-2), -0.51dBm(-3), -0.68dBm(-4), -0.85dBm(-5), -1.0dB(-6)
. More...

uint8_t **UNUSED**

UNUSED, Reserved for future use. More...